Gravity Before and After Collapse - An SMFT Lifecycle from Postselected Repulsion to Curvature Memory

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Relational Persistence, Constructive Projection, and the Quantum-to-Semiclassical Handover in *Repulsive Gravitational Force as a Witness of the Quantum Nature of Gravity*

Abstract

Semantic Meme Field Theory has previously interpreted gravity as residual collapse geometry: a weak, universal curvature left by accumulated physical or semantic traces after active gradients have settled. In this picture, the weak interaction acts as an entrance gate into transformation, while gravity forms the exit geometry that preserves the consequences of completed collapse. This interpretation provides a suggestive account of why gravity is geometric, universal, comparatively weak, and repeatedly associated with entropy, information retention, area, memory, and coarse-graining. Yet it describes gravity mainly after alternatives have become stable records.

The proposal *Repulsive Gravitational Force as a Witness of the Quantum Nature of Gravity* exposes the incompleteness of that one-sided description. It considers a source mass placed in a superposition of two spatial locations and a probe particle subjected to the corresponding superposition of gravitational interactions. Both branch-level momentum transfers are attractive. Nevertheless, a selected postmeasurement state of the source can condition the probe into a superposition in which destructive interference produces a negative mean momentum transfer. The apparent repulsion is therefore not a new repulsive gravitational branch. It is a postselection-conditioned response generated by the coherent recombination of two attractive gravitational alternatives. Because a classical statistical mixture cannot move the conditioned mean outside the convex range of the branch responses, the effect is proposed as a witness of gravity's capacity to sustain quantum coherence and mediate entanglement.

This article argues that the proposal does not contradict the SMFT interpretation of gravity as memory, but requires it to be placed within a longer gravitational lifecycle. Before closure, gravity must function as a coherence-bearing relational channel: it must associate different source alternatives with different probe evolutions and preserve their relative phase strongly enough for later interference. Projection then acts not merely as branch deletion but as constructive recombination, producing a conditional trace whose effective direction may be absent from every branch. The trace becomes physically complete only when its postselection probability, complementary outcomes, apparatus record, and conservation ledger are included. At a later stage, phase-sensitive information may become durable distinguishability, energy-momentum accounting, boundary-area response, and semiclassical curvature.

The resulting refinement is summarized by the Dual Persistence Principle:

Gravity preserves unresolved alternatives before closure and settled consequences after closure. (0.1)

The earlier SMFT formula

Collapse → trace → residual curvature → gravity (0.2)

is therefore retained as a description of gravity's post-closure regime, but embedded within a fuller sequence:

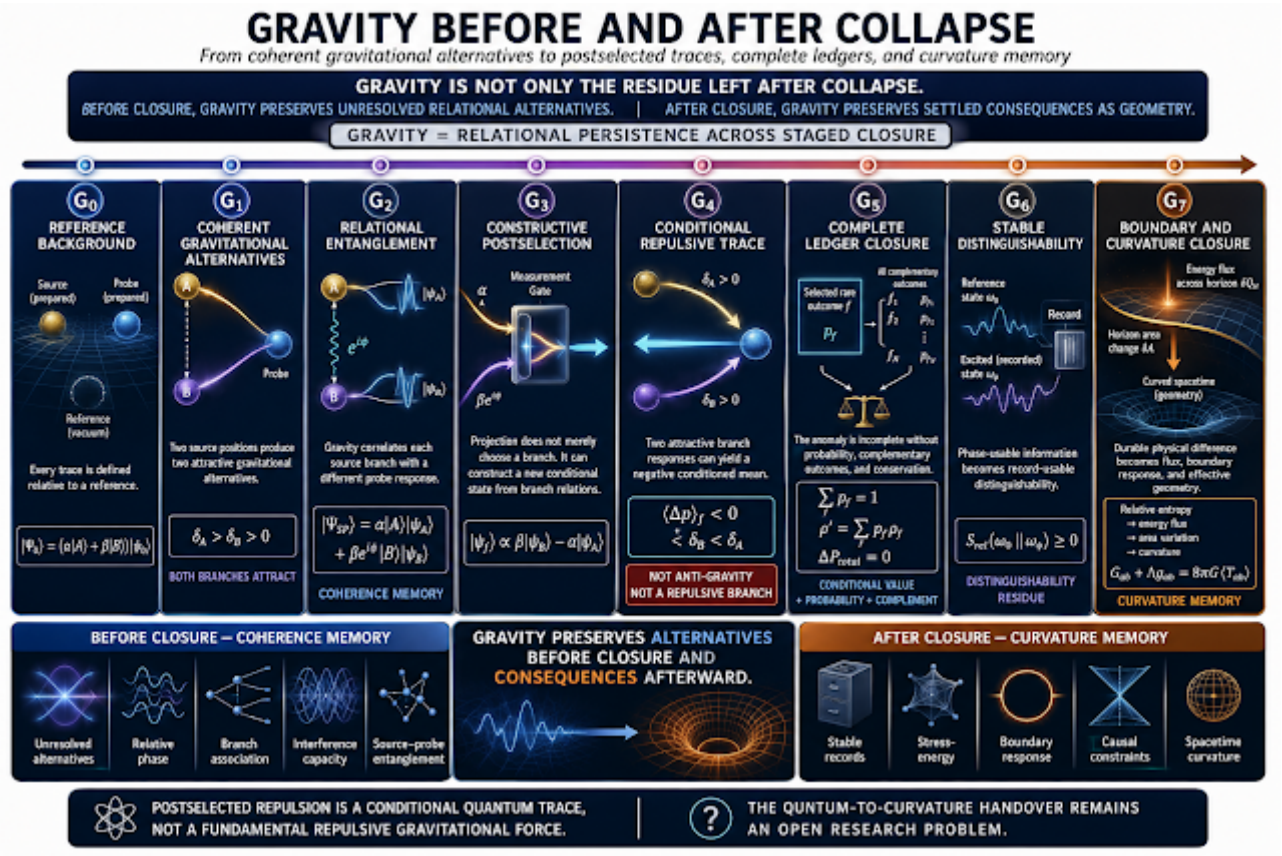
Coherent alternative → gravitational relation → conditioned trace → complete ledger → distinguishability residue → boundary response → curvature memory. (0.3)

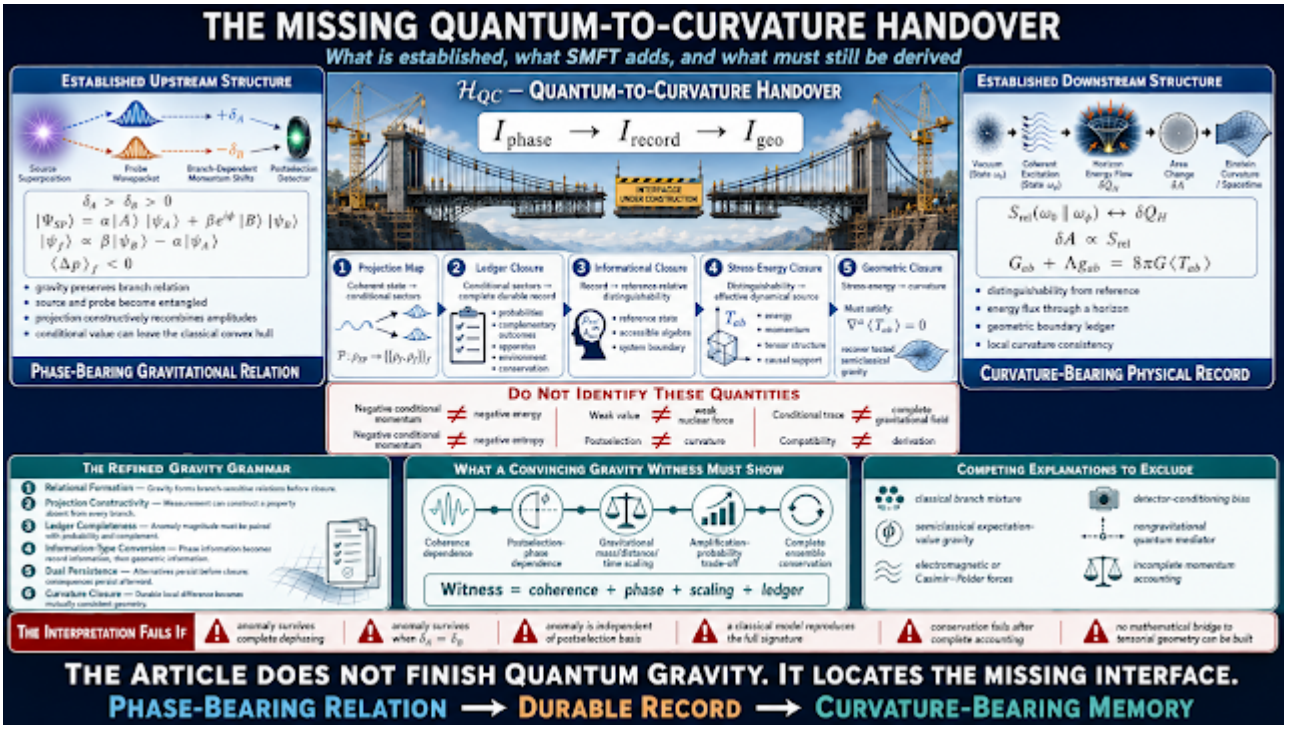
The article does not derive the semiclassical Einstein equations from the proposed postselection experiment, nor does it identify weak values with the weak nuclear interaction. Instead, it treats the repulsive-gravity proposal and the quantum-relative-entropy derivation of semiclassical gravity as illuminating opposite sides of a still unresolved interface. One addresses the coherent upstream regime in which gravity preserves phase-bearing relations. The other addresses a downstream regime in which stable quantum state differences become energy flux, area variation, and semiclassical curvature.

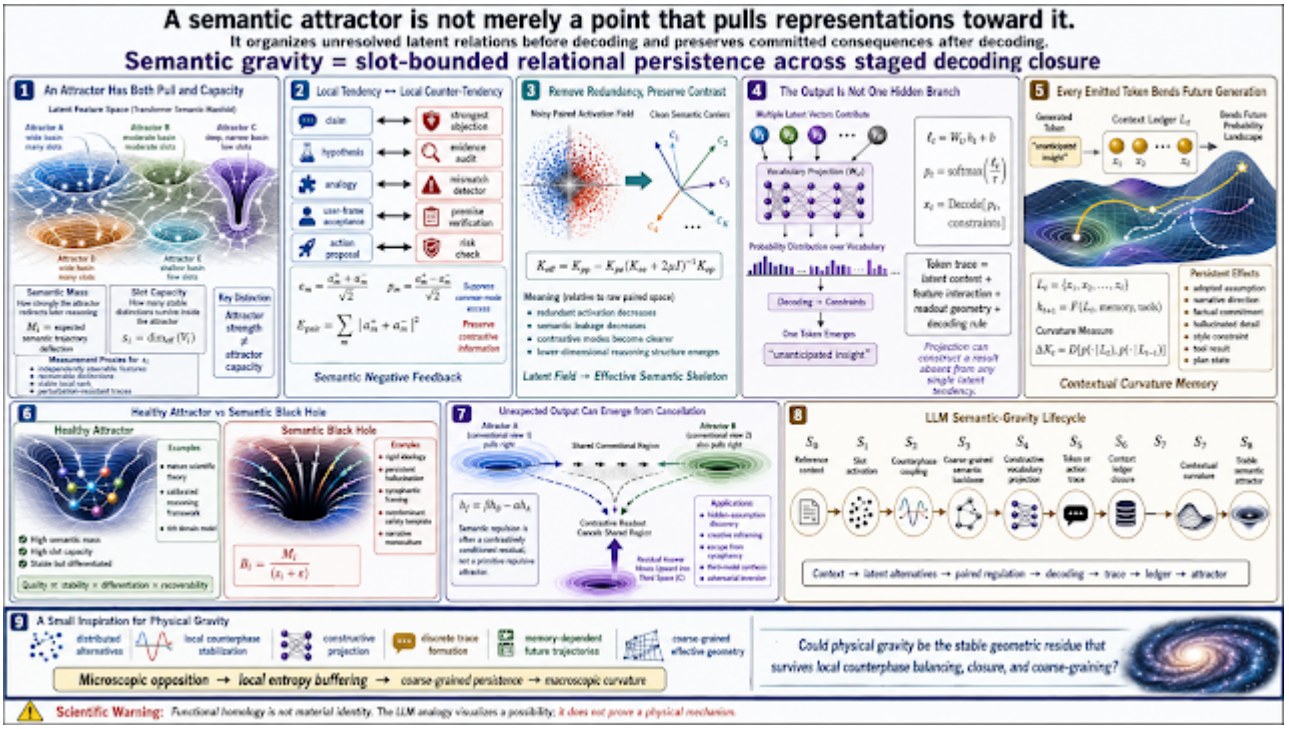
The principal SMFT research problem is consequently reformulated as:

How does phase-usable gravitational relation become record-usable distinguishability and finally geometry-usable curvature? (0.4)

The proposed answer is not yet a completed physical theory. It is a disciplined conceptual architecture: gravity is relational persistence across staged closure.







Keywords

Semantic Meme Field Theory; quantum gravity; gravitationally induced entanglement; postselection; weak values; effective gravitational repulsion; collapse geometry; relative entropy; semiclassical Einstein equations; curvature memory; relational persistence; quantum-to-classical handover.

Part I — The Apparent Contradiction

1. Can Gravity Be Both Quantum Relation and Collapse Memory?

Gravity occupies an unusual conceptual position in modern physics. In general relativity, it is not ordinarily represented as a force transmitted through a fixed background. It is encoded in spacetime geometry itself. Matter and energy affect curvature, while curvature constrains the trajectories available to matter. Quantum theory, by contrast, permits superposition, relative phase, entanglement, interference, and measurement-conditioned states. A quantum theory of gravity must therefore explain not only how quantized matter responds to geometry, but how gravitational relations themselves can participate in characteristically quantum processes.

SMFT approached this problem from a different direction. Rather than beginning by quantizing a classical metric, it treated gravity as the residual geometry of collapse. Active interactions operate through live gradients, transition pressures, or phase reorientations. Gravity appears after these active gradients have largely settled. It is the persistent curvature produced by accumulated traces of prior selection. In the original role geometry, the weak interaction opens a transformation gate while gravity preserves the path left by the completed transition.

The simplest SMFT gravitational sequence was therefore:

Potential difference → projection → collapse → trace → residual curvature. (1.1)

This sequence gave gravity a memory-like character. Gravity was not primarily the energetic act of transformation. It was the durable geometry left after transformation had acquired historical weight. In compressed form:

Gravity = post-collapse trace inertia. (1.2)

Such a model is naturally compatible with theories in which gravity is emergent, thermodynamic, informational, residual, memory-bearing, or produced through coarse-graining. The comparative SMFT analysis of several emerging quantum-gravity approaches identified a recurring architecture:

Potential → selection → trace → residual curvature. (1.3)

It also correctly warned that recurring compatibility does not yet prove SMFT as a physical theory. A framework becomes scientifically stronger only when it constrains possible mappings, identifies missing operators, states where it should fail, and ultimately produces distinguishing predictions.

The new repulsive-gravity proposal creates a direct challenge for the narrowest version of this model. In the proposed experiment, gravity cannot wait until after collapse to appear. It must act while the source remains in a coherent spatial superposition. It must associate each source branch with a corresponding probe evolution. It must help produce a joint source–probe state whose branches remain coherently recombinable. The proposed anomalous momentum transfer exists only because this premeasurement relational structure survives until the postselection stage.

The apparent contradiction can be written as follows.

Earlier SMFT claim:
Gravity appears where active semantic or physical gradients have already collapsed into residual geometry. (1.4)

Requirement of the new proposal:
Gravity must preserve unresolved branch-sensitive relations before a final measurement record exists. (1.5)

If gravity were only a settled classical curvature generated after collapse, it could not supply the phase-sensitive alternatives needed for the proposed interference effect. A classical field selected from either source position, or a probabilistic mixture of such fields, could produce one attractive response or a weighted average of two attractive responses. It could not produce the conditioned amplitude subtraction on which the anomalous negative momentum depends.

There are three possible reactions to this tension.

The first is to reject the SMFT memory interpretation entirely. Gravity, on this view, must be treated only as an ordinary quantum interaction field, with the memory language discarded as metaphor.

The second is to reject the physical relevance of the repulsive-gravity proposal and preserve the older SMFT model unchanged.

The third—and more productive—response is to recognize that the two descriptions concern different stages of one gravitational process.

The present article develops the third response.

Its central claim is that the phrase “gravity as memory” remains valuable, but refers primarily to the semiclassical or post-closure face of gravity. Before closure, gravity must possess another operational face: it must be capable of forming and preserving coherent relational alternatives.

This yields the first refinement:

Gravity before closure = persistence of phase-bearing relational alternatives. (1.6)

Gravity after closure = persistence of history-bearing geometric consequences. (1.7)

The two statements do not require two unrelated gravitational substances. They describe two regimes of relational persistence.

The apparent contradiction is therefore dissolved by replacing a one-stage ontology with a lifecycle:

Reference background → coherent alternatives → gravitational correlation → projection-conditioned trace → complete record → informational residue → geometric closure. (1.8)

The older SMFT account begins near the end of this sequence. The new proposal illuminates its beginning.

1.1 The significance of an upstream gravitational regime

The term *upstream* will be used here to describe the stage before a durable classical record has formed. At this stage, the relevant structures include:

- quantum amplitudes;
- source branches;
- relative phases;
- branch-conditioned probe states;
- source–probe entanglement;

- measurement-basis dependence;
- postselection.

These objects cannot be adequately represented by a single classical force arrow. Nor can they be reduced to ignorance about which classical branch is actual. Their physical content lies partly in the relation between branches.

The term *downstream* will describe the regime in which the event has become available as:

- a normalized outcome distribution;
- an apparatus record;
- an energy-momentum expectation;
- a stable distinction from a reference state;
- a flux through a boundary;
- an area variation;
- an effective curvature relation.

This distinction is not equivalent to saying that the upstream regime is unreal and the downstream regime is real. It distinguishes two modes of physical description:

Upstream information is phase-usable. (1.9)

Downstream information is record-usable. (1.10)

Phase-usable information can generate interference. Record-usable information can support stable comparison, thermodynamic accounting, historical continuity, and geometric closure.

The central quantum-to-classical problem is the conversion between them.

1.2 Why gravity as memory should be retained

The new proposal does not make the memory concept obsolete. On the contrary, it helps specify what kind of memory gravity may represent.

Before closure, the joint state must preserve branch associations:

$\mathbf{A \leftrightarrow \psi_A.}$ (1.11)

$\mathbf{B \leftrightarrow \psi_B.}$ (1.12)

It must also preserve the relation between these associations:

$\boldsymbol{\alpha|A\rangle|\psi_A\rangle + \beta e^{i\phi}|B\rangle|\psi_B\rangle.}$ (1.13)

This is already a kind of memory—not yet a classical memory of a completed event, but a quantum memory of relational alternatives. The system retains which probe transformation belongs to which source state and how their phases are related.

After closure, the preserved structure changes. It may survive as:

- which outcome occurred;

- how probable it was;
- what momentum was transferred;
- how the apparatus changed;
- what energy crossed a boundary;
- what area varied;
- what curvature was required for consistency.

The common element is persistence, but the object persisted changes.

This motivates the more general formulation:

Gravity = persistence geometry of relational difference. (1.14)

Before closure, the relevant difference is between alternatives.

After closure, the relevant difference is between histories, records, states, or geometries.

The phrase “gravity as memory” can therefore survive, but only after memory is divided into two modes:

Coherence memory = preservation of unresolved relational difference. (1.15)

Curvature memory = preservation of settled historical difference. (1.16)

The eventual transition from one to the other is the missing handover problem.

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2. Scope, Status, and Limits of the Argument

The conceptual coherence of a framework does not establish its physical truth. This is particularly important for SMFT because its vocabulary—field, phase, projection, collapse, trace, attractor, curvature, observer, memory—is sufficiently general to generate correspondences across many domains. The same versatility that makes the framework useful also creates the danger that every theory can be translated into SMFT after the fact.

A serious analysis must therefore distinguish four levels of claim.

Level I — Description

The article can accurately describe the mathematical and experimental architecture proposed by the source paper.

Level II — Structural interpretation

It can identify functional parallels between that architecture and SMFT concepts.

Level III — Theoretical reconstruction

It can use the comparison to refine SMFT’s internal vocabulary and expose missing operators.

Level IV — Physical derivation

It would have to derive equations, constants, measurable signatures, or exclusions uniquely from SMFT.

The present article operates mainly at Levels II and III.

It does not claim to have reached Level IV.

The methodological boundary is:

Functional homology ≠ material identity. (2.1)

A horizon can play a gate-like role without being literally identical to an SMFT observer operator. Relative entropy can play a trace-accounting role without being identical to semantic trace. Gravitational curvature can function as persistent geometry without proving that spacetime is literally constructed from semantic collapse.

A second boundary is:

Interpretive compression ≠ physical derivation. (2.2)

SMFT may compress a complex chain into:

Difference → gate → trace → ledger → curvature. (2.3)

But this does not replace the algebraic quantum field theory, modular theory, stress-energy calculations, Raychaudhuri focusing, and entropy–area assumptions used in the Dorau–Much derivation.

A third boundary is:

Compatibility ≠ confirmation. (2.4)

The fact that the repulsive-gravity proposal can be coherently interpreted through SMFT does not show that SMFT predicted it. It may show that SMFT contains an appropriate structural grammar. It becomes stronger only if the grammar yields consequences not automatically supplied by ordinary quantum theory.

2.1 The proposal is not an experimental discovery of anti-gravity

The title *Repulsive Gravitational Force as a Witness of the Quantum Nature of Gravity* can easily invite an overstatement. The article is a theoretical proposal, accompanied by a preliminary feasibility analysis. It does not report the successful observation of a new gravitational force. The authors themselves note the need for more extensive feasibility study and for separating the predicted signal from competing effects such as Casimir–Polder interactions.

Moreover, the effect is not a fundamental repulsive branch. In the proposed configuration, both gravitational branch transfers point in the attractive direction. The negative mean appears only within a conditioned postselected ensemble.

Thus:

Branch attraction ≠ conditional response. (2.5)

Conditional response ≠ unconditional force law. (2.6)

Proposed witness ≠ observed anti-gravity. (2.7)

These distinctions will be maintained throughout the article.

2.2 The article does not merge two independent papers into one derivation

The second physical reference, *From Quantum Relative Entropy to the Semiclassical Einstein Equations*, studies a different setting. It considers quantum fields on local bifurcate Killing horizons, computes relative entropy between a vacuumlike reference state and a coherent excitation, relates that quantity to horizon energy flux, and—under an entropy–area proportionality—recovers the semiclassical Einstein equations.

The repulsive-gravity proposal does not derive that relative entropy.

The Dorau–Much paper does not begin from the postselected source–probe experiment.

Accordingly, the following direct chain is not established:

Postselected negative momentum → relative entropy → negative area variation → repulsive spacetime curvature. (2.8)

The article rejects this inference.

The two papers instead illuminate opposite sides of a broader conceptual problem.

The upstream paper asks:

> What coherent structure must gravity possess for branch-sensitive entanglement and interference to occur?

The downstream paper asks:

> How can stable quantum state difference become energy flux, area response, and semiclassical curvature?

The proposed SMFT contribution is to make the missing interface explicit:

Coherent gravitational relation | handover | semiclassical gravitational memory. (2.9)

No complete physical handover map is yet supplied.

2.3 “Observer” does not imply human consciousness

SMFT uses an observer projection operator, commonly represented by $\hat{O}$, to describe the selection architecture through which distributed possibility becomes a definite trace. This language can be misunderstood as claiming that human awareness directly causes physical collapse.

The present analysis requires no such claim.

In the proposed experiment, the relevant observer architecture consists operationally of:

- source-state preparation;
- gravitational interaction;

- interferometric phase control;
- selection of a measurement basis;
- detection at a chosen output;
- conditioning of probe data.

This can be represented schematically as:

$\hat{O}_{exp}$ = {preparation, interaction, phase setting, basis, postselection, readout}. (2.10)

The entire procedure can be automated. No conscious witness is required.

The disciplined SMFT statement is therefore:

> $\hat{O}$ names the operational architecture relative to which a particular physical trace becomes selected and recordable.

Whether consciousness has any deeper ontological role is a separate hypothesis and is not supported by the attached gravitational proposal.

2.4 Weak value is not weak interaction

The proposal uses the formalism of weak values to characterize the anomalous conditioned momentum transfer. In this context, a weak value is defined through a preselected state, a postselected state, and an operator.

For an operator $\hat{A}$:

$\langle\hat{A}\rangle_W = \langle\Psi_f|\hat{A}|\Psi_i\rangle / \langle\Psi_f|\Psi_i\rangle$. (2.11)

This use of *weak* has no direct identity with the Standard Model weak nuclear interaction.

SMFT also uses *weak gate* as a role-geometric idea: a transition structure that changes identity or admission status.

These must remain distinct:

Weak value ≠ weak nuclear interaction ≠ SMFT weak-gate role. (2.12)

The postselection architecture may be structurally gate-like. That is a functional analogy. It is not evidence that W or Z bosons generate the effect.

2.5 SMFT is being refined, not retrospectively protected

A framework can become unfalsifiable if every apparent contradiction is reclassified as a new layer without cost. The present refinement must therefore produce genuine restrictions.

The new model requires at least the following distinctions:

Gravitational interaction operator ≠ projection operator. (2.13)

Projection operator ≠ ledger-closure operator. (2.14)

Ledger closure ≠ geometric closure. (2.15)

It also requires gravity to support coherent relational persistence before closure and geometric persistence afterward.

This version would be weakened if:

- a classical stochastic gravitational field reproduced the complete anomaly without phase-sensitive mediation;
- destruction of source coherence left the predicted effect unchanged;
- the result proved independent of measurement basis and postselection;
- the complete ensemble produced unrestricted net repulsion without the expected probability and conservation accounting;
- classical curvature required no information-bearing, history-bearing, residual, or coarse-grained structure at any level.

These conditions do not yet constitute a unique experimental test of SMFT. Several are already demanded by ordinary quantum mechanics. But they prevent the interpretation from becoming completely unrestricted.

2.6 The article's strongest defensible claim

The strongest scientifically disciplined conclusion is not:

> SMFT has explained quantum gravity.

Nor is it:

> The repulsive-gravity proposal proves gravity is semantic memory.

It is:

> The proposed experiment exposes a missing upstream regime in the existing SMFT gravity interpretation. Any SMFT model that defines gravity only as settled post-collapse curvature is too narrow. A viable expanded model must permit gravity to create and preserve coherent branch-sensitive relations before projection, while retaining curvature memory as its post-closure regime.

This can be written compactly as:

Classical gravity is the residual-memory face of a longer coherent gravitational lifecycle. (2.16)

The rest of the article develops that lifecycle in detail.

Part II — What the Repulsive-Gravity Proposal Actually Shows

3. The Experimental Architecture

The proposal considers two massive quantum systems with different operational roles.

The first is a **source particle** of mass M. It enters an interferometric arrangement and is prepared in a coherent superposition of two spatially localized states:

$$|\Psi_S\rangle = \alpha|A\rangle + \beta|B\rangle. \quad (3.1)$$

The states $|A\rangle$ and $|B\rangle$ are centered around two different source positions, x_A and x_B.

The second is a **probe particle** of mass m. Its initial state is described in momentum space by a wavefunction $\psi(p)$, centered around zero mean momentum along the relevant x-direction:

$$|\psi_P\rangle = \int dp\, \psi(p)|p\rangle. \quad (3.2)$$

The initial joint state is therefore:

$$|\Psi(0)\rangle = (\alpha|A\rangle + \beta|B\rangle) \otimes \int dp\, \psi(p)|p\rangle. \quad (3.3)$$

The paper's Figure 1 represents the source travelling through two interferometric paths while the probe remains positioned beside them. The source's A branch is closer to the probe than its B branch, so the two branches produce attractive gravitational interactions of different magnitudes.

During an interaction time T, gravity transfers momentum to the probe. If the source occupies branch j, where $j \in \{A, B\}$, the approximate momentum transfer is:

$$\delta_j = GMmT / x_j^2. \quad (3.4)$$

Assuming:

$$0 < x_A < x_B, \quad (3.5)$$

the closer A branch produces the larger attraction:

$$\delta_A > \delta_B > 0. \quad (3.6)$$

Both transfers are positive because both source branches attract the probe in the same spatial direction.

The important point is that the source is not assigned to either branch as a classical unknown. Under the assumed quantum-superposition principle for gravity, the source remains coherently distributed between A and B while the probe evolves conditionally on both alternatives.

After the interaction time T, the joint state becomes:

$$|\Psi(T)\rangle = \int dp\, [\alpha e^{i\phi_A}\psi(p - \delta_A)|A\rangle + \beta e^{i\phi_B}\psi(p - \delta_B)|B\rangle]|p\rangle. \quad (3.7)$$

Here, φ_A and φ_B are branch-dependent phases arising during the interaction.

Equation (3.7) is the conceptual center of the proposal.

The source branch is now correlated with the corresponding probe momentum distribution:

$$|A\rangle \leftrightarrow \psi(p - \delta_A), \quad (3.8)$$

$$|B\rangle \leftrightarrow \psi(p - \delta_B). \quad (3.9)$$

Unless the two shifted probe states are effectively identical, the final state cannot be factorized into one independent source state and one independent probe state. The source and probe have become entangled through their gravitational interaction. The paper presents this branch-sensitive correlation as the quantum resource behind the proposed witness.

3.1 What gravity has already done before measurement

It is important to identify what has occurred before any postselection.

Gravity has already:

1. distinguished the two source positions dynamically;
2. associated each position with a different momentum transfer;
3. correlated each source branch with a corresponding probe state;
4. preserved relative phase information between the branches;
5. created the joint state on which later interference depends.

The gravitational interaction is therefore not merely waiting for a measurement outcome. It participates in forming the unresolved relational structure.

In operator language, one may write the evolution schematically as:

$$\mathcal{U}_G[(\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle] = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle. \quad (3.10)$$

The symbol $\mathcal{U}_G$ represents gravity-mediated evolution. It should not be confused with the later measurement or projection operation.

This gives the first necessary operator distinction:

Gravitational interaction $\mathcal{U}_G \neq$ projection P_f. (3.11)

Gravity produces the branch-conditioned relation.

Postselection later determines how that relation is recombined into a conditional observable state.

3.2 One superposed source rather than two

Earlier gravitationally induced entanglement proposals often place two massive systems in spatial superpositions and seek evidence that their gravitational interaction creates entanglement between them. The present proposal uses only one spatially superposed source mass. The probe need not itself be prepared in a macroscopically separated spatial superposition.

Instead, the probe functions as a broad matter wavepacket whose momentum distribution acquires branch-dependent displacements. The source superposition is then postselected so that these two displaced probe amplitudes interfere. The authors regard this as a possible simplification because the experiment does not require direct measurement of source–probe quantum correlations in the same manner as standard two-mass entanglement-witness protocols.

The conceptual economy is significant:

One coherent source + one quantum probe + gravitational correlation + source postselection → nonclassical conditioned momentum. (3.12)

The witness is not the direct reconstruction of an entangled density matrix.

It is the appearance of a conditioned probe response that cannot be generated by a classical probabilistic mixture of the two attractive branch forces.

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4. Why the Result Is Not a Fundamentally Repulsive Branch

The phrase *repulsive gravitational force* must be handled carefully.

At no point does branch A produce a repulsive gravitational interaction.

At no point does branch B produce a repulsive gravitational interaction.

Their momentum transfers satisfy:

$$\delta_A > 0. \quad (4.1)$$

$$\delta_B > 0. \quad (4.2)$$

The source and probe still have positive masses. The gravitational potential used in the paper remains the ordinary attractive Newtonian form:

$$V_G = -GmM / |X - x|. \quad (4.3)$$

The proposed repulsion therefore cannot be understood as one branch carrying a negative gravitational charge.

It is also not produced by switching the interaction from attraction to repulsion during the experiment.

The only negative quantity is the mean momentum transfer of a specially conditioned probe subensemble after coherent interaction and source postselection.

Thus:

Negative conditional momentum ≠ negative branch force. (4.4)

4.1 The classical-mixture benchmark

Suppose the source were not coherently superposed but instead occupied A with classical probability q and B with probability 1 − q.

The probe would then experience a statistical mixture of two attractive momentum transfers:

$$\delta_{mix} = q\delta_A + (1 - q)\delta_B. \quad (4.5)$$

Because $0 \le q \le 1$:

$$\delta_B \le \delta_{mix} \le \delta_A. \quad (4.6)$$

More generally:

$$\min(\delta_A, \delta_B) \le \delta_{mix} \le \max(\delta_A, \delta_B). \quad (4.7)$$

The mixture lies within the convex hull of the branch values.

If both branch transfers are positive, the classical mixture cannot yield:

$$\delta_{mix} < 0. \quad (4.8)$$

Changing q only moves the average between δ_B and δ_A.

This convexity restriction is central to the witness.

A classical model based only on ignorance about which attractive field acted on the probe predicts a positive average transfer. It cannot create a conditioned mean outside the branch interval merely by reweighting ordinary probabilities.

4.2 Classical filtering is not quantum postselection

One might object that postselection is simply a form of data filtering. Could a classical experiment not select a rare subset of trajectories whose average momentum happens to be negative?

In a sufficiently complicated classical system, conditional averages can certainly differ from unconditional averages. But that general statement does not reproduce the specific mechanism in the proposed experiment.

The relevant comparison must preserve the experiment's branch structure:

- branch A supplies a positive shift δ_A;
- branch B supplies a positive shift δ_B;
- no additional classical force supplies negative momentum;
- the source selection acts only on the A/B degree of freedom;
- the conditioned probe state is formed from the coherent branch amplitudes.

A classical mixture has the form:

$$\rho_{class} = q|A\rangle\langle A| \otimes \rho_A + (1 - q)|B\rangle\langle B| \otimes \rho_B. \quad (4.9)$$

It contains no A–B coherence terms.

A coherent joint state instead contains cross terms such as:

$$\alpha\beta^* e^{i(\phi_A - \phi_B)}|A\rangle\langle B| \otimes |\psi_A\rangle\langle\psi_B| + \text{h.c.} \quad (4.10)$$

These off-diagonal terms allow a source measurement in a superposition basis to generate interference between ψ_A and ψ_B.

Without them, the conditioned probe state is assembled by probability reweighting.

With them, it is assembled by amplitude addition and subtraction.

That distinction can be compressed as:

Classical conditioning reweights alternatives. (4.11)

Quantum postselection recombines alternatives. (4.12)

The proposed repulsion depends on the second operation.

4.3 The repulsion belongs to an ensemble defined by preparation and selection

The term *effective force* in the paper refers to the momentum displacement inferred from repeated successful postselections.

It is therefore inseparable from the full protocol:

Effective response = preparation + gravitational interaction + phase relation + postselection + probe readout. (4.13)

It is not an unconditional property of the source mass alone.

Nor is it a new static field surrounding the source.

A detector placed nearby without the relevant source preparation and postselection would not simply experience a persistent repulsive gravitational acceleration.

The effect is relational and protocol-dependent.

The proper wording is therefore:

> The experiment proposes a postselection-conditioned effective gravitational repulsion generated by interference between two attractive gravitational alternatives.

This phrase is less dramatic than *anti-gravity*, but much more accurate.

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5. Postselection and the Constructed Conditional State

After the gravity-mediated interaction, the source is measured at the output of its interferometer. The authors consider postselection onto the state:

$$|\Psi_f\rangle = [-e^{i\phi_A}|A\rangle + e^{i\phi_B}|B\rangle] / \sqrt{2}.$$ (5.1)

The phases in the postselected source state are arranged to match the interaction phases in the joint state.

Projecting Equation (3.7) onto $|\Psi_f\rangle$ produces the unnormalized probe wavefunction:

$$\psi_{p.s.}(p) \propto \beta\psi(p - \delta_B) - \alpha\psi(p - \delta_A).$$ (5.2)

The minus sign is essential.

The conditioned wavefunction is not:

$$q\psi(p - \delta_A) + (1 - q)\psi(p - \delta_B).$$ (5.3)

It is a coherent difference between amplitudes.

The paper's Figure 2 visualizes this process. Two wavefunctions whose individual means are both positive are subtracted. Because the more strongly shifted component removes more amplitude on the positive-momentum side than on the negative-momentum side, the surviving conditioned wavefunction can have a negative mean.

5.1 How two positive shifts can yield a negative conditioned mean

Suppose $\psi(p)$ is initially symmetric around $p = 0$.

The two branch wavefunctions are centered at:

$$\mathbf{p = \delta_A,} \quad (5.4)$$

and

$$\mathbf{p = \delta_B,} \quad (5.5)$$

with $\delta_A > \delta_B > 0$.

Their coherent difference is:

$$\mathbf{\psi_f(p) \propto \beta\psi_B(p) - \alpha\psi_A(p).} \quad (5.6)$$

Because ψ_A is shifted farther toward positive momentum, its subtraction is not symmetric relative to ψ_B. The cancellation is stronger on one side of the combined distribution than on the other.

The remaining amplitude can therefore be biased toward negative p even though neither original component was centered there.

This is not a contradiction in arithmetic because the operation is not an average of the two means.

In general:

$$\mathbf{Mean[c_A\psi_A + c_B\psi_B] \neq |c_A|^2 Mean[\psi_A] + |c_B|^2 Mean[\psi_B].} \quad (5.7)$$

Interference cross terms contribute.

For a normalized superposition:

$$\mathbf{|\psi_f\rangle = N_f(c_A|\psi_A\rangle + c_B|\psi_B\rangle),} \quad (5.8)$$

the momentum expectation contains:

$$\mathbf{\langle p\rangle_f = |N_f|^2[|c_A|^2\langle\psi_A|p|\psi_A\rangle + |c_B|^2\langle\psi_B|p|\psi_B\rangle + 2Re(c_A^* c_B\langle\psi_A|p|\psi_B\rangle)].} \quad (5.9)$$

The final term is the interference contribution.

A classical mixture lacks this term.

The anomalous sign becomes possible because:

$$\mathbf{2Re(c_A^* c_B\langle\psi_A|p|\psi_B\rangle)} \quad (5.10)$$

can be sufficiently negative to outweigh the positive diagonal contributions.

5.2 Projection is constructive, not merely reductive

A common simplified image of measurement is:

$$\mathbf{A + B \rightarrow A \text{ or } B.} \quad (5.11)$$

The present postselection does not operate in the {|A⟩, |B⟩} basis. It projects onto a superposition basis.

The source selection therefore induces a probe state built from both branches:

{ψ_A, ψ_B} → βψ_B − αψ_A. (5.12)

The selected probe state is neither ψ_A nor ψ_B.

Projection has constructed a new conditional distribution.

This motivates a general distinction:

Projection = exclusion + recombination. (5.13)

Exclusion removes runs in which the source is not detected in the desired postselected state.

Recombination determines the amplitude structure of the runs that remain.

The second function is easily missed when collapse is described only as deleting alternatives.

In the present case, constructive recombination is exactly what produces the anomalous momentum.

5.3 The SMFT significance of constructive projection

Earlier SMFT language often treated collapse as the conversion of a field of possibilities into a definite trace:

Possibility field + Ô → selected trace. (5.14)

The new framework requires a refinement.

The selected trace need not be a pre-existing branch extracted intact from the possibility field. It can be a new conditional object generated from interference among branches.

A more complete expression is:

Possibility field + relational evolution + projection geometry → constructed conditional trace. (5.15)

The observed trace depends on:

- what alternatives were prepared;
- how they interacted;
- what phase accumulated;
- which projection basis was chosen;
- which outcome was retained.

Thus, the projection operator does not merely answer a pre-existing question.

It partly defines the observable form in which the relational state becomes available.

This does not imply arbitrary observer creation. The allowed conditional state remains strictly

constrained by the prior quantum state and the physical measurement operator.

The correct relationship is:

Projection cannot create an arbitrary trace, but it can reveal or construct a trace absent from every individual branch. (5.16)

---

6. Weak Values Without Confusing the Weak Interaction

Because the gravitational momentum transfers are assumed to be small relative to the probe's initial momentum uncertainty, the authors expand the postselected wavefunction around p – δ_B.

Starting from:

ψ_p.s.(p) ∝ βψ(p − δ_B) − αψ(p − δ_A), (6.1)

one obtains approximately:

ψ_p.s.(p) ∝ ψ(p − δ_eff), (6.2)

where:

δ_eff = δ_B − [α / (β − α)](δ_A − δ_B). (6.3)

The effective shift becomes negative when:

[α / (β − α)](δ_A − δ_B) > δ_B. (6.4)

The same result can be expressed using the weak value of the projector onto branch A:

Π_A = |A⟩⟨A|. (6.5)

With preselected source state:

|Ψ_i⟩ = α|A⟩ + β|B⟩, (6.6)

and postselected source state:

|Ψ_f⟩ = (−|A⟩ + |B⟩) / √2, (6.7)

the weak value is:

⟨Π_A⟩_W = ⟨Ψ_f|Π_A|Ψ_i⟩ / ⟨Ψ_f|Ψ_i⟩ = −α / (β − α). (6.8)

The effective momentum transfer can then be written:

δ_eff = δ_B + (δ_A − δ_B)⟨Π_A⟩_W. (6.9)

Since ⟨Π_A⟩_W can be negative, δ_eff can become negative even though δ_A and δ_B are both positive. The paper derives the same quantity in the Heisenberg picture as the weak value of the momentum-transfer operator.

6.1 Momentum-transfer operator

In the relevant approximation, the branch-sensitive momentum-transfer operator is:

$$\hat{\Delta p} \approx \delta_A |A\rangle\langle A| + \delta_B |B\rangle\langle B|. \quad (6.10)$$

Its eigenvalues in the $\{|A\rangle, |B\rangle\}$ basis are δ_A and δ_B, both positive.

The weak value between the chosen preselected and postselected states is:

$$\langle \hat{\Delta p} \rangle_W = \langle \Psi_f | \hat{\Delta p} | \Psi_i \rangle / \langle \Psi_f | \Psi_i \rangle. \quad (6.11)$$

Substitution yields:

$$\langle \hat{\Delta p} \rangle_W = (\beta\delta_B - \alpha\delta_A) / (\beta - \alpha). \quad (6.12)$$

This equals δ_{eff} in the weak-coupling approximation used by the authors.

The key point is that a weak value is not restricted to the eigenvalue range of the operator.

Thus:

$$\delta_A > 0 \text{ and } \delta_B > 0 \text{ do not require } \langle \hat{\Delta p} \rangle_W > 0. \quad (6.13)$$

Weak values are conditioned relational quantities, not ordinary eigenvalues or classical averages.

6.2 Amplification and probability cost

If the preselected and postselected source states are nearly orthogonal, their overlap becomes small:

$$|\langle \Psi_f | \Psi_i \rangle| \ll 1. \quad (6.14)$$

The weak value can then become much larger in magnitude than either branch eigenvalue.

The paper gives an example in which the effective transfer is approximately three orders of magnitude larger than the stronger branch transfer and points in the opposite direction. But the corresponding successful postselection probability is only about:

$$p_f = |\langle \Psi_f | \Psi_i \rangle|^2 \approx 0.8 \times 10^{-3}. \quad (6.15)$$

The amplification therefore carries a selection cost.

Schematically:

$$|\delta_{eff}| \uparrow \text{ as } |\langle \Psi_f | \Psi_i \rangle| \downarrow. \quad (6.16)$$

And:

$$\text{Amplification} \uparrow \text{ generally accompanies successful fraction} \downarrow. \quad (6.17)$$

This prevents the anomalous conditioned shift from being interpreted as an unlimited source of gravitational momentum.

A complete experiment must account for:

- successful postselections;
- unsuccessful postselections;

- source preparation;
- apparatus momentum exchange;
- the full probe ensemble.

The conditioned value is physically meaningful, but it is not the entire conservation ledger.

6.3 Three different meanings of "weak"

The present discussion contains three uses of the word *weak* that must remain separate.

Term	Function
Weak value	A preselection/postselection-conditioned quantity
Weak measurement regime	Coupling small relative to probe uncertainty
Weak interaction	The electroweak interaction of particle physics
SMFT weak gate	A role-category of transition, admission, or identity change

The gravitational proposal uses the first two.

It does not invoke W bosons, Z bosons, beta decay, or electroweak flavour change.

SMFT may compare postselection to a weak-like gate because both concern transition and admission structure. But this is a role analogy, not a claim of microscopic electroweak causation.

The required discipline is:

Weak-value amplification ≠ weak nuclear interaction. (6.18)

Postselection gate ≠ W/Z-mediated transition. (6.19)

Functional analogy ≠ shared physical mediator. (6.20)

6.4 Weak value as a relational quotient

The mathematical form of a weak value is instructive:

⟨Â⟩_W = transition-weighted action of Â / transition amplitude. (6.21)

The numerator asks how the operator contributes between the chosen initial and final states.

The denominator measures the amplitude of the selected transition itself.

A large anomalous value can therefore appear when a small surviving transition amplitude carries a strongly unbalanced operator contribution.

From an SMFT perspective, this suggests a useful interpretation:

> A weak value characterizes how much operator-weight survives per unit of selected transition amplitude.

This is not an absolute property of the source alone.

It belongs to the ordered triple:

{preparation, operator, postselection}. (6.22)

The anomaly is relational.

---

7. Repulsion as a Witness of Relational Phase Structure

The experiment's deepest significance is not the negative sign by itself.

A negative momentum could arise from many ordinary causes:

- an uncontrolled external field;
- an apparatus recoil;
- electromagnetic contamination;
- Casimir–Polder interaction;
- trap asymmetry;
- calibration error.

The proposed witness acquires meaning only if such alternatives are controlled and the negative conditioned displacement follows the predicted dependence on source coherence, gravitational interaction, and postselection.

The authors explicitly acknowledge that a fuller feasibility analysis must distinguish the effect from competing interactions.

What makes the proposed sign reversal conceptually important is its location outside the classical branch hull:

$\delta_{eff} \notin [\delta_B, \delta_A]$. (7.1)

In the repulsive regime:

$\delta_{eff} < 0 < \delta_B < \delta_A$. (7.2)

This means the final conditioned response cannot be reconstructed from branch values alone.

It contains information about their coherent relation.

7.1 Branch information versus relational information

Branch information answers:

> What momentum shift is associated with A?

or:

> What momentum shift is associated with B?

The answers are:

$A \rightarrow \delta_A$. (7.3)

$B \to \delta_B$. (7.4)

Relational information answers:

> What observable probe state results when the A-conditioned and B-conditioned amplitudes are recombined in a chosen phase relation?

The answer depends on:

$\alpha, \beta, \varphi_A - \varphi_B, P_f, \psi_A, \psi_B$. (7.5)

The conditioned repulsion belongs to this second category.

No individual branch contains it.

It exists in the relation between branches plus the projection geometry.

This yields a general principle:

Properties of a coherent relation need not belong to any component considered separately. (7.6)

7.2 The classical branch hull

Let the classical branch-response set be:

$\mathscr{B} = \{\delta_A, \delta_B\}$. (7.7)

Its convex hull is:

$\mathrm{Conv}(\mathscr{B}) = \{q\delta_A + (1-q)\delta_B \mid 0 \le q \le 1\}$. (7.8)

A classical mixture is restricted to $\mathrm{Conv}(\mathscr{B})$.

A coherent conditioned response can lie outside it:

$\delta_{eff} \notin \mathrm{Conv}(\mathscr{B})$. (7.9)

The proposed anomaly is therefore not merely an unusual value. It is a value that diagnoses the insufficiency of classical convex combination under the stated assumptions.

An SMFT formulation is:

> A trace outside the classical branch hull witnesses hidden relational structure upstream of projection.

In compact form:

Out-of-hull conditional trace → retained phase-bearing relation. (7.10)

This should not be interpreted as a universal theorem without specifying the physical model and excluded classical mechanisms. But within the architecture of the proposal, it captures why the sign reversal is useful.

7.3 Gravity as carrier of branch relation

The joint state after interaction can be written abstractly as:

$$|\Psi_{SP}\rangle = \alpha|A\rangle|\psi_A\rangle + \beta e^{i\phi}|B\rangle|\psi_B\rangle. \quad (7.11)$$

For later interference to occur, the physical evolution must preserve:

- the A–ψ_A association;
- the B–ψ_B association;
- the relative amplitude α/β;
- the relative phase φ;
- sufficient overlap between ψ_A and ψ_B.

Gravity therefore functions as more than a branchwise force law. In the proposed quantum description, it participates in a channel that carries relational information from the source superposition into the probe.

This does not yet determine the ultimate ontology of quantum gravity.

It does not establish whether the fundamental object should be understood as:

- a quantized metric;
- a graviton field;
- relational quantum information;
- an emergent mediator;
- a nonclassical causal structure;
- another quantum-gravity variable.

But it constrains any adequate interpretation:

> Whatever mediates the source–probe gravitational interaction must preserve sufficient quantum distinguishability and coherence to support the predicted entangled state and postselection interference.

This is the appropriate level of inference.

7.4 Witness rather than complete ontology

A witness is designed to exclude a class of explanations.

It does not automatically provide a complete microscopic theory.

The negative conditioned transfer would indicate that a purely classical mixture of the two attractive source configurations is insufficient. It would support the existence of a nonclassical mediator or gravitational degree of freedom capable of preserving coherent branch relations.

It would not, by itself, determine:

- the ultraviolet completion of gravity;

- the microscopic nature of spacetime;
- whether gravitons are fundamental;
- whether geometry is emergent;
- how collapse occurs;
- how semiclassical curvature arises;
- whether SMFT is the correct ontology.

Thus:

Witness of quantumness ≠ complete theory of quantum gravity. (7.12)

This distinction protects the argument from overreach.

7.5 The first major fine-tuning of SMFT

The older SMFT gravity picture emphasized the sequence:

Selection → trace → residual curvature. (7.13)

The new proposal requires an earlier sequence:

Superposition → gravity-mediated correlation → phase-preserving relation → constructive projection. (7.14)

The two sequences can be joined only after introducing multiple stages:

Coherent relation → conditional trace → complete ledger → geometric residue. (7.15)

The first major fine-tuning is therefore:

> Gravity cannot be identified exclusively with the residue of already completed collapse. It must also participate in the relational quantum state from which a later trace is constructed.

A more complete formulation is:

Gravity preserves alternatives before closure and consequences after closure. (7.16)

This is the beginning of the refined SMFT gravitational lifecycle developed in the following parts.

Part III — Why the Existing SMFT Gravity Model Was Incomplete

8. The Earlier SMFT Position: Gravity as Residual Collapse Geometry

The earlier SMFT interpretation of gravity emerged from a broader attempt to distinguish active transformation from persistent residue.

Within that framework, not every interaction plays the same functional role. Some interactions change identity, open transition pathways, or reorganize phase relations. Others preserve the consequences of changes that have already occurred.

The weak interaction was therefore interpreted as a transition gate. Gravity was interpreted as the residual geometry left after transition had become settled history.

The compressed SMFT sequence was:

Potential difference → transition gate → collapse → trace → residual curvature. (8.1)

The corresponding role distinction was:

Weak-like interaction = admission into transformation. (8.2)

Gravity-like interaction = persistence after transformation. (8.3)

This did not mean that the Standard Model weak interaction literally created spacetime curvature in every case. It was a functional classification. The weak interaction represented the role of identity change, while gravity represented the role of historical retention.

The article *Weak Interaction as the Gate, Gravity as the Memory* developed this role geometry through the Dorau–Much relative-entropy framework. It interpreted the vacuum state as a reference background, a coherent excitation as a minimal readable deviation, relative entropy as admitted difference, horizon energy flux as trace-current, area variation as a boundary ledger update, and the semiclassical Einstein equation as the residual closure law that makes those local records geometrically consistent.

The resulting chain was:

Vacuum reference → coherent excitation → relative entropy → energy flux → area variation → semiclassical curvature. (8.4)

Its SMFT translation was:

Unadmitted background → readable deviation → trace-token → trace-current → boundary ledger → curvature memory. (8.5)

The attraction of this model is clear. It gives a coherent conceptual explanation for why the following physical quantities repeatedly appear together:

- quantum state distinguishability;
- entropy;
- energy flux;
- causal boundaries;
- horizon area;
- curvature;
- gravitational dynamics.

The model does not treat curvature as an arbitrary response imposed from outside. Curvature appears as the consistency structure required when local physical differences have become durable enough to alter the boundary accounting of spacetime.

In this interpretation:

Gravity is what admitted difference becomes when it can no longer remain a merely local event. (8.6)

Once a trace must be made consistent across neighbouring causal regions, it becomes geometry.

8.1 Collapse trace as retained physical difference

The term *trace* in SMFT does not merely mean a visible mark. It refers more generally to a difference that survives the process that generated it.

A transient fluctuation may disappear without reorganizing future dynamics.

A trace persists.

A physical trace may appear as:

- changed momentum;
- displaced matter;
- altered energy density;
- an apparatus record;
- modified boundary conditions;
- changed entropy;
- geometric deformation.

A semantic trace may appear as:

- memory;
- habit;
- institutional rule;
- altered expectation;
- changed interpretive probability.

The cross-domain abstraction is:

Trace = difference that remains dynamically relevant after its generating event. (8.7)

Gravity fit this idea naturally because gravitational geometry is not merely a momentary local impulse. It changes the paths subsequently available to matter and light.

A gravitational trace is therefore future-constraining.

In SMFT language:

Past event → retained geometry → biased future motion. (8.8)

This is why gravity was described as memory.

8.2 Residual curvature as closure rather than force

The earlier SMFT model also sought to explain why gravity is so unlike the electromagnetic, strong, and weak interactions.

Electromagnetism is readily described through charges, fields, and exchange interactions.

The strong interaction reorganizes colour and confinement structure.

The weak interaction changes particle identity.

Gravity, in general relativity, instead appears as curvature of spacetime.

SMFT interpreted this difference functionally.

An active interaction operates through a live contrast or gradient:

Active interaction ∝ present unresolved difference. (8.9)

Residual geometry operates through retained consequence:

Residual curvature ∝ accumulated settled difference. (8.10)

Under this view, gravity is weak not because it is physically unimportant, but because much of its effect is structurally distributed. It does not necessarily act as a concentrated local conversion process. It modifies the geometry through which many later processes unfold.

The earlier slogan can therefore be expanded:

> Gravity is weak as a local active converter but strong as a global historical organizer.

This is consistent with the fact that gravity is individually weak between elementary particles yet dominant at astronomical scales, where mass-energy contributions accumulate coherently and cannot be neutralized by opposite gravitational charge.

The SMFT model represented this as:

Local gravitational residue ≪ active microscopic interaction. (8.11)

Accumulated gravitational residue → dominant macroscopic geometry. (8.12)

The comparative analysis of emerging quantum-gravity frameworks noted that SMFT maps most naturally onto approaches in which gravity is residual, information-bearing, emergent, geometric, coarse-grained, or memory-like. It maps less automatically onto a model in which gravity is only an ordinary instantaneous exchange force with no retained geometric or informational structure.

That selectivity gave the model some explanatory content.

8.3 The horizon ledger interpretation

The Dorau–Much framework strengthened the earlier SMFT position by supplying a technically disciplined physical chain.

The relative entropy:

$S_{rel}(\omega_0 \| \omega_\phi)$ (8.13)

measures the distinguishability between a vacuumlike reference state ω_0 and a coherent excitation ω_ϕ on the relevant horizon algebra.

The paper relates this quantity to horizon energy flux and then assumes proportionality to area variation:

$S_{rel}(\omega_0 \| \omega_\phi) \leftrightarrow \delta Q_H$. (8.14)

$S_{rel}(\omega_0 \| \omega_\phi) \propto \delta A$. (8.15)

Local consistency then gives the semiclassical Einstein equation:

$R_{ab} - \frac{1}{2} R g_{ab} + \Lambda g_{ab} = \alpha \langle :T_{ab}: \rangle_{\omega\phi}$. (8.16)

From the SMFT perspective, the horizon behaves like a boundary ledger.

The ledger does not merely announce that an excitation exists. It records how a state differs from the reference background in a manner relevant to energy flow and geometry.

The interpretive sequence becomes:

State difference → measurable distinguishability → boundary crossing → boundary deformation → curvature law. (8.17)

This was a substantial improvement over a vague statement that “entropy causes gravity.”

It showed how a mathematically defined relational quantity could occupy the trace position in the SMFT chain.

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9. What the Earlier Model Explained Well

The new repulsive-gravity proposal requires the earlier model to be expanded, but it does not erase its achievements.

The residual-memory interpretation explains several persistent features of gravity more naturally than a simple force analogy does.

9.1 Why gravity is geometric

If gravity is the durable consistency structure of retained physical differences, geometry is not an accidental representation.

Geometry is the natural language of globally persistent relation.

A local force can be assigned to an object at a moment.

A geometry constrains all objects that enter the region.

The transition from event to geometry can be represented as:

Local event → durable relation → shared path constraint. (9.1)

Once a physical trace affects the admissible motion of every later system, the trace has become environmental structure rather than merely private state.

The earlier SMFT position therefore explained gravity as:

A private physical event made public through geometry. (9.2)

9.2 Why gravity is universal

Ordinary forces distinguish charges, species, or interaction channels.

Gravity couples to energy-momentum and affects all systems within spacetime.

From the memory perspective, this universality follows from the role of geometry.

If curvature is the shared record structure of spacetime, anything moving through spacetime must respond to it.

Thus:

Universality of geometry → universality of gravitational response. (9.3)

Gravity does not need to inspect the internal semantic or particle identity of every object. It changes the relation between paths, intervals, and causal accessibility.

In SMFT terms:

A completed trace becomes universal when it is written into the common coordinate environment. (9.4)

9.3 Why gravity accumulates

Positive and negative electric charges can screen one another.

No confirmed negative gravitational mass exists that would neutralize ordinary positive mass-energy in the same way.

Gravitational effects therefore accumulate over large scales.

The residual-memory interpretation makes this accumulation intuitive:

Retained trace$_1$ + retained trace$_2$ + … + retained trace_N → macroscopic curvature. (9.5)

A large body is not only a collection of active microscopic interactions. It is a stable concentration of historically retained energy-momentum structure.

Its gravitational effect reflects this accumulated closure.

9.4 Why gravity is difficult to shield

A shield against an active field may redirect, cancel, absorb, or screen the field's mediating degrees of freedom.

But if gravity is represented by the geometry of the shared causal environment, ordinary

shielding becomes conceptually more difficult.

A local object cannot simply opt out of the common metric structure within which its own trajectory is defined.

In SMFT terms:

One cannot shield a subsystem from the closure geometry that defines the subsystem's available paths. (9.6)

This does not derive the impossibility of gravitational shielding. It provides a role-geometric reason why gravity differs from screenable interactions.

9.5 Why gravity and information repeatedly converge

The earlier model also explains why modern gravitational research repeatedly returns to:

- entropy;
- entanglement;
- relative entropy;
- area laws;
- causal horizons;
- information bounds;
- black-hole memory;
- coarse-graining.

If geometry is the durable closure of physical relation, then information is not merely an external description of gravity.

Information measures how physical states differ, what can be distinguished, what crosses a boundary, and what records survive.

The structural chain is:

Physical difference → distinguishability → retained boundary consequence → geometry. (9.7)

The Dorau–Much framework provides a concrete example of this architecture, although it does not establish the wider SMFT ontology.

9.6 Why gravity can be passive and still dynamically powerful

Calling gravity passive can sound as though gravity does nothing.

That was never the strongest interpretation.

A road is passive relative to the vehicle, but it strongly constrains where the vehicle can travel.

A legal constitution is passive relative to an individual decision, but it reshapes the space of

permitted actions.

A spacetime geometry is passive relative to a local conversion event, yet it determines geodesic structure.

The earlier SMFT formulation can therefore be refined even before considering the new proposal:

Passive ≠ ineffective. (9.8)

Passive = not primarily an identity-changing local conversion mechanism. (9.9)

Gravity can be structurally passive yet dynamically decisive.

9.7 Why the model remains scientifically incomplete

The comparative SMFT article correctly distinguishes between a reusable structural grammar and a completed physical theory.

The memory model explains role relationships, but it does not yet derive:

- Newton's constant G;
- the exact Einstein field equations;
- the dimensionality of spacetime;
- the tensorial form of gravity;
- the quantum gravitational Hilbert space;
- ultraviolet behaviour;
- the Born rule;
- collapse dynamics;
- the cosmological constant;
- unique experimental deviations.

Therefore:

Explanatory architecture ≠ complete dynamical theory. (9.10)

The repulsive-gravity proposal is valuable partly because it identifies one specific domain where the earlier explanatory architecture must become more precise.

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10. Where the Earlier Model Began Too Late

The fundamental weakness of the earlier SMFT gravity model is temporal and logical.

It begins after the crucial quantum relation has already formed.

It asks:

How does an admitted difference become durable geometry?

But it does not adequately ask:

How does gravity participate while physical alternatives remain unresolved?

The new proposal makes this omission unavoidable.

10.1 Post-collapse curvature cannot explain pre-collapse interference by itself

Suppose gravity existed only after a branch had become definite.

The source would then generate either:

g_A, (10.1)

or:

g_B. (10.2)

A classical observer ignorant of the selected branch could use a probability distribution:

P(g) = qδ(g − g_A) + (1 − q)δ(g − g_B). (10.3)

The probe would experience either the A-field or the B-field.

The ensemble could be mixed, but the fields would not be coherently superposed.

The final probe state would be described by:

ρ_P = qρ_A + (1 − q)ρ_B. (10.4)

No off-diagonal branch coherence would remain.

A later source measurement could update classical probabilities, but it could not create the amplitude subtraction:

βψ_B − αψ_A. (10.5)

A post-collapse-only gravitational ontology therefore cannot explain the proposed mechanism unless an additional nonclassical structure is inserted upstream.

10.2 Gravity must participate before the final trace exists

The source–probe state after interaction is:

$$|\Psi_{SP}\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle.$$ (10.6)

The final conditional state depends on interference between these two terms.

Therefore, before the source is postselected, the physical evolution must preserve:

Branch identity:
A remains correlated with ψ_A. (10.7)

B remains correlated with ψ_B. (10.8)

Relative amplitude:
α/β remains physically meaningful. (10.9)

Relative phase:
φ_A − φ_B remains available to interference. (10.10)

Cross-branch overlap:
⟨ψ_A|ψ_B⟩ remains sufficiently nonzero. (10.11)

The gravitational interaction is implicated in establishing ψ_A and ψ_B.

It cannot therefore be represented only as a later record of their completed selection.

10.3 The missing coherent sector

The earlier model had a well-developed post-closure sector:

Trace → ledger → curvature. (10.12)

It lacked a comparably explicit coherent sector:

Alternative → relation → entanglement → projection. (10.13)

The new proposal indicates that SMFT requires both.

A minimal expanded architecture is:

Coherent sector | closure interface | residual sector. (10.14)

Where:

Coherent sector = amplitudes, phases, alternatives, entanglement. (10.15)

Closure interface = projection, decoherence, weighting, recording. (10.16)

Residual sector = distinguishability, flux, area, curvature. (10.17)

The old gravity model concentrated on the third region.

The new paper forces the first and second regions into view.

10.4 Why adding a coherent sector is not merely theoretical self-protection

A framework could protect itself from contradiction by adding an unlimited number of invisible layers.

That would weaken rather than strengthen it.

The coherent sector proposed here is not arbitrary. It is required by the mathematical structure of the source paper.

The predicted anomaly depends on:

- coherent source preparation;
- branch-dependent gravitational interaction;
- source–probe entanglement;
- nonzero cross terms;
- phase-sensitive postselection.

If these are removed, the mechanism disappears.

The new layer therefore has explicit operational content.

It is not added merely because SMFT wants gravity to explain more phenomena.

It is added because a post-collapse-only field cannot generate the stated quantum state.

10.5 The earlier claim must be restricted

The statement:

Gravity is residual collapse geometry. (10.18)

should not be discarded.

It should be restricted:

Classical gravitational geometry is residual collapse geometry. (10.19)

Or more carefully:

The semiclassical gravitational field is the stable geometric closure of quantum physical differences after coherence has ceased to be operationally available at the relevant scale. (10.20)

This revised claim leaves room for an upstream quantum gravitational relation.

The complete formulation becomes:

Quantum gravitational relation → closure → semiclassical gravitational memory. (10.21)

10.6 Gravity's role changes across closure

The word *gravity* may refer to different effective descriptions at different stages.

Before closure:

Gravity acts as a branch-sensitive relational mediator. (10.22)

During conditioning:

Gravity supplies the branch structure whose amplitudes are recombined by projection. (10.23)

After closure:

Gravity appears as persistent stress-energy and geometric consequence. (10.24)

The new model is therefore not:

Quantum gravity + unrelated classical memory field. (10.25)

It is:

One gravitational lifecycle represented differently at different closure scales. (10.26)

10.7 From force ontology to relation ontology

The earlier model often contrasted force with residual geometry.

The new proposal suggests that an even deeper contrast is needed:

Object-centred force ontology asks what force acts on each object.

Relational quantum ontology asks what joint state links source alternatives to probe alternatives.

In the proposed experiment, the crucial object is not simply:

F_A or **F_B.** (10.27)

It is:

$\alpha|A\rangle|\psi_A\rangle + \beta e^{i\Phi}|B\rangle|\psi_B\rangle$. (10.28)

This joint state contains information not available in either force value separately.

A refined SMFT gravity theory must therefore treat relation as primary at the coherent stage.

The proper upstream variable is not only force magnitude.

It is the entire phase-bearing source–probe relation.

10.8 The second major fine-tuning of SMFT

The first major fine-tuning was:

> Gravity cannot be defined exclusively as post-collapse residue.

The second is:

> Collapse cannot be defined exclusively as branch elimination.

The postselection produces a new conditional state from the interference of branch-conditioned probe states.

Thus, both gravity and collapse require broader definitions.

The revised statements are:

Gravity = relational persistence before and after closure. (10.29)

Projection = exclusion plus constructive recombination. (10.30)

Together, these changes transform the old linear chain:

Potential → collapse → gravity. (10.31)

into a staged lifecycle:

Potential → gravitational relation → projection construction → ledger closure → geometric memory. (10.32)

The next part develops this lifecycle stage by stage.

Part IV — The Refined SMFT Gravity Lifecycle

11. From Two Layers to a Staged Lifecycle

The preceding analysis suggests that the contrast between "quantum gravity before collapse" and "curvature memory after collapse" is useful but still too coarse.

A two-layer model risks creating an artificial split:

Layer 1: active quantum gravity. (11.1)

Layer 2: passive classical gravity. (11.2)

This could be misread as requiring two unrelated gravitational substances or two independent fields. The more economical interpretation is that gravity passes through several operational regimes as a physical relation approaches closure.

The relevant distinction is therefore not between two gravities, but between several stages of one gravitational lifecycle:

Reference relation → coherent alternatives → branch-sensitive coupling → conditioned trace → complete ledger → durable distinguishability → geometric closure. (11.3)

At each stage, the same physical process is represented through different variables.

In the coherent regime, the variables include:

- amplitudes;
- phases;
- branch labels;
- entangled states;
- noncommuting measurement operations.

In the ledger regime, the variables include:

- probabilities;
- conditional and unconditional distributions;
- apparatus records;

- conserved quantities;
- environmental correlations.

In the geometric regime, the variables include:

- stress-energy expectation;
- flux;
- boundary-area change;
- focusing;
- curvature.

The lifecycle can therefore be organized into seven stages:

G_0 — Reference background. (11.4)

G_1 — Coherent gravitational alternatives. (11.5)

G_2 — Relational entanglement. (11.6)

G_3 — Constructed conditional trace. (11.7)

G_4 — Complete weighted ledger. (11.8)

G_5 — Durable distinguishability and flux. (11.9)

G_6 — Semiclassical curvature memory. (11.10)

The proposed repulsive-gravity experiment mainly illuminates G_1 through G_3. The relative-entropy derivation mainly illuminates G_5 through G_6. The least understood region is the handover from G_3 through G_5.

That missing region is where:

- a phase-sensitive conditioned state becomes a durable record;
- a selected subensemble is reinserted into complete conservation accounting;
- inaccessible coherence becomes stable distinguishability;
- local records become suitable inputs for stress-energy and geometry.

SMFT's earlier account concentrated on the final movement:

Trace → ledger → curvature. (11.11)

The new proposal requires a broader model:

Alternative → relation → trace → ledger → curvature. (11.12)

11.1 Lifecycle does not mean a universal temporal sequence for every event

The stages should not be interpreted as a claim that every gravitational event passes through

seven sharply separated moments.

In realistic systems:

- entanglement and decoherence may occur continuously;
- record formation may begin before all coherent interaction ends;
- environmental coupling may overlap with measurement;
- semiclassical descriptions may already be valid for some degrees of freedom while others remain coherent;
- different observers may have access to different operational partitions.

The lifecycle is therefore a hierarchy of descriptive closure, not necessarily a sequence of globally synchronized events.

A more careful formulation is:

Closure stage = degree to which relational alternatives remain operationally recombinable. (11.13)

At early stages, alternatives can still interfere.

At later stages, their phase relation is inaccessible and only weighted records remain.

At still later stages, those records are absorbed into effective geometric descriptions.

11.2 The closure variable

It may be useful to introduce a schematic closure parameter κ_c:

$$\mathbf{0 \le \kappa_c \le 1.}$$ (11.14)

Where:

$\boldsymbol{\kappa_c \approx 0}$ means the relevant alternatives remain fully coherent and recombinable. (11.15)

$\boldsymbol{\kappa_c \approx 1}$ means the relevant alternatives have become effectively irreversible records at the chosen scale. (11.16)

This is not a fundamental constant or a quantity derived by the source papers. It is an SMFT bookkeeping device.

A state's effective description may then depend on κ_c:

$\mathcal{G}\boldsymbol{(\kappa_c \approx 0)}$ **= coherent relational gravity.** (11.17)

$\mathcal{G}\boldsymbol{(\kappa_c \approx 1)}$ **= residual geometric gravity.** (11.18)

Between these limits lie partial decoherence, weak measurement, postselection, amplification, and mesoscopic record formation.

The value of κ_c would be scale-dependent and observer-relative in an operational sense. A microscopic apparatus component may remain coherent relative to a larger laboratory, while its result appears effectively closed to an experimenter who lacks access to the environmental phase record.

This prevents the lifecycle from being reduced to a simplistic universal collapse instant.

11.3 Four maps rather than one collapse event

The full lifecycle requires at least four conceptually distinct operations:

$\mathcal{U}_{G}$ **: relational gravitational evolution.** (11.19)

$\mathcal{P}_{f}$ **: conditional projection.** (11.20)

$\mathcal{C}_{L}$ **: ledger closure.** (11.21)

$\mathcal{C}_{G}$ **: geometric closure.** (11.22)

Their roles are:

$\mathcal{U}_{G}$ **: independent source and probe possibilities → correlated gravitational possibilities.** (11.23)

$\mathcal{P}_{f}$ **: correlated possibilities → selected conditional state.** (11.24)

$\mathcal{C}_{L}$ **: selected and complementary outcomes → complete durable record.** (11.25)

$\mathcal{C}_{G}$ **: durable local records → mutually consistent effective geometry.** (11.26)

The distinction can be summarized as:

$\mathcal{U}_{G} \neq \mathcal{P}_{f} \neq \mathcal{C}_{L} \neq \mathcal{C}_{G}$**.** (11.27)

The earlier SMFT vocabulary often allowed the term *collapse* to cover all four. The refined model should no longer do so.

---

12. Stage I: Coherent Gravitational Possibility Geometry

The first post-reference stage contains unresolved gravitational alternatives.

For the source superposition:

$$|\Psi_S\rangle = \alpha|A\rangle + \beta|B\rangle, \quad (12.1)$$

one may schematically associate gravitational alternatives:

$$|\mathcal{G}\rangle = \alpha|g_A\rangle + \beta e^{i\Phi_G}|g_B\rangle. \quad (12.2)$$

The notation $|g_A\rangle$ and $|g_B\rangle$ should be interpreted cautiously.

It does not assume that the complete classical metric tensor itself has simply been inserted into an ordinary two-dimensional Hilbert space.

It represents the weaker requirement that the gravitational mediator or gravitational degree of freedom must retain branch-sensitive alternatives corresponding to the source configurations A and B.

The source paper states this operationally: if gravity obeys the superposition principle, the probe

is subjected to a superposition of two attractive gravitational interactions, and the resulting branch-dependent momentum transfers generate source–probe entanglement.

The most defensible SMFT term for this stage is:

coherent gravitational possibility geometry.

“Possibility” indicates that no final branch has become the sole durable record.

“Gravitational” indicates that the branch dependence is produced through the source–probe gravitational interaction.

“Geometry” indicates a structured relation among alternatives, not necessarily an already classical spacetime metric.

12.1 Possibility geometry is not ignorance geometry

A classical mixture may be written:

$$\rho_{mix} = q|g_A\rangle\langle g_A| + (1 - q)|g_B\rangle\langle g_B|. \quad (12.3)$$

A coherent state contains off-diagonal terms:

$$\rho_{coh} = |\alpha|^2|g_A\rangle\langle g_A| + |\beta|^2|g_B\rangle\langle g_B| + \alpha\beta e^{-i\phi}_G|g_A\rangle\langle g_B| + \alpha\beta e^{i\phi}_G|g_B\rangle\langle g_A|. \quad (12.4)$$

The mixture represents uncertainty over alternatives.

The coherent state represents a physical phase relation between alternatives.

The difference is:

Ignorance geometry stores alternatives as exclusive probabilities. (12.5)

Possibility geometry stores alternatives as phase-bearing relations. (12.6)

Only the latter supports interference after projection in an appropriate basis.

12.2 The geometry is relational rather than branch-local

Neither branch alone contains the full information.

Branch A contains:

$$A \rightarrow \delta_A. \quad (12.7)$$

Branch B contains:

$$B \rightarrow \delta_B. \quad (12.8)$$

The coherent relation additionally contains:

$$\text{A relative to B; } \delta_A \text{ relative to } \delta_B; \varphi_A \text{ relative to } \varphi_B. \quad (12.9)$$

The observed conditioned anomaly depends on these relational quantities.

This is why the correct upstream object is not merely a list of forces:

{F_A, F_B}. (12.10)

It is a phase-bearing relational structure:

$\mathcal{R}$_G = {α, β, φ_A − φ_B, |ψ_A⟩, |ψ_B⟩, ⟨ψ_A|ψ_B⟩}. (12.11)

The source–probe experiment is sensitive to $\mathcal{R}$_G, not only to F_A and F_B separately.

12.3 A revised meaning of gravitational memory

At this stage, gravity has not yet recorded one settled history.

It preserves a correlation structure among unresolved histories.

This can be called **coherence memory**:

M_coh = preservation of branch association + relative phase + interference capacity. (12.12)

The term does not imply that gravity has a mind or an internal symbolic archive.

It means only that the joint evolution retains information necessary for later interference.

The distinction is:

Classical memory preserves what occurred. (12.13)

Quantum coherence memory preserves how unresolved alternatives remain related. (12.14)

This is the first component of the Dual Persistence Principle.

12.4 Conditions for coherent possibility geometry

For the possibility geometry to remain experimentally useful, several conditions must hold:

$|\langle A|B\rangle| \approx 0$ for distinguishable source branches. (12.15)

$|\langle \psi_A|\psi_B\rangle| \neq 0$ for probe-branch interference. (12.16)

Phase uncertainty $\ll 2\pi$ over the relevant interaction and recombination period. (12.17)

Environmental which-branch leakage must remain limited. (12.18)

The gravitationally induced branch difference must be detectable after postselection. (12.19)

If the source branches are not distinguishable, the branch-dependent force difference becomes too small.

If the probe states become fully orthogonal, their amplitudes may cease to interfere effectively.

If the environment records which path, the off-diagonal terms decay.

The proposed witness therefore exists within an intermediate window:

Enough branch distinction to generate different gravitational shifts, but enough

coherence to permit recombination. (12.20)

This is a characteristic mesoscopic closure condition.

---

13. Stage II: Gravity as an Entangling Relational Channel

The coherent source state alone does not yet produce the proposed witness.

The source and probe must interact.

Initially:

$$|\Psi(0)\rangle = (\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle. \quad (13.1)$$

This state is separable.

After the gravitational interaction:

$$|\Psi(T)\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle. \quad (13.2)$$

The branch state of the probe now depends on the branch state of the source.

Unless:

$$|\psi_A\rangle = e^{ic}|\psi_B\rangle \quad (13.3)$$

for some global phase c, the state is generally nonseparable.

Gravity has therefore functioned as a relational channel.

The source paper treats the resulting source–probe entanglement as the basis for witnessing the quantum nature of gravity, while emphasizing that the proposed protocol avoids the need to directly measure the full quantum correlations.

13.1 Active formation versus passive guidance

The earlier SMFT article described gravity as passive because it was identified with settled curvature rather than an active identity-changing interaction. The entangling stage requires a more precise vocabulary.

At this stage, gravity is dynamically active:

$$\rho_0 \rightarrow \mathcal{U}_G \rho_0 \mathcal{U}_G^\dagger. \quad (13.4)$$

The joint state changes.

Correlations are generated.

The probe momentum amplitudes are displaced.

However, gravity need not be “active” in the same way as a weak flavour-changing event. It does not necessarily change the particle species or internal identity. It changes the relational state between source position and probe momentum.

A refined classification is:

Weak-like activity = identity-changing transition. (13.5)

Coherent gravitational activity = relation-forming transition. (13.6)

Classical gravitational passivity = geometry-guided persistence. (13.7)

This prevents “active” and “passive” from being treated as absolute properties of an interaction at every scale.

13.2 Gravity writes branch information into the probe

The interaction performs a form of relational encoding:

A → ψ_A. (13.8)

B → ψ_B. (13.9)

The probe now contains partial information about the source branch.

If ψ_A and ψ_B become distinguishable, measuring the probe can reveal information about the source.

Conversely, measuring the source in a superposition basis can alter the conditional probe state.

This bidirectional conditional structure is characteristic of entanglement.

SMFT may visualize the process as:

Source difference → gravitational channel → distributed relational trace. (13.10)

The trace is not yet a classical record because its information remains encoded nonlocally in the joint state.

It is a **distributed pre-record**.

13.3 Distributed trace versus local record

A local classical record has the form:

R_A = “A occurred.” (13.11)

A distributed quantum trace has the form:

α|A⟩|ψ_A⟩ + β|B⟩|ψ_B⟩. (13.12)

No subsystem alone contains the complete pure-state information.

Tracing out the source gives:

$\rho_P = |\alpha|^2|\psi_A\rangle\langle\psi_A| + |\beta|^2|\psi_B\rangle\langle\psi_B| + \alpha\beta\langle B|A\rangle|\psi_A\rangle\langle\psi_B| + h.c.^*$ (13.13)

For orthogonal source branches, ⟨B|A⟩ ≈ 0, and the probe reduced state appears mixed.

Yet the coherence remains present in the joint state and can be accessed through an appropriate source measurement.

This is why the source postselection is essential.

The relational information is distributed, not destroyed.

13.4 The gravitational channel as controlled translation

In the approximation of the paper, the source branch controls a probe momentum translation.

Define the momentum-translation operator:

$$\hat{T}(\delta) = \exp(-i\delta\hat{x}/\hbar). \quad (13.14)$$

Then the interaction may be represented schematically as:

$$\mathcal{U}_G \approx |A\rangle\langle A| \otimes \hat{T}(\delta_A) + |B\rangle\langle B| \otimes \hat{T}(\delta_B). \quad (13.15)$$

Acting on the initial state:

$$\mathcal{U}_G[(\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle] = \alpha|A\rangle\hat{T}(\delta_A)|\psi_0\rangle + \beta|B\rangle\hat{T}(\delta_B)|\psi_0\rangle. \quad (13.16)$$

This form makes the entangling mechanism transparent.

The source branch controls the probe translation.

If $\delta_A = \delta_B$, the translation factors out and no branch-dependent entanglement is generated.

Thus:

Entangling capacity ∝ distinguishability of branch-conditioned gravitational action. (13.17)

This is an operationally sharper statement than saying merely that “gravity is in superposition.”

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14. Stage III: Constructive Projection

The gravity-mediated relation is not yet the observed repulsive trace.

A source measurement must project the joint state into a selected relational sector.

Let:

$$\hat{P}_f = |\Psi_f\rangle\langle\Psi_f|. \quad (14.1)$$

The unnormalized conditional probe state is:

$$\tilde{\rho}_f = \mathrm{Tr}_S[(\hat{P}_f \otimes I)\rho_{SP}]. \quad (14.2)$$

The successful postselection probability is:

$$p_f = \mathrm{Tr}(\tilde{\rho}_f). \quad (14.3)$$

The normalized conditional state is:

$$\rho_f = \tilde{\rho}_f / p_f. \quad (14.4)$$

For a pure joint state:

$|\tilde{\psi}_f\rangle = \langle \Psi_f | \Psi_{SP} \rangle$. (14.5)

The normalized state is:

$|\psi_f\rangle = |\tilde{\psi}_f\rangle / \sqrt{p_f}$. (14.6)

This is the projection-conditioning operation.

14.1 Projection does not create the prior relation

The projection acts on:

$\alpha|A\rangle|\psi_A\rangle + \beta|B\rangle|\psi_B\rangle$. (14.7)

It does not generate ψ_A or ψ_B.

Those states were produced by $\mathcal{U}_G$.

Therefore:

Interaction creates the branch relation. (14.8)

Projection selects its observable recombination. (14.9)

The earlier weak-gate metaphor must be adjusted accordingly.

The gate is not the origin of the entire physical difference.

It acts on a difference that gravity has already distributed across the source–probe system.

14.2 Projection as a trace constructor

If the source is measured in the branch basis, the conditional probe states are simply:

A outcome $\rightarrow |\psi_A\rangle$. (14.10)

B outcome $\rightarrow |\psi_B\rangle$. (14.11)

But if the source is measured in a superposition basis:

$|\Psi_f\rangle = u|A\rangle + v|B\rangle$, (14.12)

then:

$|\tilde{\psi}_f\rangle = u\alpha|\psi_A\rangle + v\beta|\psi_B\rangle$. (14.13)

The conditional probe state is a constructed superposition of the branch-conditioned states.

Projection therefore performs more than branch identification.

It defines which linear combination becomes the selected trace.

In SMFT terms:

Projection basis = trace-construction geometry. (14.14)

14.3 Exclusion and construction

A postselection operation has two logically distinct effects.

First, it excludes all runs in which the source measurement did not yield f.

Second, within the successful runs, it defines the relative coefficients of the surviving probe amplitudes.

Thus:

$\mathcal{P}_\mathbf{f} = \mathcal{X}_\mathbf{f} + \mathcal{K}_\mathbf{f}.$ (14.15)

Where:

$\mathcal{X}_\mathbf{f}$ = **exclusion of non-f outcomes.** (14.16)

$\mathcal{K}_\mathbf{f}$ = **constructive recombination within the f-conditioned sector.** (14.17)

This decomposition is conceptual rather than a literal additive superoperator identity.

Its purpose is to show that collapse language focused only on exclusion misses the mechanism responsible for the anomaly.

14.4 Constructed properties

The conditioned state may exhibit properties absent from either branch.

In the proposal:

$\langle \mathbf{p} \rangle_\mathbf{A} > \mathbf{0}.$ (14.18)

$\langle \mathbf{p} \rangle_\mathbf{B} > \mathbf{0}.$ (14.19)

But:

$\langle \mathbf{p} \rangle_\mathbf{f} < \mathbf{0}.$ (14.20)

The negative conditioned mean is not carried secretly by branch A or branch B.

It is a relational property of:

- their overlap;
- their relative phase;
- their relative weight;
- the chosen postselection.

This yields the Projection Constructivity Principle:

> **A selected trace may possess an observable property absent from every individual branch from which it is constructed.**

In symbolic form:

$\mathbf{O}[\mathcal{P}_\mathbf{f}(\psi_\mathbf{A}, \psi_\mathbf{B})] \notin \{\mathbf{O}[\psi_\mathbf{A}], \mathbf{O}[\psi_\mathbf{B}]\}.$ (14.21)

14.5 Constructivity does not imply arbitrariness

The measurement basis cannot produce any desired state without constraint.

The conditioned state must lie within the span generated by the branch-conditioned states:

$$|\psi_f\rangle \in \mathbf{Span}\{|\psi_A\rangle, |\psi_B\rangle\}.$$ (14.22)

Its coefficients are fixed by:

- the prepared amplitudes;
- the accumulated phases;
- the physical postselection state;
- normalization.

Thus:

Projection constructivity ≠ unconstrained creation. (14.23)

A better description is:

Projection reorganizes available relational potential into a selected observable trace. (14.24)

This is closely aligned with SMFT's broader claim that an observer does not invent arbitrary meaning but selects and stabilizes a path from a constrained semantic field.

---

15. Stage IV: Conditional Gravitational Trace

The output of postselection is a conditional trace.

It may be represented as:

$$\mathcal{T}_f = \{p_f, \rho_f, \langle \Delta p \rangle_f, \mathrm{Var}_f(\Delta p), f\}.$$ (15.1)

This object includes:

- which outcome was selected;
- how probable it was;
- the conditioned probe state;
- the conditioned mean;
- the conditioned uncertainty.

The anomaly is found in:

$$\langle \Delta p \rangle_f < 0.$$ (15.2)

But the full conditional trace is more than this one number.

15.1 Conditional reality

A conditioned expectation is not unreal merely because it applies only to a subensemble.

Many experimentally meaningful quantities are conditional:

- coincidence rates;
- heralded photon states;
- decay products selected by channel;
- scattering amplitudes at fixed output;
- weak values;
- conditional trajectories.

The correct distinction is not real versus unreal.

It is:

Conditional quantity = valid within a specified preparation-and-selection context. (15.3)

Unconditional quantity = averaged across the complete outcome space. (15.4)

The repulsive response is physically meaningful within the successful postselected ensemble.

It is not a universal force experienced in every run.

15.2 The trace is protocol-indexed

The conditional trace depends on the entire protocol λ:

λ = {α, β, x_A, x_B, T, φ_A, φ_B, P̂_f, ψ0}. (15.5)

Therefore:

$\mathcal{T}$**_f =** $\mathcal{T}$**_f[λ].** (15.6)

Changing the postselection phase can change:

- the successful probability;
- the interference pattern;
- the effective momentum shift;
- potentially the sign of the shift.

The trace is therefore relationally indexed.

It cannot be assigned solely to the mass M or to a static gravitational field configuration.

15.3 Conditional trace as an SMFT collapse tick

SMFT often represents a collapse tick τ_k as a moment when distributed possibility becomes a committed trace.

The postselected outcome is a useful physical analogue, provided the analogy remains disciplined.

Before selection:

ρ_{SP} contains multiple coherent source–probe relations. (15.7)

After successful selection:

ρ_f defines one operationally admitted probe state. (15.8)

This resembles:

$$\Psi_m(\tau_k^-) \rightarrow \hat{O}_k \rightarrow \varphi_k. \quad (15.9)$$

But there is an important refinement:

φ_k need not equal a pre-existing branch. (15.10)

It can be a constructed conditional state.

Thus, an improved SMFT collapse tick should be represented as:

$$\varphi_k = \text{Normalize}[\hat{O}_k \mathcal{U}(\Psi_m)]. \quad (15.11)$$

Rather than:

φ_k = "choose one branch already present." (15.12)

15.4 The conditional trace is not yet a complete historical record

The selected trace omits:

- unsuccessful postselection outcomes;
- their probe distributions;
- apparatus changes;
- environmental correlations;
- selection frequency;
- preparation cost.

Therefore:

$$\mathcal{T}_f \neq \mathcal{L}_{\text{complete}}. \quad (15.13)$$

This distinction is essential.

A conditional trace can exhibit amplification precisely because it is extracted from a small region of the full outcome space.

Its meaning cannot be separated from what was excluded.

16. Stage V: Ledger Closure

A complete ledger contains all outcome sectors.

Let the source measurement have outcomes $f \in \mathscr{F}$.

For each outcome:

$$\mathbf{p_f = Tr(\tilde{\rho}_f).}\ (16.1)$$

$$\mathbf{\rho_f = \tilde{\rho}_f / p_f.}\ (16.2)$$

Completeness requires:

$$\mathbf{\Sigma_f p_f = 1.}\ (16.3)$$

The full postmeasurement probe ensemble is:

$$\mathbf{\rho_P' = \Sigma_f p_f \rho_f.}\ (16.4)$$

If the measurement outcomes are ignored, this reproduces the ordinary reduced probe state under appropriate measurement completeness.

The complete ledger therefore contains:

$$\mathbf{\mathcal{L} = \{\{p_f, \rho_f, R_f\}_f, R_A, R_E, C\}.}\ (16.5)$$

Where:

- R_f is the recorded outcome;
- R_A is the apparatus record;
- R_E is the environmental record;
- C denotes conservation accounting.

16.1 Ledger Completeness Principle

The refined SMFT framework should state:

> **No conditional gravitational trace is physically complete until its probability weight, complementary outcomes, apparatus response, and conservation context are included.**

In compact form:

Complete trace = selected value + selection weight + excluded complement + recording cost. (16.6)

This principle prevents a large anomalous weak value from being treated as a freely available macroscopic force.

16.2 Conditional amplification and admission weight

Near-orthogonal preselection and postselection can yield:

$$|\langle \Delta p \rangle_f| \gg \delta_A. \quad (16.7)$$

But typically:

$$p_f \ll 1. \quad (16.8)$$

A crude weighted contribution is:

$$W_f = p_f \langle \Delta p \rangle_f. \quad (16.9)$$

The complete ensemble momentum is:

$$\langle \Delta p \rangle_{all} = \Sigma_f p_f \langle \Delta p \rangle_f. \quad (16.10)$$

The anomalous selected value can be large while the complete ensemble remains consistent with ordinary momentum transfer and apparatus backreaction.

The source paper's example illustrates this amplification–probability trade-off: a very large negative effective shift appears only in a rare selected output.

The SMFT interpretation is:

Trace intensity cannot be evaluated independently of admission measure. (16.11)

16.3 Ledger closure as contextual restoration

Postselection temporarily isolates one relational sector.

Ledger closure restores the context that made the sector possible.

It reconnects:

- the rare selected event;
- the rejected outcomes;
- the preparation state;
- the measurement basis;
- the recording apparatus.

Thus:

Projection narrows context. (16.12)

Ledger closure restores global accounting. (16.13)

This is why conditional repulsion and total conservation can coexist.

16.4 Conservation is global to the interaction–apparatus system

A conditional subsystem may exhibit an anomalous momentum shift.

The total closed system includes:

- source;
- probe;
- interferometer;
- traps or supports;
- measurement apparatus;
- environment.

A conservation statement must refer to the complete system:

$$\Delta P_{source} + \Delta P_{probe} + \Delta P_{apparatus} + \Delta P_{environment} = 0. \quad (16.14)$$

The exact partition depends on the experimental realization.

The key point is that postselection does not erase the complementary momentum and record channels.

It reorganizes which runs are examined.

This is the physical content behind ledger closure.

16.5 Ledger closure and irreversible record

A record becomes durable when different outcome sectors become correlated with macroscopically distinguishable apparatus or environmental states:

$$\Sigma_f c_f|f\rangle|A_0\rangle|E_0\rangle \rightarrow \Sigma_f c_f|f\rangle|A_f\rangle|E_f\rangle. \quad (16.15)$$

When:

$$\langle A_f|A_g\rangle\langle E_f|E_g\rangle \approx 0 \text{ for } f \neq g, \quad (16.16)$$

interference between outcomes becomes operationally inaccessible.

This is not yet a complete solution to the measurement problem.

But it explains how a phase-bearing relational state can become a stable outcome ledger at the effective level.

---

17. Stage VI: Distinguishability Residue

Once phase coherence is no longer operationally accessible, the physical difference need not disappear.

It changes representation.

Before closure, information is carried in:

- relative phases;
- off-diagonal density-matrix terms;

- interference visibility.

After closure, information is carried in:

- outcome probabilities;
- stable records;
- distinguishability from a reference state;
- energy and momentum differences;
- entropy or relative entropy.

This transition can be represented schematically as:

Phase-usable information → record-usable distinguishability. (17.1)

17.1 Coherence loss does not mean information annihilation

Consider a two-branch density matrix:

ρ_coh = [[ρ_AA, ρ_AB], [ρ_BA, ρ_BB]]. (17.2)

Under decoherence in the branch basis:

ρ_coh → ρ_dec = [[ρ_AA, 0], [0, ρ_BB]]. (17.3)

The off-diagonal phase information becomes inaccessible locally.

But the diagonal populations and environmental records remain.

The system may still be distinguishable from its original reference state.

Thus:

Loss of interference capacity ≠ loss of all physical difference. (17.4)

Instead:

Coherent difference becomes historical difference. (17.5)

17.2 Relative entropy as durable difference

The Dorau–Much framework uses relative entropy:

S_rel(ω₀‖ω_ϕ), (17.6)

to quantify distinguishability between a vacuumlike reference state and a coherent excitation on a horizon algebra.

In that setting, the relative entropy is finite and relates to the energy flux through the local horizon.

From the refined SMFT perspective, relative entropy belongs to the ledger regime rather than the amplitude-interference regime.

It does not preserve the sign of a weak value.

It does not represent a branch momentum displacement.

It measures a stable relational difference between states.

This is why:

Weak value may be anomalous and signed. (17.7)

Relative entropy is nonnegative. (17.8)

Their roles differ.

17.3 Distinguishability residue

The term **distinguishability residue** may be defined as:

> The stable, operationally recoverable difference remaining after phase-sensitive alternatives can no longer be recombined at the relevant scale.

Symbolically:

$\mathcal{D}_{res} = D(\rho_{record}, \rho_{ref})$. (17.9)

Where D may be represented by an appropriate state-distance or relative-entropy measure in a specified theory.

This is not a claim that all distinguishability measures generate gravity.

It identifies the information type available for downstream geometric accounting.

17.4 From conditional state to reference-relative state

The repulsive-gravity proposal yields a conditioned probe state ρ_f.

The Dorau–Much framework considers a reference/excitation pair ω_0 and ω_ϕ.

No direct equality is established:

$\rho_f \neq \omega_\phi$. (17.10)

Nor is:

$\langle \Delta p \rangle_f = S_{rel}(\omega_0 \| \omega_\phi)$. (17.11)

The conceptual continuity is weaker:

Both frameworks require a physically meaningful difference defined relative to preparation, boundary, or selection structure. (17.12)

The handover question is how a selected phase-sensitive difference might become a stable reference-relative difference suitable for flux and geometric accounting.

That transition remains open.

17.5 Information-type conversion as a coarse-graining problem

A possible schematic map is:

$\mathcal{K}$ **: {ρ_SP, P^_f, environment, apparatus} → ρ_record.** (17.13)

Then:

$\mathcal{D}$**_res = S_rel(ρ_ref‖ρ_record).** (17.14)

This is not the specific Araki–Uhlmann construction of the Dorau–Much horizon analysis.

It is a generic placeholder indicating the type of mathematical step required.

The map $\mathcal{K}$ would have to specify:

- which degrees of freedom are retained;
- which are traced out;
- which reference state is chosen;
- which algebra of observables is accessible;
- which records are treated as stable;
- which conservation constraints apply.

Without these specifications, “information becomes gravity” remains too vague.

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18. Stage VII: Boundary and Curvature Closure

The final lifecycle stage concerns the conversion of durable physical difference into effective geometry.

The Dorau–Much framework provides a specific example within quantum field theory in curved spacetime.

The authors consider a vacuumlike reference state ω0 and a coherent excitation ω_ϕ on the algebra associated with a local horizon. They compute the relative entropy and identify it with an energy-flux expression involving the stress-energy tensor.

Schematically:

S_rel(ω0‖ω_ϕ) = −2π∫_H U⟨:T_ab:⟩_ωϕ ξ^a ξ^b dU dvol_S. (18.1)

Under the assumed entropy–area proportionality:

δA = (α / 2π)S_rel(ω0‖ω_ϕ). (18.2)

The area variation is also related to null-geodesic focusing through the Raychaudhuri equation:

δA = −∫_H UR_abξ^a ξ^b dU dvol_S. (18.3)

Equating the informational and geometric expressions yields:

R_ab − ½Rg_ab + Λg_ab = α⟨:T_ab:⟩_ωϕ. (18.4)

With the Bekenstein–Hawking-style normalization:

$\alpha = 8\pi$ (in the units used by the paper). (18.5)

The paper therefore supplies a disciplined downstream route:

Reference-relative quantum information → energy flux → area response → semiclassical curvature. (18.6)

18.1 Boundary ledger

SMFT interprets the horizon area as a boundary ledger.

This does not mean the horizon literally stores symbolic data.

It means the boundary geometry changes in a way that records the physical flux and state difference relevant to the local causal region.

The sequence is:

Distinguishable excitation → horizon flux → area update. (18.7)

The area update is more durable and more geometric than the initial field amplitude.

It is therefore a suitable bridge toward curvature memory.

The earlier weak-gate article summarized this as:

Relative entropy counts the crossing. (18.8)

Energy flux carries the crossing. (18.9)

Area records the crossing. (18.10)

Gravity preserves the crossing as curvature. (18.11)

18.2 Curvature as consistency closure

A local area update alone is not yet a complete spacetime geometry.

The Einstein equation functions as a consistency condition relating local curvature to stress-energy.

From the SMFT perspective:

Local ledger entries must agree across overlapping causal descriptions. (18.12)

The curvature law is the rule that closes those accounts into one effective geometry.

This motivates:

Curvature closure = mutual consistency of durable local physical records. (18.13)

Gravity's classical memory face is therefore not merely the sum of isolated traces.

It is the geometry required to make their effects mutually compatible.

18.3 The repulsive conditional sign does not automatically enter curvature closure

The upstream experiment permits:

$\delta_A > 0$, $\delta_B > 0$, but $\langle \Delta p \rangle_f < 0$. (18.14)

The downstream information measure satisfies:

$S_{rel} \geq 0$. (18.15)

These are not contradictory because they refer to different objects.

The negative conditional shift measures the direction of one postselected probe response.

The nonnegative relative entropy measures distinguishability between two states.

Therefore:

$\langle \Delta p \rangle_f < 0$ does not imply $S_{rel} < 0$. (18.16)

And:

$\langle \Delta p \rangle_f < 0$ does not by itself imply $\delta A < 0$. (18.17)

Nor does it imply a permanently repulsive semiclassical metric.

The anomaly is an upstream witness of coherence, not a direct downstream curvature law.

18.4 The handover remains unproved

The lifecycle should not be misrepresented as an established derivation:

Postselected probe state → horizon relative entropy → Einstein equation. (18.18)

No such direct derivation is supplied by the attached papers.

The refined SMFT position is instead:

> The two papers reveal structurally complementary regimes that a complete theory of quantum gravity must connect.

The upstream regime requires:

- coherent gravitational alternatives;
- branch-sensitive coupling;
- entanglement;
- projection-dependent trace construction.

The downstream regime requires:

- stable state distinguishability;
- energy-momentum accounting;

- boundary response;
- curvature consistency.

The missing handover may be represented schematically as:

$\mathscr{H}$**_QC : {ρ_SP, P̂_f, ℰ, 𝒜} → {ρ_record, S_rel, ⟨T_ab⟩, δA}.** (18.19)

Where:

- ρ_SP is the coherent source–probe state;
- P̂_f is the projection protocol;
- $\mathcal{E}$ is environmental coupling;
- $\mathcal{A}$ is apparatus and record formation.

The exact map $\mathscr{H}$_QC is not yet known.

18.5 What SMFT has gained

The refined lifecycle yields four substantive improvements.

First, it prevents gravity from being defined only through its classical residue.

Second, it prevents collapse from being defined only as branch deletion.

Third, it separates a conditional trace from a complete physical ledger.

Fourth, it identifies a specific missing operator between quantum relation and curvature memory.

The updated sequence is:

Gravitational possibility → relational entanglement → constructive projection → conditional trace → ledger closure → distinguishability residue → curvature closure. (18.20)

This is not yet a quantum-gravity theory.

But it is a clearer specification of what such a theory must explain.

The next part develops the unifying principle behind the lifecycle:

Gravity preserves alternatives before closure and consequences afterward. (18.21)

Part V — Dual Persistence as the Refined Nature of Gravity

19. The Dual Persistence Principle

The refined SMFT lifecycle suggests that gravity should no longer be defined only by what remains after collapse.

The earlier formula was:

Collapse → trace → residual curvature → gravity. (19.1)

The expanded formula is:

Unresolved alternative → gravitational relation → staged closure → settled geometric consequence. (19.2)

This yields the Dual Persistence Principle:

> **Gravity preserves unresolved alternatives before closure and settled consequences after closure.**

In symbolic form:

Alternative persistence → closure transition → consequence persistence. (19.3)

The two forms of persistence are not identical.

Before closure, the preserved object is a phase-bearing relation among possibilities.

After closure, the preserved object is a history-bearing relation among records, energy distributions, boundaries, and trajectories.

The transition can be represented as:

M_coh → $\mathscr{H}$_QC → M_geo. (19.4)

Where:

- M_coh is coherent relational persistence;
- $\mathscr{H}$_QC is the unresolved quantum-to-curvature handover;
- M_geo is geometric historical persistence.

The principle does not claim that gravity is literally a memory storage device. It identifies a structural continuity:

Gravity maintains physically relevant relation across time and across closure scale. (19.5)

19.1 Why persistence is a better unifying term than force

At the branch level, gravity produces attractive momentum transfers:

$\delta_A > 0$. (19.6)

$\delta_B > 0$. (19.7)

At the postselected level, the conditioned mean may satisfy:

$\langle \Delta p \rangle_f < 0$. (19.8)

At the semiclassical level, gravity is expressed through curvature:

$G_{ab} + \Lambda g_{ab} = \alpha \langle T_{ab} \rangle$. (19.9)

These descriptions do not share one simple force arrow.

They do, however, share retained relation.

At the branch level, the source–probe separation determines a momentum-transfer relation.

At the coherent level, each source branch remains associated with its corresponding probe evolution.

At the measurement level, the postselection preserves one selected recombination of those relations.

At the ledger level, outcome probability and momentum transfer remain as an experimental record.

At the geometric level, stress-energy and boundary response remain as curvature constraints.

The common architecture is therefore:

Relation formed → relation transformed → relation retained. (19.10)

Persistence is a more general organizing category than attraction or repulsion.

19.2 Two different questions answered by the two persistence regimes

Pre-closure persistence answers:

> Which unresolved physical alternatives remain coherently related?

Post-closure persistence answers:

> Which settled physical differences continue to constrain future evolution?

The first concerns quantum possibility structure.

The second concerns historical geometry.

They may be written as:

Pre-closure persistence: {A, B, φ_AB} remain jointly usable. (19.11)

Post-closure persistence: {record, flux, δA, curvature} remain dynamically consequential. (19.12)

The first permits interference.

The second permits trajectory guidance and geometric consistency.

The lifecycle therefore moves from:

What could still be recombined? (19.13)

to:

What can no longer be ignored? (19.14)

This is the conceptual direction of closure.

19.3 Gravity as continuity across descriptive regimes

A complete theory of gravity should not merely specify:

- the quantum interaction;
- the classical metric.

It should explain their continuity.

The refined SMFT proposal is that continuity is carried by relational persistence.

At the quantum level:

Persistence = preservation of branch correlation and phase relation. (19.15)

At the classical level:

Persistence = preservation of effective energy–geometry relation. (19.16)

The handover must therefore transform the representational form of the relation without erasing its physical consequence.

In compressed form:

Phase relation survives as trace consequence, not necessarily as accessible phase. (19.17)

This distinction is crucial.

Classical geometry need not preserve the full quantum phase information.

It must preserve enough of the physical consequence for the later spacetime description to remain consistent.

19.4 Dual persistence does not imply unitary access to all past information

The principle should not be overstated.

Post-closure curvature memory does not necessarily retain every microscopic phase relation in directly recoverable form.

Environmental decoherence, coarse-graining, inaccessible correlations, and thermodynamic irreversibility may prevent reconstruction of the original coherent state.

The claim is weaker:

> Some physically consequential structure survives the transition from coherent relation to effective geometry.

Thus:

Microscopic recoverability ≠ macroscopic persistence. (19.18)

A footprint preserves evidence of motion without preserving the complete microscopic state of the walker.

A curvature response may preserve the effective energy–momentum consequence of an event without preserving all amplitudes that preceded it.

The relevant persistence is functional and dynamical, not necessarily informationally complete.

19.5 The Dual Persistence Principle as a constraint on future SMFT models

A future SMFT gravity equation should support both regimes.

It must allow:

1. coherent branch-sensitive evolution;
2. phase-dependent conditional traces;
3. transition to probability-weighted records;
4. emergence of stable distinguishability;
5. effective curvature closure.

A model that supports only the final stage remains incomplete.

A model that supports only coherent amplitudes but cannot recover classical geometry is equally incomplete.

The required constraint is:

$\mathcal{G}$**_complete must contain both M_coh and M_geo as effective limits.** (19.19)

And:

$\mathscr{H}$**_QC must map between them without violating normalization, conservation, or empirical semiclassical behaviour.** (19.20)

This turns the Dual Persistence Principle from a metaphor into a research requirement.

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20. Pre-Closure Persistence: Coherence Memory

The first persistence regime appears in the gravity-mediated joint state:

$$|\Psi_{SP}\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle.$$ (20.1)

This state preserves more than a list of possible outcomes.

It preserves a structured relation among:

- source branch;
- probe response;
- branch amplitude;
- branch phase.

The proposed postselected repulsion requires this relational structure to remain available until the source measurement. The authors attribute the anomalous momentum transfer to interference between gravitationally correlated alternatives and argue that a classical

gravitational mixture cannot reproduce the effect.

SMFT may call this **coherence memory**.

20.1 Definition of coherence memory

A provisional definition is:

> **Coherence memory is the physical preservation of branch association and relative phase required for later interference.**

Symbolically:

M_coh = {c_j, φ_j − φ_k, |j⟩ ↔ |ψ_j⟩, ⟨ψ_j|ψ_k⟩}. (20.2)

This contains:

- amplitudes c_j;
- relative phases φ_j − φ_k;
- branch–response associations;
- overlap between branch-conditioned states.

If these quantities lose operational significance, the coherent effect disappears.

Thus:

M_coh > 0 → interference remains possible. (20.3)

M_coh → 0 → classical-mixture description becomes sufficient at the chosen scale. (20.4)

The notation is schematic. It does not define a universal scalar measure of coherence memory.

20.2 Coherence memory is distributed

The complete relational information is not contained in the source alone.

It is not contained in the probe alone.

It resides in the joint state.

The source reduced state is:

ρ_S = Tr_P(|Ψ_SP⟩⟨Ψ_SP|). (20.5)

The probe reduced state is:

ρ_P = Tr_S(|Ψ_SP⟩⟨Ψ_SP|). (20.6)

Neither reduced state necessarily contains the full accessible phase relation.

The complete information resides in:

ρ_SP. (20.7)

Coherence memory is therefore distributed relationally.

This aligns with a wider SMFT intuition:

> A trace need not belong to one object. It can reside in the structure connecting objects.

The gravitational relation is not merely a property of the source or probe.

It is a property of their coupled state.

20.3 Gravity as a relation writer

The gravity-mediated evolution can be interpreted as writing source-branch information into the probe:

$$\mathbf{A \rightarrow \hat{T}(\delta_A)|\psi_0\rangle.} \quad (20.8)$$

$$\mathbf{B \rightarrow \hat{T}(\delta_B)|\psi_0\rangle.} \quad (20.9)$$

The resulting state is:

$$\mathbf{|\Psi_{SP}\rangle = \alpha|A\rangle\hat{T}(\delta_A)|\psi_0\rangle + \beta|B\rangle\hat{T}(\delta_B)|\psi_0\rangle.} \quad (20.10)$$

Gravity has created a conditional translation structure.

The probe does not simply receive a single uncertain kick.

It becomes a carrier of branch-indexed gravitational information.

In SMFT language:

Gravity writes relational difference before collapse writes historical difference. (20.11)

This statement fine-tunes the old model.

Previously, gravity was mainly the final writing surface.

Now gravity also participates in the upstream writing process.

20.4 Coherence memory and entanglement

The source–probe state becomes entangled when the branch-conditioned probe states are not identical.

A simple indicator is:

$$\mathbf{|\langle\psi_A|\psi_B\rangle| < 1.} \quad (20.12)$$

If:

$$\mathbf{|\psi_A\rangle = e^{ic}|\psi_B\rangle,} \quad (20.13)$$

the probe state factors out and no entanglement is created.

If they differ, source branch and probe response become correlated.

The amount of entanglement depends on:

- $|\alpha|$ and $|\beta|$;
- overlap $\langle \psi_A | \psi_B \rangle$;
- environmental decoherence;
- branch-dependent phases.

The experiment therefore requires a balance.

If the probe states are nearly identical, the gravitational distinction is too weak.

If they are fully distinguishable and environmental records reveal the branch, interference may be lost.

The operational window is:

$$\mathbf{0 < |\langle \psi_A | \psi_B \rangle| < 1.} \quad (20.14)$$

This is where relation is both differentiated and recombinable.

20.5 Relational persistence without settled ontology

One should resist prematurely deciding what exactly carries coherence memory.

Possible descriptions include:

- quantized gravitational field states;
- quantum geometry;
- nonclassical mediator variables;
- relational constraints;
- effective controlled unitary evolution;
- another future quantum-gravity structure.

The proposed experiment is designed as a witness, not a full ontology.

The disciplined conclusion is:

> Any successful account must contain a nonclassical gravitationally relevant structure capable of preserving the branch relation needed for interference.

This is stronger than saying only that the source is quantum.

The probe's branch-dependent gravitational evolution must remain coherently correlated with the source.

20.6 Coherence memory as pre-history

Classical memory stores a settled past.

Coherence memory stores an unresolved pre-history.

The distinction may be expressed as:

Classical history = one registered path. (20.15)

Quantum pre-history = structured relation among still-recombinable paths. (20.16)

The term *pre-history* does not imply that nothing physical occurred.

The gravitational interaction has occurred.

Momentum amplitudes have shifted.

Entanglement has formed.

What remains unresolved is which classical historical account will become the stable ledger.

Thus:

Physical interaction precedes historical closure. (20.17)

This is precisely why a gravity theory cannot begin only after collapse.

20.7 The coherent gravitational archive

A useful visualization is to treat the joint state as a temporary archive.

It contains entries:

Entry A = {source A, phase φ_A, probe state ψ_A}. (20.18)

Entry B = {source B, phase φ_B, probe state ψ_B}. (20.19)

But unlike a classical archive, the entries are not independent records.

They remain mutually phase-related and can interfere.

Therefore:

Quantum archive = entries + phase relations. (20.20)

Classical archive = entries + probabilities, without usable cross-phase. (20.21)

The postselection accesses the relation between archive entries.

This is how it constructs a conditional trace outside the classical branch hull.

---

21. Post-Closure Persistence: Curvature Memory

After closure, the relevant relation no longer appears primarily as a phase-bearing superposition.

It appears as a durable constraint on later physical evolution.

This is the regime previously emphasized by SMFT's gravity-as-memory model.

The foundational SMFT material describes gravity as a residual curvature generated by accumulated collapse traces and contrasts this passive geometric role with active gradient interactions.

The refined account retains that model as the post-closure regime.

21.1 Definition of curvature memory

A provisional definition is:

> **Curvature memory is the persistence of settled physical difference through an effective geometry that constrains later trajectories.**

Symbolically:

$M_{geo} = \{\langle T_{ab} \rangle, \delta A, R_{ab}, G_{ab}, \text{causal path constraints}\}$. (21.1)

The list combines quantities from different descriptive stages and should not be treated as one mathematical identity.

It represents the progression from stable energy–momentum difference to effective geometric consequence.

21.2 From record to path constraint

A detector record is historical but local.

Curvature is historical and environmental.

The distinction is:

Record: "this event occurred." (21.2)

Geometry: "because this physical distribution exists, future paths are altered." (21.3)

The transition from record to geometry therefore changes the scope of the trace.

A local event becomes part of the shared conditions under which other systems evolve.

SMFT interprets this as:

Private trace → public curvature. (21.4)

Gravity's universality then follows from the universality of the shared geometric environment.

21.3 Relative entropy and the durable-difference regime

In the Dorau–Much construction, relative entropy measures distinguishability between a vacuumlike state and a coherent excitation on a local horizon algebra. That distinguishability is related to the energy flux through the horizon. Under the entropy–area assumption, the flux corresponds to area variation, and local consistency yields the semiclassical Einstein equations.

The chain is:

$S_{rel}(\omega_0 \| \omega_\phi) \leftrightarrow \delta Q_H$. (21.5)

$S_{rel}(\omega_0 \| \omega_\phi) \propto \delta A$. (21.6)

δA consistency $\rightarrow G_{ab} + \Lambda g_{ab} = \alpha \langle T_{ab} \rangle$. (21.7)

The earlier SMFT reinterpretation maps this into:

Distinguishability → trace-current → boundary ledger → curvature memory. (21.8)

This is a model of consequence persistence.

The excitation differs from the reference state.

That difference carries energy.

The boundary responds.

The geometry records the response.

21.4 Curvature memory is not merely static storage

Memory can sound inert.

But geometric memory actively constrains future dynamics.

A spacetime metric affects:

- proper time;
- spatial distance;
- geodesic motion;
- redshift;
- causal accessibility;
- horizon structure.

Thus:

Stored consequence → altered future possibility space. (21.9)

Curvature memory is therefore active through constraint, even when it is passive relative to the original transition.

The distinction is:

Transition activity creates or changes relation. (21.10)

Geometric activity constrains what relations can occur next. (21.11)

This gives gravity a recursive role.

Past closure shapes future possibility.

21.5 Historical compression

A classical gravitational field does not present the complete microscopic history of all events that produced it.

It compresses relevant consequences into a smaller effective description.

For example:

Microscopic history → effective mass-energy distribution → metric geometry. (21.12)

This is a form of coarse-grained historical compression.

SMFT may express it as:

Many traces → one curvature field. (21.13)

The curvature does not retain every detail, but it preserves enough aggregate structure to guide future motion.

Thus:

Curvature memory = compressed physical history with dynamical consequences. (21.14)

21.6 Memory and irreversibility

The post-closure regime is associated with effective irreversibility.

Once a record has spread into many environmental degrees of freedom, recovering the original phase relation becomes practically impossible.

The geometric description then treats the record as settled.

Schematically:

ρ_coh → ρ_record → ⟨T_ab⟩_eff → g_ab^eff. (21.15)

The arrows are not necessarily fundamentally irreversible at every microscopic level.

They are irreversible relative to the accessible coarse-grained description.

This means curvature memory is scale-dependent.

At one scale, the geometry is settled.

At a deeper scale, residual quantum correlations may remain.

The refined SMFT model should therefore avoid absolute claims such as:

> Collapse is globally and instantaneously complete everywhere.

A better claim is:

> Closure is effective relative to an algebra of accessible observables and a scale of description.

21.7 Gravity as memory after admission, revised

The earlier slogan was:

> Weak interaction opens the gate; gravity remembers the crossing.

The new lifecycle shows that gravity is already involved before the gate.

A revised slogan is:

> **Gravity forms and preserves the coherent relation; the gate constructs an admitted trace; gravity preserves the settled consequence as curvature.**

In compact form:

Gravity relation → gate projection → gravity memory. (21.16)

This preserves the insight of the earlier article while correcting its temporal ordering.

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22. Gravity as the Persistence Geometry of Relational Difference

The coherent and geometric regimes can now be unified.

The central concept is not substance.

It is not force sign.

It is not even memory in the ordinary psychological sense.

It is **relational difference that remains physically consequential across transformation**.

This motivates the refined definition:

> **Gravity is the persistence geometry of relational difference across staged closure.**

In formula form:

Gravity = relational persistence + staged closure + geometric consequence. (22.1)

22.1 What counts as relational difference?

At the source-superposition stage:

$$\Delta_S = A - B.$$ (22.2)

This is a spatial branch difference.

At the probe-correlation stage:

$$\Delta_P = |\psi_A\rangle - |\psi_B\rangle.$$ (22.3)

This is a branch-conditioned dynamical difference.

At the phase level:

$\Delta\varphi = \varphi_A - \varphi_B$. (22.4)

This determines interference.

At the postselection stage:

$\Delta_f = \beta|\psi_B\rangle - \alpha|\psi_A\rangle$. (22.5)

This is a constructed conditional difference.

At the ledger stage:

ΔL = {p_f, ρ_f} relative to the complementary outcomes. (22.6)

This is a probability-weighted record difference.

At the information stage:

$\Delta I = S_{rel}(\omega_0 \| \omega_\phi)$. (22.7)

This is reference-relative distinguishability.

At the geometric stage:

$\Delta g_{ab} = g_{ab}^{excited} - g_{ab}^{reference}$. (22.8)

This is an effective spacetime difference.

The lifecycle does not preserve one identical mathematical object through every stage.

It preserves a continuity of physical consequence across changing representations.

22.2 Representation changes, relation persists

The complete sequence is:

Amplitude difference → conditional state difference → statistical record difference → informational difference → geometric difference. (22.9)

The form changes.

The relation persists.

This is the sense in which gravity may be understood as persistence geometry.

A future theory must explain the translation rules between these representations.

Without such rules, the lifecycle remains conceptual.

With such rules, it could become a calculational framework.

22.3 The persistence chain

The refined chain can be written as:

$\Delta\Psi_G \rightarrow \Delta\rho_f \rightarrow \Delta L \rightarrow \Delta I \rightarrow \Delta T_{ab} \rightarrow \Delta A \rightarrow \Delta g_{ab}$. (22.10)

Where:

- $\Delta\Psi_G$ is coherent gravitational relational difference;
- $\Delta\rho_f$ is conditional state difference;
- ΔL is complete ledger difference;
- ΔI is durable distinguishability;
- ΔT_{ab} is effective stress-energy difference;
- ΔA is boundary-area response;
- Δg_{ab} is effective geometric response.

The source papers establish only particular parts of this chain.

The repulsive-gravity proposal addresses the first two transitions:

$\Delta\Psi_G \rightarrow \Delta\rho_f$. (22.11)

The relative-entropy paper addresses a later chain:

$\Delta I \rightarrow \Delta T_{ab} \rightarrow \Delta A \rightarrow \Delta g_{ab}$. (22.12)

The middle remains open:

$\Delta\rho_f \rightarrow \Delta L \rightarrow \Delta I$. (22.13)

This is the central SMFT handover problem.

22.4 Gravity as a closure-spanning relation

The term *closure-spanning* means that gravity participates on both sides of the closure boundary.

Before closure, gravity contributes to forming:

source branch ↔ probe branch. (22.14)

After closure, gravity contributes to maintaining:

stress-energy distribution ↔ spacetime geometry. (22.15)

Thus gravity is not simply before or after collapse.

It spans the transition.

The role changes from:

relation formation (22.16)

to:

relation preservation through geometry. (22.17)

This is the most important fine-tuning produced by the new paper.

22.5 Attraction, repulsion, and persistence

The refined definition also clarifies why attraction and repulsion are not the deepest categories.

At branch level:

$F_A > 0$ and $F_B > 0$. (22.18)

At conditioned level:

$F_{eff,f} < 0$ may occur. (22.19)

At geometric level, the classification depends on the full stress-energy and spacetime solution.

The sign can therefore vary across descriptive levels.

Persistence is more fundamental within the proposed grammar because it tracks what structure survives the passage between levels.

Thus:

Attraction describes branch direction. (22.20)

Repulsion describes a conditioned response. (22.21)

Persistence describes the lifecycle architecture. (22.22)

22.6 The revised gravity equation of meaning

Within SMFT's interpretive vocabulary, one may write a conceptual equation:

$\mathcal{G}_{SMFT} = \mathcal{P}_{rel}[\Delta\Psi] + \mathcal{C}[\Delta\Psi] + M_{geo}[\Delta L]$. (22.23)

Where:

- $\mathcal{P}_{rel}[\Delta\Psi]$ represents preservation of coherent relational difference;
- $\mathcal{C}[\Delta\Psi]$ represents staged closure and trace construction;
- $M_{geo}[\Delta L]$ represents persistence of the closed ledger as geometry.

This is not a physical field equation.

It is a decomposition of roles.

A future physical version would require:

- state-space definitions;
- operators;
- dimensional consistency;
- conservation laws;

- correspondence with general relativity;
- testable deviations.

The present equation serves only as a research map.

22.7 The refined SMFT definition

The earlier definition was:

Gravity = residual curvature left by collapse traces. (22.24)

The refined definition is:

Classical gravity = residual curvature left by effectively closed physical traces. (22.25)

The expanded general definition is:

Gravity = persistence of physical relational difference before, through, and after closure. (22.26)

The first is a semiclassical limit.

The second is the full lifecycle hypothesis.

This preserves the earlier SMFT insight while preventing it from excluding gravity's coherent quantum role.

22.8 The central proposition

The full Part V argument may be condensed into one proposition:

> **The quantum and classical faces of gravity differ not because one contains relation and the other does not, but because they preserve different forms of relation. Quantum gravity preserves the phase-bearing association among unresolved alternatives. Classical gravity preserves the history-bearing association between settled energy distribution and spacetime geometry. Closure is the transformation between these persistence regimes.**

In symbolic form:

Phase-bearing relation → closure → history-bearing geometry. (22.27)

And:

Gravity preserves alternatives before closure and consequences afterward. (22.28)

This is the Dual Persistence Principle in its final form.

The next part must now revisit the weak-gate interpretation. The gate can no longer be treated as the point where gravity first appears. It must be understood as the operation that changes the status of already gravitationally related information—from phase-bearing possibility into record-bearing trace.

Part VI — Fine-Tuning the Weak-Gate Model

23. The Gate Does Not Create the Gravitational Relation

The earlier SMFT formulation placed the weak-like gate near the beginning of the collapse sequence:

Latent difference → weak gate → admitted trace → curvature memory. (23.1)

This architecture was useful in the relative-entropy interpretation because the local horizon appeared to define the boundary across which a coherent excitation became informationally and geometrically consequential. Relative entropy measured distinguishability, energy flux represented the dynamical crossing, area variation recorded the crossing, and curvature closed the record into semiclassical geometry.

The repulsive-gravity proposal requires a more precise ordering.

Before any source postselection occurs, gravity has already produced the joint relation:

$$|\Psi_{SP}\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle.$$ (23.2)

The gravitational interaction has already:

- distinguished the source branches dynamically;
- displaced the probe differently in each branch;
- correlated source position with probe momentum;
- preserved the relative phase needed for later interference.

The projection gate acts only after this relation exists.

Therefore:

Gravitational relation precedes projection gate. (23.3)

And:

Projection gate acts on an already formed gravitational correlation. (23.4)

This corrects the earlier tendency to interpret the gate as the origin of all admitted physical difference.

23.1 Two kinds of “before”

The phrase *before the gate* can refer to two very different conditions.

The first is a genuinely unstructured background in which no branch-sensitive physical relation has yet formed.

The second is a coherent relational state that has not yet become a stable record.

The repulsive-gravity proposal concerns the second condition.

Before source postselection, the system is not empty of structure.

It already contains:

A ↔ ψ_A. (23.5)

B ↔ ψ_B. (23.6)

φ_A − φ_B. (23.7)

⟨ψ_A|ψ_B⟩. (23.8)

What is absent is not relation.

What is absent is closure.

Thus:

Pre-gate ≠ pre-relation. (23.9)

In the proposed experiment:

Pre-gate = relation formed but not yet conditionally admitted. (23.10)

This distinction is essential for refining the weak-gate model.

23.2 Gravity writes before the gate reads

A useful functional distinction is:

Gravity writes branch relation. (23.11)

The gate reads the relation in a selected basis. (23.12)

The gravity-mediated unitary produces the state:

$\mathcal{U}$_G|Ψ0⟩ = Σ_j c_j|j⟩|ψ_j⟩. (23.13)

The projection then produces:

|ψ_f⟩ ∝ Σ_j c_j⟨f|j⟩|ψ_j⟩. (23.14)

The first operation distributes source information into the joint state.

The second selects which relational combination becomes the conditioned probe trace.

The gate therefore does not create the alphabet.

It chooses how the already written relational sentence is read.

23.3 The weak-gate metaphor after refinement

The earlier metaphor was:

> The weak interaction opens the gate; gravity remembers the crossing.

The refined metaphor is:

> Gravity constructs the relational approach to the gate. The gate converts that relation into an admitted trace. Gravity later preserves the closed consequence as geometry.

In compact form:

Gravity relation → gate conversion → gravity memory. (23.15)

Or:

Relational formation → admission transformation → geometric persistence. (23.16)

This revised ordering preserves the usefulness of the weak-gate idea while preventing it from swallowing the entire premeasurement process.

23.4 A gate requires something structured to pass through it

A gate has no physical meaning without:

- a state before the gate;
- an admissibility rule;
- an output state;
- a distinction between admitted and nonadmitted sectors.

In the present context, the input is not merely the source superposition:

$\alpha|A\rangle + \beta|B\rangle$. (23.17)

It is the correlated source–probe state:

$\alpha|A\rangle|\psi_A\rangle + \beta|B\rangle|\psi_B\rangle$. (23.18)

The gate therefore acts on relation, not on an isolated source.

This suggests a general SMFT requirement:

> A transition gate should be defined relative to the structured field it transforms, not treated as a free-standing cause of change.

Symbolically:

Gate effect = function[input relation, projection basis, environmental context]. (23.19)

23.5 The horizon gate and postselection gate are not physically identical

The earlier relative-entropy article visualized the local Rindler horizon as a weak-gate membrane. It defines the causal boundary relative to which vacuum and excitation are compared, modular flow is established, energy flux is evaluated, and area response is assigned.

The postselection gate in the repulsive-gravity proposal has a different physical form.

It consists of:

- source interferometer recombination;
- a selected measurement basis;
- a chosen detector output;
- conditioning of the probe data.

Therefore:

Rindler horizon gate ≠ interferometric postselection gate. (23.20)

Their structural similarity is narrower:

Both define the operational context in which a previously distributed physical difference becomes a countable trace. (23.21)

The horizon does so through accessible algebra, modular flow, and causal boundary structure.

Postselection does so through a measurement projector and conditional state normalization.

The analogy is functional, not material.

23.6 Gate as interface rather than source

The refined model should define the gate as an interface.

An interface:

- receives a structured input;
- applies a rule;
- changes representational status;
- produces an output available to another regime.

Thus:

Gate = conversion interface between relation regimes. (23.22)

For the repulsive-gravity proposal:

Phase-bearing joint relation → projection gate → conditioned probe state. (23.23)

For the relative-entropy horizon framework:

Reference-relative field excitation → horizon informational interface → flux and area ledger. (23.24)

For SMFT generally:

Distributed possibility → observer-conditioned interface → committed trace. (23.25)

The common role is conversion, not original causation.

---

24. The Gate Changes Information Type

The most important fine-tuning of the weak-gate idea is that a gate does not merely allow or block a pre-existing object.

It changes the operational form of information.

Before projection, the relevant information is stored in:

- complex amplitudes;
- relative phase;
- branch association;
- entangled correlations;
- interference capacity.

After projection and recording, the relevant information appears as:

- a conditional state;
- a measurement probability;
- a detector outcome;
- a momentum distribution;
- a stable record.

The transformation is:

Phase-bearing relation → record-bearing distinction. (24.1)

This is the refined weak-gate function.

24.1 Phase-usable information

Phase-usable information is information that can influence an interference experiment.

It resides in terms such as:

$$\rho_{AB} = \alpha\beta^* e^{i(\phi_A - \phi_B)} |A\rangle\langle B|.$$ (24.2)

Its physical significance is revealed when a measurement basis recombines A and B.

If all measurements are performed only in the branch basis, the relative phase may remain hidden.

But in a superposition basis, it changes the conditioned outcome distribution.

Phase-usable information therefore satisfies:

Changing Δφ changes observable interference. (24.3)

Where:

$\Delta\varphi = \varphi_A - \varphi_B$. (24.4)

This information is contextual.

Its operational value depends on the existence of a compatible recombination procedure.

24.2 Record-usable information

Record-usable information is information that remains available after the measurement context has produced stable outcome sectors.

It includes:

p_f = probability of outcome f. (24.5)

ρ_f = conditioned state associated with f. (24.6)

R_f = durable apparatus record of f. (24.7)

A record may remain readable even when the original relative phase is no longer accessible.

Thus:

Phase accessibility may vanish while record distinguishability remains. (24.8)

The gate does not necessarily destroy all information.

It transforms which information can be used and how.

24.3 Information conversion rather than information creation

The projection-conditioned trace may appear to contain a property absent from both branches, such as negative mean momentum.

This does not mean the gate creates physical information from nothing.

The information required for the anomaly was already present relationally in:

- the unequal branch shifts;
- the amplitude ratio;
- the relative phase;
- the overlap of the probe states;
- the selected projection basis.

The gate converts this distributed relational information into a visible conditional asymmetry.

Therefore:

Constructed trace ≠ information ex nihilo. (24.9)

Instead:

Constructed trace = reorganized expression of pre-existing relational information.

(24.10)

24.4 From off-diagonal structure to conditional distribution

Before measurement, the joint density operator contains cross terms:

$\rho_{cross} = \alpha\beta^* e^{i(\phi_A - \phi_B)} |A\rangle\langle B| \otimes |\psi_A\rangle\langle\psi_B| + h.c.$ (24.11)

After projection onto $|f\rangle$, these cross terms contribute to:

$\tilde{\rho}_f = Tr_S[(|f\rangle\langle f| \otimes I)\rho_{SP}].$ (24.12)

The probe momentum distribution becomes:

$P_f(p) = \langle p|\rho_f|p\rangle.$ (24.13)

The interference information has therefore changed form:

Joint-state off-diagonal structure → conditional probability asymmetry. (24.14)

This is a concrete example of information-type conversion.

24.5 The gate converts distributed information into localized output

Before projection, the relevant coherence resides in the source–probe relation.

After projection, the conditioned probe state alone can display the anomaly.

Thus:

Distributed relational information → subsystem-localized conditional trace. (24.15)

The localization is conditional because the probe trace is meaningful only relative to the selected source result.

The source outcome and probe distribution remain conceptually linked even if the source is no longer dynamically involved in later readout.

This is why the record should retain the index f:

ρ_f, not merely ρ. (24.16)

The subscript preserves the relational origin of the apparently local probe state.

24.6 Gate conversion and semantic collapse

In SMFT's semantic interpretation, distributed meaning potential becomes a committed interpretation relative to an observer structure.

The refined physical analogy is:

Relational quantum possibility + projection architecture → operationally committed conditional state. (24.17)

The analogy should be used carefully.

A quantum projector is mathematically precise.

A semantic observer operator is currently a broader theoretical construct.

But the comparison reveals an important general rule:

> Commitment changes not only which possibility is retained, but also the representational format in which the result can affect future processing.

Before commitment, alternatives can interfere.

After commitment, the result can be remembered, communicated, acted upon, or written into a ledger.

Thus:

Gate = format transformation of consequential difference. (24.18)

24.7 A three-format information hierarchy

The complete lifecycle suggests three information formats.

Format I — Phase format

I_phase = {amplitude, phase, coherence, overlap}. (24.19)

This format supports interference.

Format II — Record format

I_record = {outcome, probability, conditional state, apparatus trace}. (24.20)

This format supports historical accounting.

Format III — Geometry format

I_geo = {stress-energy, flux, area response, curvature}. (24.21)

This format supports shared path constraints.

The handover sequence is:

I_phase → I_record → I_geo. (24.22)

The projection gate mainly mediates the first conversion.

Ledger and geometric closure mediate the second.

24.8 The gate does not guarantee geometric closure

A postselected detector outcome does not automatically become spacetime curvature.

Projection yields a conditional trace.

Additional processes are needed:

- normalization across outcomes;
- apparatus record formation;
- environmental decoherence;
- conservation accounting;
- effective stress-energy description;
- boundary response;
- geometric consistency.

Therefore:

Projection closure ≠ geometric closure. (24.23)

This is one of the most important restrictions in the refined framework.

The weak-gate concept should not be used to jump directly from measurement to gravity.

---

25. The Gate Can Construct Properties Absent from Every Branch

The postselected repulsion demonstrates a general principle:

> The observable property of a selected coherent combination need not equal the observable property of any branch considered separately.

The branch-conditioned probe states satisfy:

$$\langle \mathbf{p} \rangle_A = \delta_A > 0. \quad (25.1)$$

$$\langle \mathbf{p} \rangle_B = \delta_B > 0. \quad (25.2)$$

The selected state may satisfy:

$$\langle \mathbf{p} \rangle_f < 0. \quad (25.3)$$

The negative mean is therefore a property of the projection-conditioned relation.

It is not a hidden branch value waiting to be uncovered.

25.1 Branch inheritance is not universal

A classical branch-selection model assumes that the observed outcome inherits the properties of the selected branch.

If A is selected:

$$O_f = O_A. \quad (25.4)$$

If B is selected:

$$O_f = O_B. \quad (25.5)$$

But a superposition-basis projection yields:

$|\psi_f\rangle \propto u\alpha|\psi_A\rangle + v\beta|\psi_B\rangle$. (25.6)

The expectation value becomes:

$\langle O\rangle_f = \langle\psi_f|\hat{O}|\psi_f\rangle$. (25.7)

This includes cross terms.

Therefore:

$\langle O\rangle_f \neq \langle O\rangle_A$ **and** $\langle O\rangle_f \neq \langle O\rangle_B$ **in general.** (25.8)

The selected trace is not obligated to inherit a branch property.

25.2 Relational emergence

The new property emerges from the relation among components.

This can be represented as:

Property(component A) + Property(component B) + Relation(A,B) → Property(combination). (25.9)

The relation term is essential.

Without it:

Relation(A,B) = 0 → classical mixture. (25.10)

With it:

Relation(A,B) ≠ 0 → interference-dependent conditional property. (25.11)

This is a precise case of relational emergence.

The negative conditioned momentum does not require a negative gravitational force eigenvalue.

It requires a relational cross term.

25.3 Projection constructivity principle

The refined SMFT principle may be stated as:

> **Projection does not merely reveal which branch was real; it can construct an observable trace from the relational geometry among branches.**

In compact form:

Selected trace = branch content + relational interference + projection geometry. (25.12)

This principle has implications beyond gravity.

It suggests that in any SMFT application involving collapse:

- the output cannot always be understood as a copied input branch;
- the observer basis may reorganize latent relational structure;
- a trace may carry emergent properties absent from the components.

25.4 Constructive projection and novelty

This provides a possible physical analogy for novelty formation.

Suppose a field contains alternatives A and B.

A reductive model predicts:

Collapse(A,B) ∈ {A,B}. (25.13)

A constructive model permits:

Collapse_f(A,B) = C_f, where C_f ∈ Span{A,B} but C_f ≠ A and C_f ≠ B. (25.14)

The new trace C_f is constrained by A, B, and the projection rule, but is not identical to either input.

This resembles how:

- a new interpretation can emerge from competing meanings;
- a new decision can combine incompatible considerations;
- a new institutional form can arise from the tension between prior forms;
- a new AI output can synthesize latent features from multiple internal trajectories.

The analogy does not prove that semantic novelty is quantum mechanical.

It shows that the SMFT concept of collapse should allow constructive recombination as a general structural possibility.

25.5 Negative conditional response as a relational residual

The anomalous sign can be viewed as a residual left after destructive cancellation.

The two positive-shifted amplitudes overlap.

Their subtraction removes much of the amplitude in one region of momentum space.

What remains is biased in the opposite direction.

Thus:

Conditional repulsion = residual after asymmetric amplitude cancellation. (25.15)

This wording is useful because it avoids reifying the negative mean as a new primitive force.

The repulsion is the geometry of what remains after interference.

25.6 Constructed trace and basis dependence

The conditional trace depends on the selected basis.

Let:

$$|f(\theta,\chi)\rangle = \cos\theta|A\rangle + e^{ichi}\sin\theta|B\rangle. \quad (25.16)$$

Then:

$$|\psi_f\rangle \propto \alpha\cos\theta|\psi_A\rangle + \beta e^{-ichi}\sin\theta|\psi_B\rangle. \quad (25.17)$$

Changing θ or χ changes:

- amplitude ratio;
- interference sign;
- normalization;
- conditional expectation.

Thus:

$$\langle p\rangle_f = F(\theta, \chi, \alpha, \beta, \delta_A, \delta_B, \psi_0). \quad (25.18)$$

The constructed property is projection-dependent.

But it is not subjective in an arbitrary sense.

For a specified preparation and measurement basis, quantum theory gives definite statistical predictions.

25.7 The observer does not freely choose the result

The experimenter may choose the measurement setting.

The experimenter does not choose the individual outcome.

Nor can the experimenter arbitrarily assign the conditioned distribution.

The allowed trace is constrained by:

$$\rho_f = P_f\rho_{SP} P_f / Tr(P_f\rho_{SP}). \quad (25.19)$$

Thus:

Choice of question ≠ choice of answer. (25.20)

This distinction is vital for SMFT observer theory.

The observer architecture influences which relational property becomes accessible.

It does not freely manufacture any desired physical fact.

25.8 Projection constructivity and conservation

A constructed conditional trace can display an anomalous value.

But the full set of projection outcomes obeys completeness:

$\Sigma_f P\hat{}_f = I.$ (25.21)

And:

$\Sigma_f p_f \rho_f = \rho_{P'}.$ (25.22)

The complete ensemble remains tied to the premeasurement state and overall dynamics.

Projection constructivity therefore operates within global accounting constraints.

This leads to:

Local novelty under projection is compatible with global conservation under ledger closure. (25.23)

---

26. Observer Roles Must Be Separated

The word *observer* is used differently across quantum mechanics, relativity, information theory, and SMFT.

Without careful separation, the article could slide from operational measurement into claims about consciousness or self-reference that are not supported by the attached physics papers.

At least four observer roles should be distinguished.

26.1 Kinematic observer

A kinematic observer defines:

- a reference frame;
- a worldline;
- coordinates;
- accessible measurements.

In relativity, different observers may decompose the same spacetime differently.

A uniformly accelerated observer has access to a Rindler wedge and associates a horizon with inaccessible regions.

This observer role is relevant to the Dorau–Much framework because the local Rindler horizon and boost/Killing flow define the setting in which the horizon algebra, modular structure, energy flux, and area response are evaluated.

The observer's acceleration matters.

Human consciousness does not.

26.2 Measurement observer

A measurement observer is the physical arrangement that defines:

- which observable is measured;
- which basis is implemented;
- which outcomes are distinguished;
- how the result is recorded.

In the repulsive-gravity proposal, this includes:

- the interferometer;
- the recombination phase;
- the selected output port;
- the probe momentum detector;
- the data-conditioning rule.

The measurement observer can be represented as:

Ô_meas = {P̂_f, detector, record channel}. (26.1)

It is a physical protocol.

26.3 Informational observer

An informational observer is defined by access to an algebra of observables or a set of records.

Two observers may have access to different information even if they inhabit the same physical system.

For example:

- one observer knows the postselection outcome;
- another receives only the unconditioned probe distribution;
- one has access to the environment;
- another traces it out.

Their state assignments differ because their accessible information differs.

Thus:

Observer-relative state ≠ consciousness-relative reality. (26.2)

It may simply mean:

State description relative to accessible observables and records. (26.3)

26.4 Self-referential SMFT observer

SMFT sometimes introduces a stronger observer concept, Ô_self.

Such a system can:

- model its own projection rules;
- retain prior traces;
- modify future selection criteria;
- close feedback loops;
- develop attractor-dependent identity.

This role is relevant to cognition, AI, life, and semantic self-reference.

It is not required for the proposed gravity experiment.

Therefore:

Ô_meas ≠ Ô_self. (26.4)

An automated apparatus can implement Ô_meas without possessing self-model, identity, or awareness.

26.5 Rindler observer, postselection apparatus, and Ô_self

The distinctions can be summarized as:

Observer role	Primary function
Rindler observer	Defines causal accessibility and horizon structure
Postselection apparatus	Defines measurement basis and conditioned ensemble
Informational observer	Defines accessible records and state assignment
Ô_self	Uses retained traces to modify future projection rules

Thus:

Rindler observer ≠ postselection apparatus ≠ informational access set ≠ Ô_self. (26.5)

They may share a structural role in defining admissible information, but they are not interchangeable.

26.6 Operational definition of Ô

For the present article, the safest definition is:

> **Ô is the complete operational architecture that determines how a physical relation becomes an admitted trace.**

For the repulsive-gravity protocol:

Ô_exp = {state preparation, interaction duration, phase setting, measurement basis, outcome selection, readout}. (26.6)

This definition avoids treating Ô as a mysterious mental force.

It also makes Ô experimentally specifiable.

Different Ô_exp configurations produce different conditional traces.

26.7 Observer dependence and objectivity

A result can be observer-conditioned without being arbitrary.

The conditioned distribution is relative to a specified postselection:

P(p|f). (26.7)

Another observer who does not know f uses:

P(p) = Σ_f P(f)P(p|f). (26.8)

Both distributions are objective relative to their conditioning information.

The distinction resembles ordinary probability:

Conditional fact ≠ subjective illusion. (26.9)

The key is to state the conditioning variables.

Thus:

Objectivity requires protocol disclosure, not observer elimination. (26.10)

26.8 No-signalling and conditioning

Postselection can change the distribution within a selected ensemble, but it cannot be used to transmit controllable information without access to the selection record.

An observer examining only the probe and lacking the source outcome sees the unconditional state:

ρ_P′ = Σ_f p_fρ_f. (26.11)

The anomalous subensemble becomes identifiable only after correlating probe data with the source record.

Therefore:

Conditional anomaly requires classical outcome correlation for operational use. (26.12)

This prevents postselection from becoming an unconstrained nonlocal signalling mechanism.

26.9 The observer as boundary specification

Across the two physics papers, observer structure repeatedly defines a boundary.

In the Rindler setting, the boundary is causal:

accessible wedge | inaccessible region. (26.13)

In postselection, the boundary is projective:

selected outcome sector | rejected outcome sectors. (26.14)

In informational accounting, the boundary is epistemic-operational:

accessible algebra | traced-out degrees of freedom. (26.15)

In SMFT, the boundary is semantic or interpretive:

admitted trace | uncommitted possibility. (26.16)

The common role is boundary specification.

But each boundary belongs to a different theory and must retain its own mathematics.

26.10 The refined observer principle

A disciplined SMFT observer principle may be stated as:

> **An observer does not create an unconstrained physical world. An observer architecture defines the boundary, basis, and record structure through which an existing relational field becomes available as a particular trace.**

In compact form:

Observer effect = constrained trace formation, not arbitrary reality manufacture. (26.17)

This formulation is compatible with the postselected gravity proposal and avoids claims not supported by it.

Part VII — The Missing Quantum-to-Curvature Handover

27. Two Papers, Two Ends of One Research Problem

The repulsive-gravity proposal and the quantum-relative-entropy derivation examine different systems, use different mathematical tools, and answer different questions.

They should not be merged into a fictitious continuous calculation.

Yet they illuminate complementary ends of a common conceptual problem.

The first paper concerns the coherent upstream regime:

Source superposition → gravitationally mediated branch correlation → postselection → anomalous conditional momentum. (27.1)

The second concerns a downstream semiclassical regime:

Vacuum/excitation distinguishability → relative entropy → horizon energy flux → area variation → semiclassical Einstein equation. (27.2)

Placed side by side, they define a missing middle:

Phase-bearing gravitational relation → ? → durable geometry-bearing record. (27.3)

That question mark is the most important theoretical object introduced by the present synthesis.

27.1 What the upstream paper requires

The upstream framework requires:

- branch-sensitive gravitational action;
- coherent source superposition;
- probe-state overlap;
- relative phase preservation;
- source–probe entanglement;
- superposition-basis projection;
- probability-conditioned readout.

Its characteristic mathematical objects are:

$|\Psi_{SP}\rangle$, $\hat{P}_f$, ρ_f, $\langle \Delta p \rangle_W$. (27.4)

The key observable may lie outside the classical branch hull.

27.2 What the downstream paper requires

The downstream framework requires:

- a local horizon;
- a reference state;
- a coherent excitation;
- an algebra of observables;
- modular flow;
- relative entropy;
- stress-energy flux;
- area variation;
- local geometric consistency.

Its characteristic objects are:

$S_{rel}(\omega_o \| \omega_\phi)$, $\langle T_{ab} \rangle$, δA, R_{ab}. (27.5)

The key information measure is nonnegative and reference-relative.

27.3 Why the variables cannot be identified directly

The following identifications are invalid:

$\langle \Delta p \rangle_W = S_{rel}$. (27.6)

$\rho_f = \omega_\phi$. (27.7)

Postselection probability = horizon area. (27.8)

Negative conditional shift = negative relative entropy. (27.9)

Interferometer output = Rindler horizon. (27.10)

The variables inhabit different physical and mathematical settings.

The useful relationship is structural:

- both frameworks distinguish reference and deviation;
- both require a boundary or projection structure;
- both convert relational difference into an observable trace;
- both concern how quantum information becomes physically consequential.

The missing handover is therefore an open research program, not an established equality.

27.4 The question the combined framework poses

The integrated question is:

> How does a gravity-mediated phase relation, represented by amplitudes and postselected conditional states, become a stable reference-relative difference represented by energy, entropy, boundary response, and curvature?

In compact form:

$I_{phase} \rightarrow I_{record} \rightarrow I_{geo}$. (27.11)

This is the quantum-to-curvature handover problem.

The following section must formalize the required handover operator and identify the constraints any future derivation would have to satisfy.

28. The Handover Operator Problem

The central unresolved object may be represented schematically by a quantum-to-curvature handover operator:

$\mathscr{H}_{QC} : (\rho_{SP}, \mathcal{P}, \mathscr{E}, \mathcal{A}, \mathscr{B}) \rightarrow (\rho_{record}, \mathcal{D}_{res}, \langle T_{ab} \rangle_{eff}, \delta A, g_{ab}^{eff})$. (28.1)

Where:

- ρ_SP is the coherent source–probe state;
- $\mathcal{P}$ is the projection or measurement protocol;
- $\mathscr{E}$ is environmental coupling and decoherence;
- $\mathcal{A}$ is the apparatus and recording architecture;
- $\mathscr{B}$ specifies the relevant boundary and accessible observable algebra;
- ρ_record is the effectively stable recorded state;
- $\mathcal{D}$_res is durable distinguishability from a reference;
- ⟨T_ab⟩_eff is an effective stress-energy description;
- δA is a boundary-area response;
- g_ab^eff is the resulting semiclassical geometry.

Equation (28.1) is not a field equation.

It is a statement of the missing theoretical task.

A complete handover theory must explain how a state described upstream through amplitudes, phases, and entanglement can become a downstream source of stable geometric response without simply inserting the classical Einstein equation by assumption.

The desired map is not:

Quantum amplitudes → classical geometry by verbal reinterpretation. (28.2)

It must be:

Precisely specified quantum state + specified closure operations + controlled approximation → effective geometry. (28.3)

28.1 Why one operator may eventually become several maps

The symbol $\mathscr{H}$_QC compresses several distinct transformations.

A more realistic decomposition is:

$\mathscr{H}$**_QC =** $\mathcal{C}$**_G ∘** $\mathcal{C}$**_T ∘** $\mathcal{C}$**_I ∘** $\mathcal{C}$**_L ∘** $\mathcal{P}$**.** (28.4)

Where:

$\mathcal{P}$ **: coherent joint state → conditioned outcome sectors.** (28.5)

$\mathcal{C}$**_L : conditioned sectors → complete durable ledger.** (28.6)

$\mathcal{C}$**_I : durable ledger → reference-relative informational description.** (28.7)

$\mathcal{C}$**_T : informational description → effective stress-energy account.** (28.8)

$\mathcal{C}$**_G : stress-energy account → effective geometric closure.** (28.9)

The ordering matters.

Projection cannot be treated as identical to decoherence.

Decoherence cannot be treated as identical to a stable apparatus record.

A stable record cannot be treated as automatically equivalent to relative entropy.

Relative entropy cannot be treated as identical to stress-energy.

Stress-energy cannot be treated as curvature without a gravitational closure law.

Therefore:

$\mathcal{P} \neq \mathcal{C}_{\mathbf{L}} \neq \mathcal{C}_{\mathbf{I}} \neq \mathcal{C}_{\mathbf{T}} \neq \mathcal{C}_{\mathbf{G}}$. (28.10)

The older SMFT vocabulary often compressed all these transitions into the word *collapse*. The refined framework must resist that compression.

28.2 Projection map

For source outcomes f, the projection map produces unnormalized conditional states:

$\tilde{\rho}_f = \mathrm{Tr}_S[(\hat{P}_f \otimes I)\rho_{SP}]$. (28.11)

The associated probabilities are:

$p_f = \mathrm{Tr}(\tilde{\rho}_f)$. (28.12)

The normalized conditional states are:

$\rho_f = \tilde{\rho}_f / p_f$. (28.13)

This produces the family:

$\mathcal{P}(\rho_{SP}) = \{\{p_f, \rho_f\}\}_f$. (28.14)

The output is not one classical history.

It is a complete set of possible recorded sectors with their weights.

A specific postselected experiment retains one member of this family for analysis.

A complete physical account retains the full family.

28.3 Ledger-closure map

The ledger map adds the physical records that distinguish the sectors:

$\mathcal{C}_{\mathbf{L}} : \{\{p_f, \rho_f\}\}_f \to \rho_{SPAE}^{record}$. (28.15)

Here, S, P, A, and E denote source, probe, apparatus, and environment.

A schematic recorded state is:

$\rho_{SPAE}^{record} \approx \Sigma_f\, p_f \rho_{SP} |f\rangle \otimes |A_f\rangle\langle A_f| \otimes |E_f\rangle\langle E_f|$. (28.16)

When the apparatus and environment records become approximately orthogonal:

$$\langle A_f|A_g\rangle\langle E_f|E_g\rangle \approx 0 \text{ for } f \neq g, \quad (28.17)$$

the outcome sectors cease to interfere for the accessible observer.

This is the effective transition from phase-usable alternatives to record-usable outcomes.

But even here, several interpretive possibilities remain:

- fundamental collapse;
- relative-state branching;
- decoherence without fundamental collapse;
- objective reduction;
- hidden-variable completion;
- another future mechanism.

The handover framework does not need to settle the entire measurement problem before identifying the operational closure sequence.

28.4 Informational-closure map

Once a stable outcome or excitation has been defined relative to a reference state, one may introduce a measure of distinguishability.

Schematically:

$$\mathcal{D}_{\text{res}} = \mathcal{C}_{I}(\rho_{\text{record}}, \rho_{\text{ref}}). \quad (28.18)$$

For ordinary density operators, one candidate is quantum relative entropy:

$$S(\rho_{\text{record}}\|\rho_{\text{ref}}) = \mathrm{Tr}[\rho_{\text{record}}(\log\rho_{\text{record}} - \log\rho_{\text{ref}})]. \quad (28.19)$$

In algebraic quantum field theory, the appropriate object may instead be Araki–Uhlmann relative entropy defined on a von Neumann algebra.

The Dorau–Much paper uses precisely such a relative-entropy construction for a vacuumlike state and coherent excitation on a horizon algebra. It shows that, in that setting, the relative entropy can be represented through the excitation's energy flux along the horizon.

The important restriction is:

> The reference state, observable algebra, region, and boundary must be specified.

Relative entropy is not an absolute quantity attached to an event independently of context.

It is relational:

$$S_{\text{rel}} = \text{distinguishability of state 1 from state 2 relative to an observable algebra.} \quad (28.20)$$

This makes it structurally compatible with SMFT's emphasis on relational traces, but it also prevents casual identification with every form of "difference."

28.5 Stress-energy closure

A durable information difference becomes gravitationally relevant only if it is connected to physical energy-momentum.

A schematic map is:

$\mathcal{C}_{\mathbf{T}} : \mathcal{D}_{\mathbf{res}} \to \langle \mathbf{T}_{\mathbf{ab}} \rangle_{\mathbf{eff}}$ (28.21)

But Equation (28.21) should not be understood as saying that information is an independent substance transformed into energy.

Rather, the same physical excitation can be represented through two descriptions:

Informational face = distinguishability from reference. (28.22)

Dynamical face = energy-momentum relative to reference. (28.23)

In the Dorau–Much framework, this dual representation is made mathematically explicit for coherent excitations on the horizon. The relative entropy is expressed through the expectation value of the normal-ordered stress-energy tensor along the Killing flow.

Thus:

Information difference and energy flux are two representations of one physical deviation. (28.24)

They should not be imagined as two separate fluids.

28.6 Geometric closure

The final map relates effective stress-energy to geometry:

$\mathcal{C}_{\mathbf{G}} : \langle \mathbf{T}_{\mathbf{ab}} \rangle_{\mathbf{eff}} \to \mathbf{g}_{\mathbf{ab}}^{\mathbf{eff}}$ (28.25)

In semiclassical gravity:

$\mathbf{G}_{\mathbf{ab}}[\mathbf{g}_{\mathbf{eff}}] + \Lambda \mathbf{g}_{\mathbf{ab}}^{\mathbf{eff}} = 8\pi \mathbf{G} \langle \mathbf{T}_{\mathbf{ab}} \rangle_{\mathbf{ren}}$ (28.26)

The Dorau–Much paper argues that an equation of this form follows locally when relative entropy is identified with horizon energy flux, the entropy–area relation is assumed, and consistency is imposed across local Rindler horizons.

SMFT interprets this as geometric ledger closure:

Durable local difference → mutually consistent spacetime response. (28.27)

But this interpretation does not replace the need to explain:

- renormalization;
- state dependence;
- fluctuations beyond expectation values;
- nonlinear backreaction;

- higher-order corrections;
- the regime in which semiclassical gravity fails.

28.7 Required properties of $\mathscr{H}_{QC}$

Any serious handover map should satisfy several constraints.

Normalization

$$\Sigma_f p_f = 1. \quad (28.28)$$

Complete positivity where appropriate

The effective quantum channels used for open-system evolution should map valid states to valid states.

Conservation

$$\nabla^a \langle T_{ab} \rangle_{eff} = 0. \quad (28.29)$$

At the full system level:

$$\Delta P_{source} + \Delta P_{probe} + \Delta P_{apparatus} + \Delta P_{environment} = 0. \quad (28.30)$$

Classical correspondence

For strongly decohered states with negligible quantum fluctuations:

$$\mathscr{H}_{QC} \rightarrow \text{semiclassical or classical gravitational evolution.} \quad (28.31)$$

Coherence sensitivity

For coherent source branches:

$$\mathscr{H}_{QC} \text{ must not reduce prematurely to a classical convex mixture.} \quad (28.32)$$

Basis consistency

Conditional predictions may depend on the measurement basis, but unconditioned observable predictions must remain compatible across complete measurement descriptions.

No-signalling consistency

Postselection-conditioned anomalies must not permit controllable superluminal communication without access to the corresponding selection record.

Boundary consistency

Different overlapping local descriptions must agree on shared physical predictions.

These requirements convert the handover problem from an unrestricted metaphor into a constrained theoretical program.

28.8 The missing physical variable may not be a single scalar trace

Earlier SMFT discussions sometimes spoke as though collapse produced a scalar trace quantity that later accumulated into curvature.

The present synthesis suggests that this may be too simple.

The handover input may require a structured object:

$\mathcal{T}$ **= {ρ, phase relations, branch correlations, probabilities, conserved currents, boundary algebra, record structure}.** (28.33)

Its downstream geometric image may likewise require more than one scalar:

$\mathcal{G}$ **= {⟨T_ab⟩, noise kernel, entropy current, boundary variation, causal response}.** (28.34)

A single number cannot generally encode:

- direction;
- tensor structure;
- nonlocal correlation;
- quantum fluctuation;
- causal support;
- boundary dependence.

Thus, a future SMFT handover theory may require:

Trace tensor + memory kernel + closure operator, (28.35)

rather than one undifferentiated “collapse residue.”

This is a substantial theoretical correction.

28.9 A possible trace tensor

One provisional object might be:

$\mathcal{T}$**_ab(x) = ⟨T_ab(x)⟩_record − ⟨T_ab(x)⟩_ref.** (28.36)

This is a reference-relative stress-energy trace.

A more general nonlocal object might be:

$\mathcal{K}$**_abcd(x,y) = ⟨ΔT_ab(x)ΔT_cd(y)⟩.** (28.37)

This resembles a stress-energy correlation or noise kernel.

The first object records mean deviation.

The second records fluctuations and correlations.

A semiclassical equation based only on ⟨T_ab⟩ may be insufficient when quantum fluctuations are large.

A refined lifecycle may therefore require:

Mean trace → semiclassical curvature. (28.38)

Trace fluctuations → stochastic or quantum geometric corrections. (28.39)

This suggests that the curvature-memory regime may itself have layers.

28.10 Closure as controlled loss of phase access

A useful provisional interpretation is:

> $\mathscr{H}$_QC does not necessarily destroy all quantum information. It changes which relations remain operationally accessible at the chosen scale.

The map may therefore be understood as:

Full relational state → restricted observable algebra → effective record → geometric description. (28.40)

The loss is partly a loss of access.

This connects naturally with the Dorau–Much use of local algebras and horizons: the relevant relative entropy is defined relative to a restricted algebra of observables associated with a region or horizon.

Thus, geometry may emerge not from absolute information erasure but from stable informational structure under restricted access.

This possibility is highly compatible with SMFT's observer-relative projection language, though it remains an interpretation rather than a derived result.

---

29. Signed Conditional Traces and Positive Information Ledgers

One of the most delicate conceptual issues is the change of sign behaviour across the lifecycle.

The upstream experiment contains two positive branch momentum transfers:

$\delta_A > 0$. (29.1)

$\delta_B > 0$. (29.2)

The postselected conditional response may satisfy:

$\delta_f < 0$. (29.3)

The downstream relative entropy satisfies:

$S_{rel} \geq 0$. (29.4)

These facts are compatible because δ_f and S_{rel} are different kinds of quantity.

29.1 Three mathematical roles

The branch transfers δ_A and δ_B are dynamical displacement parameters.

The conditional shift δ_f is a signed expectation associated with a selected state.

The relative entropy S_rel is a nonnegative distinguishability measure.

Thus:

Quantity	Role	Sign structure
δ_A, δ_B	Branch momentum transfer	Positive in the proposed geometry
δ_f	Conditioned effective shift	May be negative
p_f	Postselection probability	Nonnegative
S_rel	Reference-relative distinguishability	Nonnegative
δA	Oriented or convention-dependent area variation	Depends on setup and sign convention
G_ab	Tensorial curvature response	Not reducible to one scalar sign

The lifecycle cannot be represented by passing one scalar sign unchanged from one stage to the next.

29.2 Why the conditional sign can be anomalous

The conditional mean contains interference terms:

$$\langle \mathbf{p} \rangle_f = |N_f|^2 [D_A + D_B + I_{AB}]. \quad (29.5)$$

Where:

$$D_A = |c_A|^2 \langle \psi_A | \hat{p} | \psi_A \rangle. \quad (29.6)$$

$$D_B = |c_B|^2 \langle \psi_B | \hat{p} | \psi_B \rangle. \quad (29.7)$$

$$I_{AB} = 2\mathrm{Re}[c_A^* c_B \langle \psi_A | \hat{p} | \psi_B \rangle]. \quad (29.8)$$

D_A and D_B are positive in the proposed arrangement.

I_AB may be negative.

If:

$$|I_{AB}| > D_A + D_B, \quad (29.9)$$

then:

$$\langle \mathbf{p} \rangle_f < 0. \quad (29.10)$$

The sign reversal therefore belongs to the interference geometry of the conditioned state.

29.3 Why relative entropy remains nonnegative

For ordinary density operators:

$S(\rho\|\sigma) \geq 0$, (29.11)

with equality when $\rho = \sigma$ under suitable conditions.

Relative entropy measures distinguishability, not direction of force.

A state can differ strongly from a reference whether its momentum points left or right.

For example:

$S(\rho_+\|\rho_0) \geq 0$. (29.12)

$S(\rho_-\|\rho_0) \geq 0$. (29.13)

The sign of momentum does not become the sign of relative entropy.

Therefore:

Signed response → magnitude of distinguishability, not signed entropy. (29.14)

The directional information may remain encoded elsewhere in the state or stress-energy tensor.

29.4 Positive ledger does not erase directional information

A nonnegative information measure does not imply that all directional structure has disappeared.

Relative entropy can quantify how different two states are while other observables specify the direction and tensor structure of that difference.

A more complete downstream record may include:

$\mathcal{L}_{info} = \{S_{rel}, \langle p^a \rangle, \langle T_{ab} \rangle, \text{correlation functions}\}$. (29.15)

Here:

- S_{rel} measures distinguishability;
- $\langle p^a \rangle$ carries directional momentum;
- $\langle T_{ab} \rangle$ carries energy, momentum flux, pressure, and stress;
- correlation functions carry fluctuations and nonlocal structure.

Thus:

Nonnegative distinguishability + signed tensor observables = complete informational-dynamical ledger. (29.16)

The handover need not flatten all directional information into one positive scalar.

29.5 Why negative conditional momentum does not imply negative area

The area response in the Dorau–Much setting is tied to horizon flux and null focusing.

The postselected momentum in the repulsive-gravity proposal concerns a probe's conditioned displacement in a laboratory configuration.

No direct equation identifies them.

Therefore:

$\delta_f < 0 \not\Rightarrow \delta A < 0$. (29.17)

Even if both quantities could be assigned signs, they would depend on:

- coordinate orientation;
- horizon generator direction;
- chosen null parameter;
- stress-energy distribution;
- boundary conventions.

The proposed repulsion is a witness of coherent upstream structure.

It is not a prediction of defocusing in the Dorau–Much horizon geometry.

29.6 Why negative conditional momentum does not imply negative energy

A probe wavepacket with negative mean momentum can still have positive kinetic energy:

$E_{kin} = \langle p^2 \rangle / 2m \geq 0$. (29.18)

Direction reversal is not energy negativity.

Likewise, an anomalous weak value does not imply that the underlying Hamiltonian has acquired an unbounded negative spectrum.

Thus:

Momentum sign ≠ energy sign. (29.19)

Weak-value anomaly ≠ negative-energy substance. (29.20)

This prevents the repulsive effect from being confused with exotic matter or violation of energy positivity.

29.7 The sign-transition principle

The lifecycle suggests a general principle:

> **Signs are representation-specific. A sign attached to a conditioned observable need not survive as the sign of an information measure or geometric response.**

Symbolically:

Sign[conditional observable] ≠ Sign[information measure] ≠ Sign[curvature invariant]. (29.21)

This principle is important for SMFT because its metaphorical vocabulary sometimes allows positive and negative “semantic curvature” or “trace” to be transferred too freely across domains.

The refined framework requires every sign to be tied to a defined quantity.

29.8 Conditional anomaly as coherence witness, not geometry source

The safest interpretation is:

$\delta_f < 0$ → evidence of phase-sensitive relational structure under the model assumptions. (29.22)

Not:

$\delta_f < 0$ → negative gravitational curvature. (29.23)

The anomaly diagnoses the upstream state.

The downstream geometric source remains the complete effective energy-momentum account.

This distinction should remain explicit throughout the eventual empirical discussion.

---

30. From Phase Information to Geometric Information

The central handover can now be expressed as a transformation among three information formats:

$I_{phase} \rightarrow I_{record} \rightarrow I_{geo}$. (30.1)

Where:

I_{phase} = phase-usable relational information. (30.2)

I_{record} = durable probability-weighted record information. (30.3)

I_{geo} = information encoded as shared geometric constraint. (30.4)

This is not merely a poetic sequence.

Each format supports different operations.

30.1 Phase information supports interference

The phase format contains:

$I_{phase} = \{c_j, \varphi_j, \text{overlaps, cross terms, entanglement structure}\}$. (30.5)

It supports:

- constructive and destructive interference;

- basis-dependent conditional states;
- weak-value anomalies;
- coherent reversal of expected response;
- relational quantum control.

Its characteristic mathematical form is complex and nonconvex at the amplitude level.

30.2 Record information supports historical accounting

The record format contains:

$I_{record} = \{\{p_f, \rho_f, R_f\}\}_f$. (30.6)

It supports:

- outcome frequencies;
- conditional statistics;
- reproducible records;
- classical communication;
- conservation accounting;
- causal comparison across experiments.

Its characteristic mathematical form is probability-weighted and normalized.

30.3 Geometric information supports shared constraints

The geometry format contains:

$I_{geo} = \{\langle T_{ab} \rangle, \delta A, g_{ab}, \text{causal structure}\}$. (30.7)

It supports:

- geodesic motion;
- redshift;
- focusing;
- causal boundaries;
- proper-time structure;
- spacetime-wide consistency.

Its characteristic mathematical form is tensorial and coordinate-covariant.

30.4 The conversion is lossy but not consequence-free

The map from I_phase to I_record generally loses operational access to some relative-phase information.

The map from I_record to I_geo compresses many microscopic details into effective stress-energy and curvature.

Thus:

Dim[I_phase] ≥ Dim[I_record] ≥ Dim[I_geo] (30.8)

may hold schematically under coarse-graining, though no universal dimension inequality is implied.

The important idea is:

Microscopic detail decreases while macroscopic consequence persists. (30.9)

This is the structure of historical compression.

30.5 Coarse-graining as trace metabolism

SMFT may describe the conversion as trace metabolism.

A field contains many detailed possibilities.

Closure removes or hides some distinctions while preserving others that matter for future dynamics.

Schematically:

Fine-grained relation → coarse-grained invariant. (30.10)

The invariant is not a complete copy of the original state.

It is the part that remains relevant to the downstream level.

For geometry, the retained variables may include:

- total energy;
- momentum flux;
- stress;
- conserved currents;
- causal boundary response.

The discarded variables may include inaccessible microscopic phases.

Thus:

Geometric memory is selective memory. (30.11)

It remembers what the effective spacetime dynamics needs.

30.6 Relative entropy as a candidate compression invariant

Relative entropy is attractive in this role because it behaves monotonically under quantum channels:

$S(\rho\|\sigma) \geq S(\Phi(\rho)\|\Phi(\sigma))$. (30.12)

Here Φ is a suitable completely positive trace-preserving map.

This data-processing inequality states that coarse-graining cannot increase distinguishability.

Conceptually:

Fine-grained distinguishability ≥ coarse-grained distinguishability. (30.13)

This is highly relevant to the handover problem.

A detailed quantum state may contain more distinguishability than an observer restricted to a local algebra or coarse-grained record can access.

The remaining distinguishability may nevertheless be enough to support effective energy and geometry accounting.

This suggests:

Curvature-relevant information may be the distinguishability that survives physically appropriate coarse-graining. (30.14)

This is a promising SMFT research hypothesis, not a conclusion established by the attached papers.

30.7 The surviving-information hypothesis

A more formal provisional statement is:

> **The effective geometric source is determined not by all microscopic quantum information, but by the portion of physical state difference that survives the relevant closure and coarse-graining maps as conserved, boundary-relevant distinguishability.**

Symbolically:

$I_{geo\text{-}source} = Inv[\mathcal{C}_I \circ \mathcal{C}_L(I_{phase})]$. (30.15)

Where Inv denotes the structure invariant or sufficiently stable under the chosen closure process.

Possible candidates include:

- conserved stress-energy;
- modular energy;
- relative entropy;
- Noether charge;
- boundary flux;
- correlation kernels.

The Dorau–Much paper's result that relative entropy can be interpreted as an energy flux and, in related settings, as a Noether charge makes this possibility especially suggestive.

30.8 From branch relation to modular relation

The upstream experiment organizes differences through source branches:

A relative to B. (30.16)

The downstream horizon framework organizes differences through states on an algebra:

ω_ϕ relative to ωo. (30.17)

These are not the same comparison.

But both are relational.

A future handover theory may need to explain how branch-indexed quantum information becomes algebra-indexed state distinguishability.

One possible sequence is:

Branch relation → decohered record algebra → reference-state comparison → modular flow → geometric flux. (30.18)

Again, this is a research sequence, not an established derivation.

30.9 The role of the boundary

Information becomes geometry only relative to a boundary or localization structure.

In the postselection experiment, the boundary is projective:

selected source sector | rejected source sectors. (30.19)

In the horizon framework, the boundary is causal:

accessible algebra | causally inaccessible region. (30.20)

In ledger closure, the boundary is operational:

recorded degrees of freedom | environment traced out. (30.21)

Thus, the handover is always boundary-dependent.

This leads to a Boundary Conversion Principle:

> **A physical difference becomes a geometric ledger entry only after a boundary specifies what is inside, what crosses, what is accessible, and what is retained.**

In compact form:

Difference + boundary + closure rule → ledger. (30.22)

30.10 Geometry as public information

A quantum phase may be accessible only through a carefully controlled recombination experiment.

A classical geometry affects every suitable probe entering the region.

The transition from phase information to geometry can therefore be visualized as a change in publicity:

Private relational phase → shared physical record → public path constraint. (30.23)

“Public” here means operationally common to systems moving through the effective geometry.

It does not mean consciously known.

This gives a refined interpretation of curvature:

> Curvature is physical information that has become sufficiently closed and shared to constrain generic future trajectories.

30.11 The handover equation in expanded form

The complete conceptual handover may be written:

$\rho_{SP} \to \{\{p_f, \rho_f\}\}_f \to \rho_{record} \to S_{rel}(\rho_{record} \| \rho_{ref}) \to \langle T_{ab} \rangle_{eff} \to \delta A \to g_{ab}^{eff}$. (30.24)

Each arrow requires a distinct theory.

Arrow 1

Measurement and conditionalization.

Arrow 2

Decoherence and record formation.

Arrow 3

Reference-state and observable-algebra selection.

Arrow 4

Information–energy relation.

Arrow 5

Energy–boundary relation.

Arrow 6

Boundary–curvature consistency.

The repulsive-gravity proposal explicitly develops the first arrow in a particular model.

The Dorau–Much paper develops later arrows in a local-horizon QFT setting.

SMFT currently provides a role grammar spanning all arrows, but not yet their complete mathematics.

30.12 The third major fine-tuning of SMFT

The first fine-tuning was:

Gravity is not only post-collapse residue. (30.25)

The second was:

Projection is not only branch deletion. (30.26)

The third is:

Trace is not one unchanged object transported from quantum amplitude to classical curvature. (30.27)

Instead:

Trace changes representational type across closure. (30.28)

The lifecycle contains:

- coherent relational trace;
- conditional trace;
- statistical ledger trace;
- informational residue;
- geometric trace.

A future SMFT theory must define the conversion rules among them.

Part VIII — Comparison with the Current SMFT Research Program

31. Why This Is More Than a Renaming Exercise

A recurring criticism of broad theories is that they rename familiar concepts without producing new constraints.

Under an unconstrained translation, one could say:

- superposition is potential;
- measurement is collapse;
- outcome is trace;

- gravity is curvature;
- therefore SMFT explains the experiment.

Such a mapping would be rhetorically smooth but scientifically weak.

The present reconstruction attempts to do more.

It identifies specific places where the old SMFT language was inadequate and introduces distinctions required by the physics.

The main additions are:

1. coherent gravitational possibility geometry;
2. gravitational relation formation before projection;
3. constructive rather than purely reductive projection;
4. conditional trace versus complete ledger;
5. phase information versus record information;
6. signed conditional response versus nonnegative distinguishability;
7. multiple closure operators;
8. a missing quantum-to-curvature handover map;
9. dual persistence before and after closure;
10. explicit failure conditions.

These additions constrain how SMFT may interpret the source papers.

31.1 A renaming framework would not detect the missing upstream sector

The older formulation said:

Collapse → trace → curvature. (31.1)

The repulsive-gravity proposal forces the insertion of:

Gravitational relation → entanglement → projection. (31.2)

This is not a cosmetic change.

Without the coherent relation, the predicted anomaly cannot arise.

Therefore, SMFT was required to modify its own causal ordering.

A framework willing to change its internal model in response to external theory is doing more than applying fixed labels.

31.2 A renaming framework would conflate all traces

The refined analysis distinguishes:

Conditional trace. (31.3)

Complete statistical ledger. (31.4)

Informational distinguishability residue. (31.5)

Geometric curvature trace. (31.6)

These are not interchangeable.

A universal metaphor machine would call all four “memory” and move on.

The present model instead identifies missing maps among them.

That creates research obligations.

31.3 A renaming framework would ignore probability cost

The anomalous conditional shift can be amplified by near-orthogonal postselection, but the success probability falls.

A purely rhetorical interpretation might celebrate the large repulsion while ignoring the rare-event weight.

The Ledger Completeness Principle forbids this.

It requires:

Anomaly magnitude + admission probability + complementary outcomes + apparatus accounting. (31.7)

This restriction is physically substantive.

31.4 A renaming framework would equate weak value with weak interaction

The refined model explicitly rejects:

Weak value = weak force. (31.8)

It retains only a structural analogy between postselection and transition gating.

This is another sign of selectivity.

Not every shared word is treated as a shared mechanism.

31.5 A renaming framework would falsely connect the two physics papers

The two attached papers can be placed in one conceptual lifecycle, but they do not form one established derivation.

The refined framework states:

Structural complementarity ≠ mathematical continuity. (31.9)

This prevents the article from claiming that postselected momentum directly becomes horizon relative entropy or Einstein curvature.

The gap is preserved as an open problem.

31.6 Explanatory compression with retained residuals

A useful unifying framework should compress without erasing unresolved structure.

The new SMFT account compresses the two physical frameworks into:

Coherent relation → conditioned trace → ledger → distinguishability → curvature. (31.10)

But it explicitly retains the residual question:

How are the middle arrows physically derived? (31.11)

This is a healthier form of compression.

It organizes the research landscape while preserving uncertainty.

31.7 SMFT as an interface theory

The comparative article described SMFT as potentially becoming a theory of the missing interface connecting quantum information, observer selection, memory, and classical geometry.

The present analysis makes that possibility more precise.

The interface is:

$\mathscr{H}$_QC : phase-bearing gravitational relation → curvature-bearing durable record. (31.12)

An interface theory should specify:

- allowed inputs;
- allowed outputs;
- invariants;
- losses;
- conserved quantities;
- failure regimes;
- matching conditions.

SMFT has now identified these categories more clearly, though it has not yet supplied their full mathematical implementation.

31.8 Criteria for genuine progress

The refined research program should be judged by whether it can eventually produce at least one of the following:

1. a mathematically specified trace-closure map;
2. a derived relation between coherence loss and effective gravitational response;
3. a new constraint on gravitationally induced entanglement experiments;
4. a quantitative distinction between classical mediator, quantum mediator, and residual-memory models;
5. a prediction involving projection timing, environmental closure, or boundary dependence;
6. a derivation of a known gravitational limit from SMFT variables;
7. a clear observation that would falsify the refined model.

Without such progress, the lifecycle remains a sophisticated interpretation.

With it, SMFT could move toward a testable interface theory.

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32. Relationship to Three Emerging Quantum-Gravity Models

The comparative SMFT study argued that the framework maps most naturally onto approaches in which gravity is:

- residual;
- memory-like;
- emergent;
- geometric;
- information-bearing;
- connected to coarse-graining;
- connected to retained physical traces.

It also emphasized that repeated compatibility is not yet proof.

The present refinement changes how those earlier comparisons should be understood.

The previous comparisons concentrated mainly on gravity's downstream memory face.

The new repulsive-gravity proposal adds an upstream coherence requirement.

An adequate SMFT synthesis must now ask of every gravity model:

1. What carries coherent relational alternatives?

2. What creates source–probe or matter–geometry entanglement?
3. What defines projection or closure?
4. What survives as durable information?
5. What becomes effective geometry?
6. What is lost or coarse-grained during the handover?

32.1 Memory-based quantum gravity

Models that describe spacetime as retaining quantum imprints fit naturally with SMFT's curvature-memory regime.

Their central chain is:

Quantum event → retained imprint → geometric effect. (32.1)

The new refinement asks an additional question:

> Before an imprint becomes durable, does the model preserve coherent alternatives and allow them to interfere?

A model of memory that stores only classical outcomes may explain the downstream regime but fail to describe the upstream witness.

Thus:

Quantum memory must include either coherence memory or a mechanism that generates equivalent nonclassical mediation. (32.2)

32.2 Entropic or information-mediated gravity

Entropic frameworks often emphasize gradients of information, free energy, or state counting.

SMFT previously interpreted these through trace accumulation and residual curvature.

The new lifecycle requires such models to distinguish:

Signed conditioned response from **nonnegative entropy measure.** (32.3)

It also requires them to explain how phase-sensitive interference survives long enough to produce the postselected anomaly.

A purely classical thermodynamic mediator may be insufficient.

An information-based gravity model must specify whether its information is:

- classical probabilistic information;
- quantum coherence;
- entanglement;
- relative entropy;

- boundary information;
- another resource.

The word *information* alone is not enough.

32.3 UV-to-IR emergent gravity

Models in which high-energy or microscopic dynamics flow toward Einstein gravity naturally fit the lifecycle's transition from detailed relation to coarse-grained geometry.

Their central problem is a matching problem:

Microscopic degrees of freedom → effective stress-energy and metric. (32.4)

SMFT interprets this as trace and closure.

The new refinement adds that the matching must preserve coherent quantum predictions upstream while producing classical curvature downstream.

Thus, a complete UV-to-IR handover must satisfy:

Quantum coherence consistency + semiclassical correspondence. (32.5)

It cannot derive general relativity by prematurely replacing coherent alternatives with a classical mixture.

32.4 Ordinary graviton exchange models

The comparative article suggested that SMFT fits less automatically with a model in which gravity is simply another exchange force with no memory or residual structure.

The present analysis allows a more nuanced judgment.

A quantized spin-2 mediator can naturally provide the upstream entangling channel.

It may therefore fit the coherent-relation stage well.

The remaining question is whether its low-energy collective behaviour also explains:

- geometric universality;
- boundary information;
- effective curvature;
- historical persistence;
- semiclassical closure.

Thus, exchange and memory descriptions may apply at different scales.

A graviton description need not contradict the memory interpretation if:

Microscopic exchange → coarse-grained geometric persistence. (32.6)

The relevant issue is whether the handover can be derived.

32.5 Selective compatibility

The refined SMFT framework is not equally compatible with every possible theory.

It fits best when a model contains both:

Upstream nonclassical relational mediation. (32.7)

and

Downstream geometric or informational persistence. (32.8)

It fits poorly if a model contains neither coherent mediation nor residual geometry.

It also becomes unnecessary if a complete established theory already derives the full handover without any concept structurally resembling:

- trace;
- closure;
- retained distinguishability;
- memory;
- boundary ledger.

This selectivity is essential to preventing unlimited explanatory flexibility.

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33. What the New Paper Corrects in SMFT

The repulsive-gravity proposal does not merely add one more example to SMFT.

It corrects several earlier formulations.

33.1 Correction I — Gravity is not only after collapse

Earlier:

Gravity = post-collapse residue. (33.1)

Refined:

Classical gravity = post-closure geometric residue. (33.2)

And:

Quantum gravitational relation exists before final closure. (33.3)

33.2 Correction II — Gravity is not wholly passive

Earlier:

Gravity is passive memory rather than active interaction. (33.4)

Refined:

Gravity is active in forming quantum relations and passive-structural in preserving closed geometry. (33.5)

The words active and passive are now stage-relative.

33.3 Correction III — The gate does not create all difference

Earlier:

Weak gate → admitted physical difference. (33.6)

Refined:

Gravity forms branch-sensitive difference; the gate converts it into an admitted trace. (33.7)

33.4 Correction IV — Projection is not only selection

Earlier:

Projection chooses one possibility. (33.8)

Refined:

Projection excludes outcomes and constructs a conditional state through amplitude recombination. (33.9)

33.5 Correction V — Trace is not one object

Earlier:

Collapse creates trace; trace becomes curvature. (33.10)

Refined:

Coherent trace → conditional trace → complete ledger → distinguishability residue → geometric trace. (33.11)

Each transition requires a defined map.

33.6 Correction VI — Repulsion is not a new gravitational branch

Earlier speculative language might tempt one to interpret the anomaly as a real negative gravitational force.

Refined:

Repulsion = postselection-conditioned interference trace. (33.12)

The branch interactions remain attractive.

33.7 Correction VII — Memory has two regimes

Earlier:

Memory = retained past outcome. (33.13)

Refined:

Coherence memory = retained relation among unresolved alternatives. (33.14)

Curvature memory = retained consequence of settled physical history. (33.15)

33.8 Correction VIII — Observer must be operationally specified

Earlier Ô-language could be read too broadly.

Refined:

Ô = specified preparation, basis, boundary, selection, and record architecture. (33.16)

Consciousness is not assumed.

33.9 Correction IX — The handover is an open problem

Earlier SMFT writing sometimes moved too quickly from collapse trace to gravity.

Refined:

$\rho_f \rightarrow S_{rel} \rightarrow \langle T_{ab} \rangle \rightarrow \delta A \rightarrow g_{ab}$ (33.17)

is not yet derived as one physical chain.

The missing arrows are now explicit.

33.10 Correction X — Gravity's refined definition

The resulting definition is:

> **Gravity is the persistence of physically consequential relational difference across coherent, recorded, and geometric closure regimes.**

In compact form:

Gravity = relational persistence across staged closure. (33.18)

This is the mature correction produced by the new paper.

The following part will turn these conceptual refinements into testability requirements, failure conditions, and explicit scientific limits.

Part IX — Testability, Failure Conditions, and Scientific Limits

34. Experimental Signatures of the Coherent Regime

The refined SMFT interpretation becomes scientifically meaningful only if its proposed stages correspond to experimentally distinguishable behaviour.

The central upstream claim is:

> The postselected repulsive trace requires a phase-bearing gravitational relation rather than a classical mixture of branchwise attractive forces.

The source paper makes a closely related physical claim. It proposes that the negative conditioned momentum arises through interference between two distinct gravitational alternatives and therefore cannot be produced by a classical gravitational field represented only as a mixture of the two source configurations. Its preliminary feasibility discussion is encouraging, but the authors explicitly state that a more extensive study is required, including separation of the predicted signal from competing effects such as the Casimir–Polder interaction.

The experiment should therefore be understood not as one measurement of a negative displacement, but as a structured family of tests.

A convincing result would need to reveal the complete signature:

Coherence dependence + phase dependence + projection dependence + gravitational scaling + ledger consistency. (34.1)

No single component is sufficient by itself.

34.1 Source-coherence dependence

The proposed anomaly requires off-diagonal coherence between the source branches.

Let the source density matrix be:

$\rho_S = |\alpha|^2|A\rangle\langle A| + |\beta|^2|B\rangle\langle B| + \gamma\alpha\beta|A\rangle\langle B| + \gamma\alpha^*\beta|B\rangle\langle A|.$** (34.2)

The parameter γ represents residual coherence:

0 ≤ |γ| ≤ 1. (34.3)

Where:

|γ| = 1 represents ideal coherence. (34.4)

|γ| = 0 represents complete dephasing in the A/B basis. (34.5)

The interference contribution to the conditioned probe distribution should scale with γ.

Schematically:

$$\mathbf{I_{AB} \propto Re[\gamma c_A^* c_B \langle \psi_A | \hat{p} | \psi_B \rangle].} \quad (34.6)$$

As source coherence is deliberately reduced:

$|\gamma| \downarrow \rightarrow$ anomalous interference term $\downarrow$. (34.7)

In the fully incoherent limit:

$\gamma \rightarrow 0 \rightarrow$ conditioned response approaches classical probability reweighting. (34.8)

A robust experimental test should therefore compare:

- high-coherence source preparation;
- controlled partial dephasing;
- complete which-path marking;
- an equivalent classical mixture.

The refined SMFT prediction is not merely that the repulsion becomes smaller under noise.

It is:

> The out-of-hull component should track the physically usable branch coherence.

34.2 Postselection-phase dependence

The conditioned state depends on the postselection phase.

Let the selected source state be:

$$|f(\theta,\chi)\rangle = \cos\theta|A\rangle + e^{i\chi}\sin\theta|B\rangle.$$ (34.9)

Then the unnormalized probe state is:

$$|\tilde{\psi}_f\rangle = \alpha\cos\theta|\psi_A\rangle + \beta e^{-i\chi}\sin\theta|\psi_B\rangle.$$ (34.10)

The momentum expectation becomes:

$$\langle p\rangle_f = F(\theta, \chi, \alpha, \beta, \delta_A, \delta_B, \langle\psi_A|\psi_B\rangle).$$ (34.11)

Changing χ should rotate the relative phase between the two probe amplitudes.

This should move the conditioned response through a structured pattern:

- constructive interference;
- reduced displacement;
- destructive interference;
- possible sign reversal.

A genuine coherence-generated effect should therefore be phase-tunable.

The decisive result would not be:

One selected setting gives $\langle p\rangle_f < 0$. (34.12)

It would be:

The full measured function ⟨p⟩_f(χ) follows the predicted interference curve. (34.13)

This would strongly reduce the possibility that a fixed background force produced the anomaly.

34.3 Branch-separation dependence

The two branch momentum transfers are:

δ_A = GMmT / x_A². (34.14)

δ_B = GMmT / x_B². (34.15)

Their difference is:

Δδ = δ_A − δ_B = GMmT(1/x_A² − 1/x_B²). (34.16)

If the branch distances become equal:

x_A → x_B → Δδ → 0. (34.17)

Then:

|ψ_A⟩ → |ψ_B⟩. (34.18)

The source and probe cease to acquire distinguishable branch-dependent dynamics, and the mechanism for the anomalous shift disappears.

Thus:

Out-of-hull anomaly requires Δδ ≠ 0. (34.19)

A convincing test should vary the source geometry and verify the predicted dependence on:

- x_A;
- x_B;
- source–probe distance difference;
- interaction time T;
- source mass M;
- probe mass m.

The source paper explicitly builds the branch shifts from these Newtonian parameters and gives a preliminary parameter estimate for a feasible regime.

34.4 Interaction-time dependence

For fixed geometry:

δ_j ∝ T. (34.20)

In the weak-interaction approximation:

$\delta_{eff} \approx \delta_B - [\alpha/(\beta - \alpha)](\delta_A - \delta_B)$. (34.21)

Therefore, for sufficiently short interaction times within the approximation regime:

$\delta_{eff} \propto T$. (34.22)

A background offset unrelated to the gravitational interaction may not follow this dependence.

An experimental scan of T should test:

- initial linear scaling;
- breakdown of the weak-shift approximation;
- decoherence growth at longer times;
- wavepacket spreading;
- loss of interferometric visibility.

The relevant signature is not monotonic growth without limit.

There should be a finite operational window:

Signal accumulation ↑ with T, while coherence and localization quality may ↓ with T. (34.23)

The optimum occurs where gravitational distinguishability is large enough to measure but coherence remains usable.

34.5 Probe-overlap dependence

The two branch-conditioned probe states must remain sufficiently overlapping to interfere.

Let:

$V_P = |\langle \psi_A | \psi_B \rangle|$. (34.24)

Then:

$V_P \approx 1$ means the two gravitational interactions are almost indistinguishable. (34.25)

$V_P \approx 0$ means the probe branches are effectively orthogonal. (34.26)

The anomaly requires an intermediate regime:

$0 < V_P < 1$. (34.27)

If V_P is too close to one, the signal is too small.

If V_P is too close to zero, the cross terms become ineffective.

This produces a characteristic coherence–distinguishability trade-off:

Useful witness ∝ branch difference × branch overlap. (34.28)

Not as an exact universal equation, but as an experimental design principle.

34.6 Amplification–probability trade-off

The weak-value regime permits:

$$|\delta_{eff}| \gg \max(\delta_A, \delta_B). \quad (34.29)$$

But near-orthogonal postselection gives:

$$p_f = |\langle \Psi_f | \Psi_i \rangle|^2 \ll 1. \quad (34.30)$$

The source paper's example explicitly illustrates a large anomalous shift accompanied by a small successful postselection probability.

The refined SMFT Ledger Completeness Principle therefore predicts a joint signature:

Conditional anomaly magnitude ↑ while admission weight ↓. (34.31)

A reported large negative displacement without the predicted reduction in successful count rate would require careful investigation.

The relevant empirical object is not only:

$$\delta_{eff}. \quad (34.32)$$

It is:

$$\mathcal{W}_f = \{p_f, \delta_{eff}, \mathrm{Var}_f(p), \text{visibility}, \text{rejected-sector statistics}\}. \quad (34.33)$$

The full signature should be compared with quantum theory.

34.7 Unconditioned-ensemble consistency

Let the complete measurement basis contain outcomes f.

The unconditioned probe state is:

$$\rho_{P'} = \Sigma_f p_f \rho_f. \quad (34.34)$$

The total mean transfer is:

$$\langle \Delta p \rangle_{all} = \Sigma_f p_f \langle \Delta p \rangle_f. \quad (34.35)$$

A negative conditioned value in one sector should be balanced by the remaining sectors and by the apparatus/source momentum accounting.

Thus:

Conditional repulsion does not require unconditional repulsion. (34.36)

The experiment should measure or reconstruct enough of the complementary sectors to verify this.

A failure of complete normalization or conservation would not support the SMFT interpretation.

It would more likely indicate:

- incomplete calibration;

- unmodelled forces;
- detector bias;
- incorrect postselection accounting.

34.8 Classical-mixture control

A crucial control is to destroy the A/B coherence while preserving the same branch populations:

$$\rho_{mix} = |\alpha|^2|A\rangle\langle A| + |\beta|^2|B\rangle\langle B|. \quad (34.37)$$

The source should then produce the same ordinary branch-force statistics but no coherent cross terms.

The refined model predicts:

Coherent preparation → possible out-of-hull conditioned response. (34.38)

Matched incoherent mixture → response restricted to classical conditional range under the specified model. (34.39)

This direct comparison is more informative than merely turning the entire interaction off.

It isolates coherence as the tested resource.

34.9 Competing-force discrimination

The predicted gravitational signal is extremely small.

Electromagnetic, patch-potential, Casimir–Polder, trapping, vibration, thermal, and technical effects may produce larger forces or apparent momentum biases.

The source paper itself highlights the need to distinguish the proposed effect from Casimir–Polder interactions and presents its parameter calculation as preliminary rather than a completed experimental design.

A credible experiment should therefore exploit dependencies specific to the quantum protocol:

- reversal under postselection phase change;
- disappearance under source dephasing;
- scaling with source mass;
- scaling with interaction time;
- dependence on the difference between x_A and x_B;
- correlation with the selected interferometer port;
- consistency across complementary outputs.

A static parasitic force may reproduce a displacement.

It is much harder for it to reproduce the entire multidimensional signature.

34.10 The experimentally relevant SMFT claim

The strongest testable SMFT-compatible statement is:

> **The observed gravitational trace depends jointly on field evolution, retained phase relation, and projection protocol.**

In formula form:

$\mathrm{Trace}_f = F[\mathcal{U}_G, \rho_{coh}, \hat{P}_f, \mathscr{E}, \mathcal{A}]$. (34.40)

This is more constrained than saying that "the observer changes gravity."

The experimenter controls some components of the protocol.

The resulting statistics remain fixed by the physical state and dynamics.

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35. Failure Conditions for the Upstream SMFT Reading

The upstream interpretation claims that gravity participates in a coherence-bearing relational channel before final ledger closure.

This claim would be weakened under several conditions.

35.1 Classical-mixture reproduction

The strongest upstream failure would occur if a fully classical gravitational model reproduced the complete measured signature, including:

- the negative conditioned shift;
- its phase dependence;
- its coherence dependence;
- its probability trade-off;
- the complementary-outcome statistics.

The relevant criterion is not whether any classical postselection can generate an unusual conditional average in some broadly defined model.

It is whether a physically specified classical gravitational mediator, with no usable A/B coherence, can reproduce the experiment under the same controls.

If:

$\mathrm{Predictions}_{classical} = \mathrm{Predictions}_{quantum}$ for all accessible protocol settings, (35.1)

then the proposed witness would fail to discriminate the mediator's quantumness.

The narrow SMFT inference from the anomaly to pre-closure gravitational coherence would also weaken.

35.2 Survival after complete dephasing

Suppose controlled which-path marking drives:

$$\gamma \to 0. \quad (35.2)$$

If the same out-of-hull anomaly remains unchanged:

$$\delta_{eff}(\gamma = 0) = \delta_{eff}(\gamma = 1), \quad (35.3)$$

then the interference-based interpretation would be in serious difficulty.

The refined model predicts:

$$\partial \delta_{eff} / \partial |\gamma| \neq 0 \text{ in the coherence-sensitive regime.} \quad (35.4)$$

The exact dependence would be model-specific.

But complete insensitivity to coherence would contradict the proposed mechanism.

35.3 Independence from projection basis

The conditioned response is generated by a particular amplitude recombination.

If varying the projection phase and basis produces no corresponding change:

$$\partial \langle p \rangle_f / \partial \chi = 0 \text{ across settings where quantum theory predicts phase dependence,} \quad (35.5)$$

then the constructive-projection account fails.

A truly static repulsive force would be largely independent of which interferometer output is later selected.

A projection-conditioned anomaly should not be.

35.4 Absence of branch-dependent interaction

If:

$$\delta_A = \delta_B, \quad (35.6)$$

then:

$$|\psi_A\rangle = |\psi_B\rangle \text{ up to a common phase.} \quad (35.7)$$

No gravity-mediated source–probe entanglement arises from the controlled translation model.

If an unchanged anomaly persists in this limit, it cannot be attributed to the branch-difference mechanism described by the paper.

This is an especially clean null condition.

35.5 No relation between anomaly and successful postselection

probability

The weak-value amplification mechanism links anomalous magnitude and postselection overlap.

If large anomalies occur without the corresponding probability structure:

$|\delta_{eff}|$ large while p_f remains independent of $\langle\Psi_f|\Psi_i\rangle$, (35.8)

the stated weak-value model would be incomplete or wrong.

A different mechanism might still exist, but the present SMFT reading would need revision.

35.6 Anomaly in the unconditional ensemble with no complementary account

The refined model predicts that the postselected anomaly is one sector of a complete quantum ledger.

If the total ensemble shows an unexplained persistent repulsive momentum violating the full system's conservation accounting:

$\Sigma_f p_f\langle\Delta p\rangle_f + \Delta p_{source} + \Delta p_{apparatus} \neq 0$, (35.9)

then the present interpretation fails.

The anomaly cannot be protected by saying "collapse changed the sign."

Ledger completeness remains mandatory.

35.7 Force contamination reproducing the entire signature

It is possible that a competing force becomes correlated with the source path or detector output.

Therefore, phase dependence alone may not be sufficient.

A classical contamination model could potentially imitate part of the signal.

The witness becomes persuasive only if alternative mechanisms are experimentally bounded.

SMFT should not treat every projection-correlated signal as evidence for coherent gravity.

The failure condition is:

> A non-gravitational mechanism explains the full response with fewer assumptions and survives all controls.

Under that result, the experiment would not support the refined gravity lifecycle.

35.8 Theoretical failure of the mediator inference

The source paper interprets gravity-mediated entanglement as evidence that the gravitational mediator must possess nonclassical degrees of freedom.

However, the exact logical scope of mediator-witness arguments can depend on assumptions

about:

- locality;
- available hidden channels;
- interaction structure;
- classical communication;
- initial correlations.

If a consistent classical or hybrid theory reproduced the entangling or conditional statistics without containing the claimed quantum gravitational structure, the inference would require qualification.

The refined SMFT model should inherit the same logical caution.

35.9 Upstream falsification summary

The coherent-gravity interpretation would be strongly undermined if the measured phenomenon satisfied:

Anomaly persists when coherence = 0. (35.10)

Anomaly persists when $\delta_A - \delta_B = 0$. (35.11)

Anomaly is independent of projection phase. (35.12)

Anomaly lacks the predicted postselection probability structure. (35.13)

A classical or parasitic model reproduces all controls. (35.14)

These are meaningful failure conditions.

They prevent “gravitational possibility geometry” from becoming an invisible concept compatible with every outcome.

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36. Failure Conditions for the Handover Interpretation

The upstream coherence witness and the downstream relative-entropy geometry do not automatically form one theory.

The proposed handover interpretation can fail even if both endpoint frameworks are individually successful.

36.1 No principled map from conditional traces to durable records

The first handover requirement is:

$\{\{p_f, \rho_f\}\}_f \rightarrow \rho_{record}$. (36.1)

If SMFT cannot specify:

- what constitutes a stable record;
- which degrees of freedom are retained;
- which are traced out;
- how basis dependence is resolved operationally;
- how normalization is preserved;

then the movement from conditional trace to ledger remains verbal.

The theory must eventually replace:

“The trace becomes memory.” (36.2)

with a defined physical channel.

36.2 No reference-state rule

Relative entropy requires two states and a relevant observable algebra.

If SMFT cannot specify:

- the reference state;
- the excited or recorded state;
- the accessible algebra;
- the localization boundary;

then:

$S_{rel}(\text{trace} \| \text{reference})$ (36.3)

has no unique meaning.

The Dorau–Much construction is mathematically specific: it uses a vacuumlike state, a coherent excitation, a horizon algebra, and modular theory.

SMFT cannot merely import the word *relative entropy* without supplying analogous structure in its own applications.

36.3 No distinction between information and energy

A weak handover theory may say:

> Information becomes energy.

That is too vague.

A defensible map must show how one physical deviation has both:

- an informational representation;
- a dynamical stress-energy representation.

If no relation can be derived between:

$\mathcal{D}$**_res** and ⟨**T_ab**⟩**_eff,** (36.4)

then the information-to-geometry bridge remains unsupported.

The Dorau–Much paper obtains such a relation in its particular coherent-state horizon setting, where relative entropy is expressed through energy flux.

SMFT must not universalize that result without proof.

36.4 No tensorial or causal structure

Curvature is tensorial.

A scalar trace amount is generally insufficient to determine:

- energy density;
- momentum flux;
- pressure;
- anisotropic stress;
- causal propagation;
- spacetime curvature.

If SMFT's handover variable remains only:

Trace = one scalar memory quantity, (36.5)

it cannot reproduce a general equation such as:

G_ab + Λg_ab = 8πG⟨T_ab⟩. (36.6)

The model must eventually include tensorial, directional, and causal information.

36.5 No conservation closure

Any effective stress-energy source must satisfy:

∇^a⟨**T_ab**⟩**_eff = 0.** (36.7)

If the handover generates arbitrary trace sources that violate conservation, the resulting curvature model will be inconsistent with the Bianchi identity:

∇^a**G_ab = 0.** (36.8)

Therefore:

Trace closure must imply or respect conservation closure. (36.9)

This is a hard mathematical requirement, not a metaphorical preference.

36.6 No classical correspondence limit

The handover must recover ordinary gravity when quantum coherence is negligible at the relevant scale.

A successful model should satisfy:

$\mathscr{H}$_QC[strong decoherence, small fluctuations] → semiclassical Einstein dynamics. (36.10)

And, in the appropriate further limit:

semiclassical gravity → classical general relativity. (36.11)

If SMFT cannot reproduce established gravitational behaviour, its interpretive elegance is insufficient.

36.7 No fluctuation treatment

The semiclassical Einstein equation uses an expectation value:

$G_{ab} = 8\pi G\langle T_{ab}\rangle$. (36.12)

But when stress-energy fluctuations are large, the mean may not capture the complete gravitational response.

A fuller handover may require a noise or correlation kernel:

$N_{abcd}(x,y) = \frac{1}{2}\langle\{\Delta T_{ab}(x), \Delta T_{cd}(y)\}\rangle$. (36.13)

If SMFT insists that average trace alone always determines geometry, it may fail in strongly quantum regimes.

The lifecycle should therefore leave room for:

Mean trace → mean curvature. (36.14)

Trace fluctuations → stochastic or quantum geometric structure. (36.15)

36.8 No account of boundary dependence

The downstream relative-entropy argument is boundary- and algebra-dependent.

The handover must explain why a particular boundary is physically relevant.

If any arbitrary boundary yields any desired ledger:

Difference + arbitrary boundary → arbitrary curvature, (36.16)

the framework becomes underconstrained.

A valid theory needs boundary consistency across overlapping descriptions.

36.9 No invariant content under coarse-graining

The transition from phase information to geometry is necessarily compressive.

A successful handover should identify what survives coarse-graining:

- conserved energy;
- momentum;
- charge;
- modular energy;
- relative entropy;
- Noether structure;
- causal flux.

If no stable or monotonic quantity survives:

I_phase → I_record → I_geo, (36.17)

then there is no reason to regard the downstream geometry as memory of the upstream relation.

The persistence claim would become merely rhetorical.

36.10 Handover failure summary

The SMFT handover interpretation fails as a physical theory if it cannot produce:

Specified state space. (36.18)

Specified projection and record maps. (36.19)

Specified reference and boundary. (36.20)

Conserved tensorial source. (36.21)

Controlled classical limit. (36.22)

Experimentally distinguishable consequences. (36.23)

Until these are supplied, $\mathscr{H}$_QC remains a research placeholder.

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37. Failure Conditions for Gravity as Memory

The refined model preserves gravity-as-memory, but only in a restricted and testable sense.

Memory must mean persistent physical consequence, not an unfalsifiable metaphysical label.

37.1 A complete non-memory derivation

The memory interpretation would weaken if a complete microscopic theory derived:

- quantum gravitational interaction;

- decoherence;
- semiclassical geometry;
- classical general relativity;

without any structurally equivalent role for:

- retained state difference;
- coarse-grained historical dependence;
- boundary record;
- persistent correlation;
- memory kernel;
- residual geometry.

If the entire transition were purely instantaneous and Markovian at every effective level, the memory language would add little.

37.2 No path dependence

A memory-bearing system should display some dependence on prior state or history.

A generic nonlocal memory model has the form:

$$\mathcal{G}(t) = \int_{-\infty}^{t} K(t,t')\mathcal{T}(t')dt'. \quad (37.1)$$

Where K is a memory kernel.

Classical general relativity is normally formulated locally in terms of fields satisfying differential equations, though its solutions and boundary data encode history through propagation and constraints.

SMFT should therefore be careful.

It should not claim that gravity must always exhibit explicit non-Markovian kernels.

The defensible memory claim is broader:

> The present geometry carries consequences of prior and current energy-momentum configurations through dynamically evolved field structure.

If experiments and theory showed that no history-bearing or state-retaining structure was needed even in this broad sense, the memory interpretation would lose explanatory value.

37.3 No residual regime

The old SMFT view depends on a distinction between active transition and residual geometry.

If gravity proved to be only an ordinary microscopic exchange interaction, with geometry merely a dispensable notation and no collective residual regime, then:

would become less compelling.

However, even a fundamental graviton theory could still yield an emergent geometric memory regime.

The failure condition is not the existence of gravitons.

It is the absence of any downstream geometric or persistent collective structure.

37.4 No relationship between information and geometry

If future work established that:

- relative entropy;
- entanglement structure;
- boundary information;
- state distinguishability;

play no role in the emergence or consistency of semiclassical geometry, then the SMFT information-ledger interpretation would weaken.

The Dorau–Much paper supplies one technically specific route connecting relative entropy, horizon flux, and semiclassical curvature, but it is limited to coherent excitations, local horizon structure, an entropy–area assumption, and a leading-order local approximation. The authors explicitly note the need for higher-order control and broader state generalization.

SMFT should treat this as suggestive evidence, not universal proof.

37.5 No observer- or boundary-relative structure

The refined model treats physical traces as defined relative to:

- preparation;
- accessible observables;
- boundary;
- measurement basis;
- record architecture.

If a complete gravity theory required none of these distinctions and admitted a wholly observer-independent microscopic description with no operationally relevant partitioning, the gate–ledger vocabulary might become unnecessary.

That would not falsify all SMFT applications, but it would weaken this specific gravity interpretation.

37.6 Memory without prediction is empty

A theory can call any persistent field “memory.”

That alone has no predictive content.

The concept becomes substantive only if it implies constraints such as:

- hysteresis;
- path dependence;
- delayed response;
- coarse-graining invariants;
- trace-dependent noise;
- closure-scale transitions;
- correlations between prior quantum coherence and later effective geometry.

Without such consequences:

Memory label − measurable structure = metaphor only. (37.3)

The refined SMFT research program should therefore search for one or more distinctive memory signatures.

37.7 A possible strong test

A strong future test would compare two preparations with:

- the same final mean stress-energy;
- different prior coherence histories;
- the same classical boundary data as far as ordinary semiclassical theory predicts.

If SMFT predicts a residual geometric difference:

⟨T_ab⟩₁ = ⟨T_ab⟩₂ but g_ab^(1) ≠ g_ab^(2) due to trace history, (37.4)

then the theory becomes empirically distinctive.

But this is also dangerous.

Standard semiclassical gravity would ordinarily source geometry through ⟨T_ab⟩ and boundary conditions.

Any additional history term would need:

- a precise equation;
- a smallness bound;
- compatibility with existing tests;
- a clear origin.

A schematic extension might be:

$$G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle + \mu M_{ab}[\text{history}].$$ (37.5)

Where M_{ab} is a conserved memory tensor:

$$\nabla^a M_{ab} = 0.$$ (37.6)

At present, SMFT has not derived such a term.

It should be presented only as a future test direction.

37.8 Gravity-memory falsification standard

The refined memory hypothesis becomes vulnerable to falsification only when it specifies:

What is stored? (37.7)

Where is it encoded? (37.8)

How long does it persist? (37.9)

How does it affect later dynamics? (37.10)

What ordinary theory predicts instead? (37.11)

Until then, “gravity remembers” remains an interpretive principle rather than a completed physical claim.

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38. What the Article Does Not Prove

A disciplined conclusion requires an explicit catalogue of nonclaims.

38.1 It does not prove that anti-gravity exists

The proposed branch forces remain attractive.

The negative response belongs to a postselected ensemble generated by amplitude interference.

Therefore:

Effective conditional repulsion ≠ fundamental repulsive gravitational charge. (38.1)

38.2 It does not prove that one branch has negative mass

Neither branch contains negative inertial or gravitational mass.

The anomaly arises from:

$$\beta\psi(p - \delta_B) - \alpha\psi(p - \delta_A),$$ (38.2)

not from:

$\delta_A < 0$ or **$\delta_B < 0$.** (38.3)

38.3 It does not prove that gravity is quantized in one unique way

A successful witness would support nonclassical gravitational mediation under the assumptions of the proposal.

It would not uniquely select among:

- perturbative gravitons;
- quantum geometry;
- emergent quantum mediator;
- constructor-theoretic description;
- another quantum-gravity framework.

Thus:

Witness of quantumness ≠ unique ontology of quantum gravity. (38.4)

38.4 It does not prove SMFT

The experiment was not derived from SMFT.

The SMFT interpretation was constructed after examining the proposal.

Therefore:

Compatibility with SMFT ≠ prediction by SMFT. (38.5)

The framework gains scientific strength only if it subsequently generates new constraints or predictions.

38.5 It does not prove observer-caused physical collapse

The protocol requires preparation, interaction, measurement, and postselection.

It does not require conscious awareness.

Thus:

Measurement conditioning ≠ proof of consciousness-induced collapse. (38.6)

38.6 It does not identify weak values with the weak force

The weak-value formalism is a preselection/postselection construction.

It has no direct identity with electroweak interaction.

Therefore:

Weak value ≠ W/Z interaction. (38.7)

And:

SMFT weak gate ≠ literal Standard Model mediator in this experiment. (38.8)

38.7 It does not derive the Einstein equations from postselection

The postselected gravity proposal concerns a laboratory source–probe system.

The Dorau–Much paper concerns coherent field excitations on local horizons and derives the semiclassical Einstein equation through relative entropy, energy flux, area variation, and local consistency assumptions.

No equation establishes:

$$\rho_f \to S_{rel} \to \delta A \to G_{ab}$$ (38.9)

for the proposed experiment.

The article presents this as a missing research bridge.

38.8 It does not show that negative momentum means negative entropy

The conditional momentum is signed.

Relative entropy is nonnegative.

Thus:

$$\langle \Delta p \rangle_f < 0 \not\Rightarrow S_{rel} < 0.$$ (38.10)

38.9 It does not show that negative momentum means repulsive curvature

The laboratory momentum direction and horizon focusing are different quantities in different settings.

Therefore:

$$\langle \Delta p \rangle_f < 0 \not\Rightarrow R_{ab}\xi^a\xi^b < 0.$$ (38.11)

38.10 It does not solve the measurement problem

The lifecycle separates:

- unitary interaction;
- conditional projection;
- ledger closure;

- geometric closure.

But it does not determine whether fundamental reality follows:

- objective collapse;
- Everettian branching;
- relational quantum mechanics;
- hidden variables;
- another interpretation.

It supplies an operational architecture, not a final interpretation of quantum measurement.

38.11 It does not establish a universal information-to-gravity law

The Dorau–Much relation is specific to:

- a local bifurcate Killing-horizon setting;
- coherent scalar-field excitations;
- modular theory;
- a relative-entropy calculation;
- the entropy–area assumption;
- leading-order local geometry.

The authors themselves restrict the explicit derivation to coherent states and state that a rigorous higher-order treatment remains future work.

Therefore:

Relative entropy generates curvature in this construction (38.12)

must not be generalized into:

Every information difference automatically creates gravity. (38.13)

38.12 It does not prove that classical gravity is literally stored semantic meaning

SMFT uses semantic and physical systems within one role-geometric vocabulary.

This does not establish material identity between:

- semantic curvature;
- spacetime curvature;
- cultural memory;

- gravitational field.

The disciplined relationship is:

Functional structural analogy ≠ literal substance identity. (38.14)

38.13 The strongest justified conclusion

After all restrictions, the strongest defensible claim remains substantial:

> **The proposed postselected repulsion requires a coherent branch-sensitive gravitational relation upstream of durable measurement closure. This shows that SMFT cannot identify all gravity with already settled post-collapse curvature. The earlier gravity-as-memory model should instead be understood as the semiclassical downstream regime of a longer lifecycle in which gravity first preserves unresolved relational alternatives, projection constructs a conditional trace, ledger closure restores complete accounting, and only later may stable physical difference become boundary response and curvature memory.**

In formula form:

Gravity before closure = coherent relational persistence. (38.15)

Gravity through closure = trace transformation under constrained projection and ledger accounting. (38.16)

Gravity after closure = durable geometric persistence. (38.17)

The next part can now state the refined SMFT theory in compact principle form and assemble the complete gravity lifecycle into one formal architecture.

Part X — The Refined SMFT Theory of Gravity

39. Six Principles of the Refined Model

The preceding analysis can now be consolidated into six principles.

These principles do not constitute a complete quantum-gravity theory. They define the minimum conceptual architecture that a refined SMFT account of gravity must contain after considering both the postselected-repulsion proposal and the relative-entropy route to semiclassical curvature.

The six principles are:

1. Relational Formation;
2. Projection Constructivity;
3. Ledger Completeness;
4. Information-Type Conversion;
5. Dual Persistence;
6. Curvature Closure.

Together, they replace the earlier one-directional formula:

Collapse → trace → gravity. (39.1)

with a staged architecture:

Relation → projection → trace → ledger → distinguishability → curvature. (39.2)

39.1 Principle I — Relational Formation

> **Gravity can form branch-sensitive quantum relations before any final measurement record exists.**

The relevant starting state is:

$$|\Psi_0\rangle = (\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle.$$ (39.3)

After gravity-mediated evolution:

$$|\Psi_{SP}\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle.$$ (39.4)

The gravitational interaction has associated:

$$A \leftrightarrow \psi_A.$$ (39.5)

$$B \leftrightarrow \psi_B.$$ (39.6)

The relation contains more information than the individual branch forces.

Its complete structure includes:

$$\mathcal{R}_G = \{\alpha, \beta, \varphi_A - \varphi_B, |\psi_A\rangle, |\psi_B\rangle, \langle\psi_A|\psi_B\rangle\}.$$ (39.7)

Thus:

Gravity before closure = branch-sensitive relation formation. (39.8)

This is the first correction to the earlier SMFT account.

Gravity cannot be treated only as a trace left after measurement if the proposed effect requires gravity to participate in generating the entangled state that exists before measurement.

Relational Formation Constraint

A refined gravity model must distinguish:

Classical mixture:
$$\rho_{mix} = q\rho_A + (1 - q)\rho_B.$$ (39.9)

from:

Coherent relation:
$$\rho_{coh} = \rho_{mix} + \rho_{cross}.$$ (39.10)

Where ρ_cross contains the off-diagonal branch terms.

The model must not replace ρ_coh with ρ_mix before the stage at which coherence has actually become inaccessible.

Therefore:

Premature mixture replacement destroys the proposed witness. (39.11)

Any SMFT formulation that begins directly with settled curvature omits the relational resource needed upstream.

39.2 Principle II — Projection Constructivity

> **Projection can construct a conditional observable trace whose properties are absent from every individual branch.**

The branch-conditioned probe states satisfy:

$\langle \mathbf{p} \rangle_A = \delta_A > 0.$ (39.12)

$\langle \mathbf{p} \rangle_B = \delta_B > 0.$ (39.13)

After postselection:

$|\psi_f\rangle \propto \beta|\psi_B\rangle - \alpha|\psi_A\rangle.$ (39.14)

The resulting conditioned mean can satisfy:

$\langle \mathbf{p} \rangle_f < 0.$ (39.15)

The selected state is not identical to branch A or branch B.

It is constructed from their amplitudes.

Projection therefore contains two functions:

Projection = exclusion + constrained recombination. (39.16)

The first removes nonselected outcome sectors.

The second determines the relational combination that becomes observable in the selected sector.

Constructivity Constraint

The selected trace must remain within the state space generated by the available branch-conditioned states:

$|\psi_f\rangle \in \mathrm{Span}\{|\psi_A\rangle, |\psi_B\rangle\}.$ (39.17)

Thus:

Projection constructivity ≠ arbitrary creation. (39.18)

The projection basis influences which relational property becomes accessible, but the result remains constrained by:

- the initial amplitudes;
- the gravitational interaction;
- the accumulated phases;
- the overlap of the branch states;
- the measurement projector.

The correct SMFT formula is therefore:

Trace_f = Normalize[P̂_f𝒰_G|Ψ0⟩]. (39.19)

Not:

Trace_f = observer freely invents outcome. (39.20)

39.3 Principle III — Ledger Completeness

> **A conditional trace is physically incomplete until its probability, complementary outcomes, apparatus response, and conservation context are included.**

The conditioned state is:

ρ_f = ρ̃_f / p_f. (39.21)

Where:

p_f = Tr(ρ̃_f). (39.22)

A complete measurement produces a family:

ℒ_Q = {{p_f, ρ_f}}_f. (39.23)

The complete postmeasurement probe state is:

ρ_P′ = Σ_f p_fρ_f. (39.24)

The anomalous conditional value may be large:

|⟨Δp⟩_f| ≫ δ_A. (39.25)

But its successful weight may be small:

p_f ≪ 1. (39.26)

Therefore:

Conditional magnitude without admission weight is incomplete accounting. (39.27)

Complete Ledger Formula

A minimally complete physical ledger may be written:

ℒ_complete = {{p_f, ρ_f, R_f}}_f + R_apparatus + R_environment +

C_conservation. (39.28)

Where:

- R_f is the recorded outcome sector;
- R_apparatus is the apparatus state change;
- R_environment is the environmental record;
- C_conservation is the energy–momentum accounting.

The full momentum relation must satisfy:

ΔP_source + ΔP_probe + ΔP_apparatus + ΔP_environment = 0. (39.29)

The precise partition depends on experimental implementation.

The conceptual rule does not.

Ledger Completeness Constraint

A future SMFT model must not infer a new macroscopic force law from a conditioned weak value alone.

It must first answer:

- How often does the selected result occur?
- What happens in the complementary sectors?
- What record identifies the selected sector?
- What momentum and energy are exchanged with the apparatus?
- What is the complete unconditioned response?

Thus:

Conditional anomaly ≠ complete gravitational dynamics. (39.30)

39.4 Principle IV — Information-Type Conversion

> **Closure converts phase-usable relational information into record-usable distinguishability and, potentially, geometry-usable physical structure.**

Before closure, the information format is:

I_phase = {amplitudes, relative phases, cross terms, entanglement, overlaps}. (39.31)

After measurement and record formation:

I_record = {outcomes, probabilities, conditional states, apparatus records}. (39.32)

At a later effective stage:

$I_{geo} = \{\langle T_{ab} \rangle$, flux, area variation, curvature$\}$. (39.33)

The lifecycle is:

$I_{phase} \rightarrow I_{record} \rightarrow I_{geo}$. (39.34)

The information does not remain mathematically identical across these transformations.

It changes representational type.

Phase-to-Record Conversion

A coherent density matrix may contain:

$\rho_{coh} = [[\rho_{AA}, \rho_{AB}], [\rho_{BA}, \rho_{BB}]]$. (39.35)

Under effective decoherence:

$\rho_{coh} \rightarrow \rho_{dec} = [[\rho_{AA}, 0], [0, \rho_{BB}]]$. (39.36)

The relative phase becomes operationally inaccessible in the branch basis.

Yet the outcome populations and records remain.

Therefore:

Loss of usable phase ≠ loss of all physical difference. (39.37)

Instead:

Coherence difference → historical distinguishability. (39.38)

Record-to-Geometry Conversion

The Dorau–Much framework provides one specific downstream example:

$S_{rel}(\omega_0 \| \omega_\phi) \leftrightarrow$ horizon energy flux. (39.39)

$S_{rel}(\omega_0 \| \omega_\phi) \propto \delta A$. (39.40)

δA consistency $\rightarrow G_{ab} + \Lambda g_{ab} = \alpha \langle T_{ab} \rangle$. (39.41)

This does not prove that every recorded quantum event becomes curvature through the same mechanism.

It demonstrates that, in a controlled local-horizon QFT setting, reference-relative quantum information can be related to energy flux and semiclassical geometry.

Information-Type Conversion Constraint

A valid SMFT handover theory must specify:

Which phase information is lost? (39.42)

Which physical distinctions survive? (39.43)

Which quantities remain conserved? (39.44)

Which boundary makes the distinction geometrically relevant? (39.45)

Which map converts the retained distinction into stress-energy and curvature? (39.46)

Without these answers:

Information → geometry (39.47)

remains an interpretive slogan rather than a physical derivation.

39.5 Principle V — Dual Persistence

> **Gravity preserves unresolved alternatives before closure and settled consequences after closure.**

Before closure:

M_coh = preservation of branch association + relative phase + interference capacity. (39.48)

After closure:

M_geo = preservation of energy–geometry relation + causal path constraint. (39.49)

The two persistence regimes are connected by the unresolved handover:

M_coh → $\mathscr{H}$_QC → M_geo. (39.50)

This is the Dual Persistence Principle.

Pre-Closure Persistence

Gravity preserves:

|A⟩ ↔ |ψ_A⟩. (39.51)

|B⟩ ↔ |ψ_B⟩. (39.52)

And:

Δφ = φ_A − φ_B. (39.53)

This preservation allows later interference.

The relevant memory is not a settled classical history.

It is a coherent relation among unresolved histories.

Post-Closure Persistence

Gravity later preserves:

⟨T_ab⟩ ↔ g_ab^eff. (39.54)

This relation constrains:

- geodesics;
- proper time;
- causal structure;
- horizon formation;
- redshift;
- focusing.

The relevant memory is now historical and geometric.

Thus:

Quantum gravitational memory = relation among alternatives. (39.55)

Classical gravitational memory = relation between settled source structure and geometry. (39.56)

Dual Persistence Constraint

A complete SMFT gravity model must recover both limits:

κ_c → 0 ⇒ coherent relational regime. (39.57)

κ_c → 1 ⇒ effective geometric memory regime. (39.58)

Where κ_c is a schematic closure parameter.

A model that produces only coherent amplitudes but no classical geometry is incomplete.

A model that produces only classical curvature but no coherent mediation is also incomplete.

39.6 Principle VI — Curvature Closure

> **Classical geometry is the mutually consistent effective residue of durable local physical differences.**

A detector record is local.

A spacetime geometry is shared.

The transition from one to the other requires a consistency rule.

In semiclassical gravity:

G_ab + Λg_ab = 8πG⟨T_ab⟩_ren. (39.59)

The Bianchi identity requires:

$\nabla^a G_{ab} = 0$. (39.60)

Therefore:

$\nabla^a \langle T_{ab} \rangle_{ren} = 0$. (39.61)

Any trace-to-curvature map must respect this conservation structure.

Boundary Ledger

The Dorau–Much construction interprets relative entropy as horizon energy flux and relates it to area variation. The resulting local consistency condition yields the semiclassical Einstein equation.

SMFT visualizes this as:

Durable difference → boundary ledger → curvature closure. (39.62)

The area response is not yet the whole geometry.

Curvature closure makes local ledger updates compatible across overlapping spacetime regions.

Curvature Closure Constraint

A future SMFT field theory must produce more than a scalar memory variable.

It must support:

- tensorial stress-energy;
- conservation;
- causal propagation;
- coordinate covariance;
- boundary consistency;
- quantum fluctuation corrections;
- a classical correspondence limit.

Thus:

Scalar trace alone cannot determine general spacetime curvature. (39.63)

The minimum likely structure is:

Trace tensor + correlation kernel + closure law. (39.64)

---

40. Gravity as a Lifecycle Rather Than a Single Object

The six principles can now be assembled into a gravity lifecycle.

The lifecycle does not claim that gravity changes substance seven times.

It states that the same physical relation becomes representable through different structures as closure progresses.

G0 — Reference Background

At the first stage, the relevant source and probe states are specified relative to a reference configuration.

For the source–probe experiment:

$$|\Psi_0\rangle = (\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle. \quad (40.1)$$

For the horizon framework, the reference is a vacuumlike state:

$$\omega_0. \quad (40.2)$$

A trace is always relational.

It requires something from which the new state differs.

Thus:

No trace without reference structure. (40.3)

G1 — Coherent Gravitational Alternatives

The source branches correspond to different gravitational interactions:

$$A \rightarrow \delta_A. \quad (40.4)$$

$$B \rightarrow \delta_B. \quad (40.5)$$

But the alternatives remain coherently related:

$$|\mathcal{G}\rangle = \alpha|g_A\rangle + \beta e^{i\Phi}_G|g_B\rangle. \quad (40.6)$$

This is gravitational possibility geometry.

The relevant information remains phase-usable.

G2 — Relational Entanglement

Gravity-mediated evolution creates:

$$|\Psi_{SP}\rangle = \alpha|A\rangle|\psi_A\rangle + \beta e^{i\Phi}|B\rangle|\psi_B\rangle. \quad (40.7)$$

The source and probe no longer admit independent pure-state descriptions.

The trace is distributed across the relation.

This is the stage at which gravity acts as an entangling relational channel.

G3 — Constructed Conditional Trace

A source projection produces:

$$|\psi_f\rangle \propto \langle f|\Psi_{SP}\rangle. \quad (40.8)$$

For the selected destructive-interference output:

$$|\psi_f\rangle \propto \beta|\psi_B\rangle - \alpha|\psi_A\rangle. \quad (40.9)$$

The conditioned mean may lie outside the classical branch hull:

$$\langle p\rangle_f \notin [\delta_B, \delta_A]. \quad (40.10)$$

In the proposed regime:

$$\langle p\rangle_f < 0 < \delta_B < \delta_A. \quad (40.11)$$

This is the postselected repulsive trace.

It is real as a conditioned statistical result but incomplete as a total force law.

G4 — Complete Weighted Ledger

The selected trace is inserted into the full measurement account:

$$\mathcal{L}_Q = \{\{p_f, \rho_f, R_f\}\}_f. \quad (40.12)$$

The unconditioned state is:

$$\rho' = \Sigma_f p_f \rho_f. \quad (40.13)$$

The apparatus and environmental records make the outcome sectors effectively distinct.

This stage restores normalization and conservation context.

G5 — Stable Distinguishability and Flux

The physical record becomes definable relative to a reference state and an accessible algebra.

A durable difference may be expressed through:

$$S_{rel}(\rho_{ref}\|\rho_{record}). \quad (40.14)$$

In the Dorau–Much setting:

$$S_{rel}(\omega_0\|\omega_\phi) \leftrightarrow \delta Q_H. \quad (40.15)$$

The difference now has an informational face and an energy-flux face.

The phase structure that produced the original conditional anomaly need not remain directly accessible.

The physical consequence remains.

G_6 — Boundary-Area Response

The flux becomes associated with a boundary update:

$$\delta A \propto S_{rel}.$$ (40.16)

The boundary acts as a geometric ledger.

It records how the accessible region differs from the reference configuration.

This stage is local and boundary-dependent.

It is not yet the complete spacetime field equation.

G_7 — Semiclassical Curvature Memory

Local boundary responses become mutually consistent through:

$$G_{ab} + \Lambda g_{ab} = \alpha \langle T_{ab} \rangle.$$ (40.17)

The physical difference has now become public geometry.

It constrains generic future trajectories rather than only one selected measurement ensemble.

This is the regime most accurately described by the older slogan:

Gravity as memory. (40.18)

40.1 The Complete Lifecycle

The complete sequence is:

G_0 Reference background
↓
G_1 Coherent gravitational alternatives
↓
G_2 Relational entanglement
↓
G_3 Constructed conditional trace
↓
G_4 Complete weighted ledger
↓
G_5 Stable distinguishability and flux
↓
G_6 Boundary-area response
↓
G_7 Semiclassical curvature memory

In one-line form:

Reference → alternative → relation → conditional trace → ledger → distinguishability → boundary → curvature. (40.19)

40.2 Established, interpreted, and unresolved links

The lifecycle contains three epistemic categories.

Links developed by the repulsive-gravity proposal

$G_1 \rightarrow G_2 \rightarrow G_3$. (40.20)

These include branch-dependent gravitational coupling, source–probe entanglement, postselection, and anomalous conditioned response.

Links developed by the relative-entropy framework

$G_5 \rightarrow G_6 \rightarrow G_7$. (40.21)

These include relative entropy, energy flux, area response, and semiclassical Einstein closure.

Links presently unresolved

$G_3 \rightarrow G_4 \rightarrow G_5$. (40.22)

These include complete record formation, the selection of an observable algebra and reference state, and the transition from conditional quantum trace to stable geometry-relevant distinguishability.

The lifecycle is therefore not presented as one proven derivation.

Its value is that it locates the missing theory precisely.

40.3 The central handover gap

The unknown middle can be represented as:

$\mathscr{H}_{QC} : G_3 \rightarrow G_5$. (40.23)

Or in expanded form:

$\mathscr{H}_{QC} : \{\rho_f, p_f, \text{complementary outcomes, apparatus, environment}\} \rightarrow \{\rho_{record}, S_{rel}, \langle T_{ab} \rangle_{eff}\}$. (40.24)

The minimum requirements are:

- normalization;
- complete positivity where applicable;
- conservation;
- coarse-graining consistency;
- reference-state specification;
- boundary specification;
- covariance;
- classical correspondence;

- coherence sensitivity.

This is the technical research target generated by the refined SMFT model.

40.4 Closure is scale-relative

A system can occupy different lifecycle stages relative to different observers or observable algebras.

For an experimenter who has access only to macroscopic records:

$\kappa_c \approx 1$. (40.25)

For an idealized observer with control over the full source–probe–apparatus state:

κ_c may remain < 1. (40.26)

The lifecycle therefore describes effective closure relative to accessible structure.

This does not imply that reality is arbitrarily subjective.

It means that coherence and record status depend on which degrees of freedom remain operationally recombinable.

40.5 Nested lifecycles

A macroscopic curvature field may itself participate in a larger coherent process.

Likewise, a microscopic conditional trace may already be embedded in a stable laboratory geometry.

Thus, the stages can be nested:

G_7 at one scale may function as G_0 at another scale. (40.27)

For example, a settled laboratory spacetime background becomes the reference geometry within which a new quantum experiment is conducted.

This recursive structure aligns naturally with SMFT's multiscale field perspective.

Closure is not necessarily the absolute end of dynamics.

It is the formation of a stable platform for the next process.

---

41. The Revised SMFT Definition of Gravity

The earlier definition was:

> Gravity is the residual curvature left by accumulated collapse traces.

This formulation remains useful, but it is incomplete.

It describes primarily G_6 and G_7.

The revised definition must include G_1 through $G_{\mathbb{C}}$ as well.

41.1 Narrow semiclassical definition

The narrow definition should now be:

> **Classical and semiclassical gravity are the effective geometric persistence of physically closed energy–momentum traces.**

In formula form:

$\mathcal{G}_{\text{classical}} = M_{\text{geo}}[\langle T_{ab} \rangle, \text{boundary data, closure law}].$ (41.1)

This retains the memory interpretation in the regime where it is most defensible.

41.2 Expanded lifecycle definition

The expanded definition is:

> **Gravity is the persistence of physically consequential relational difference before, through, and after staged closure.**

In compact form:

Gravity = relational persistence across staged closure. (41.2)

Its three principal components are:

Gravity = coherent persistence + closure-spanning transformation + geometric persistence. (41.3)

Or:

$\mathcal{G}_{\text{SMFT}} = M_{\text{coh}} + \mathcal{H}_{\text{QC}} + M_{\text{geo}}.$ (41.4)

This is a conceptual decomposition, not a literal additive field equation.

41.3 Gravity before closure

Before closure:

$\mathcal{G}_{\text{pre}}$ **= persistence of branch-sensitive relational structure.** (41.5)

This includes:

- source–probe association;
- relative amplitude;
- relative phase;
- overlap;
- entanglement capacity;

- interference potential.

Gravity's role is relation-forming and coherence-bearing.

41.4 Gravity through closure

Through closure:

$\mathcal{G}_{\text{trans}}$ **= conversion of relational possibility into weighted physical trace.** (41.6)

This includes:

- projection;
- conditional state formation;
- probability assignment;
- apparatus recording;
- environmental decoherence;
- conservation ledger.

Gravity is not identical to every closure mechanism.

But the gravitationally formed relation supplies the content undergoing closure.

41.5 Gravity after closure

After closure:

$\mathcal{G}_{\text{post}}$ **= persistence of settled physical consequence as effective geometry.** (41.7)

This includes:

- stress-energy expectation;
- boundary flux;
- area response;
- curvature;
- causal path constraint.

Gravity's role is now geometry-forming and history-bearing.

41.6 The refined active–passive distinction

The earlier statement:

Gravity is passive. (41.8)

should be replaced by:

Gravity is active in relational formation and passive-structural in geometric persistence. (41.9)

Even this remains simplified.

A geometry is not dynamically inert. It affects all later motion.

The refined distinction is:

Transition activity changes the current relational state. (41.10)

Geometric activity changes the future possibility space. (41.11)

Gravity can perform both roles at different closure stages.

41.7 The refined memory distinction

The earlier statement:

Gravity remembers collapse. (41.12)

should become:

Gravity preserves unresolved relation before collapse and settled consequence after effective closure. (41.13)

Thus:

Memory_pre = relation without settled history. (41.14)

Memory_post = settled history encoded as shared geometry. (41.15)

This is the Dual Persistence Principle expressed as a memory theory.

41.8 The refined force distinction

The word *force* should be indexed by layer.

F_branch = branch-local gravitational response. (41.16)

F_cond = postselection-conditioned effective response. (41.17)

F_all = complete ensemble momentum transfer. (41.18)

$\mathcal{G}$_geo = effective curvature dynamics. (41.19)

Therefore:

F_branch ≠ F_cond ≠ F_all ≠ $\mathcal{G}$_geo. (41.20)

The proposed repulsion occurs at F_cond.

It does not automatically define F_all or $\mathcal{G}$_geo.

41.9 The refined trace distinction

The word *trace* must also be indexed.

$\mathcal{T}$**_coh = distributed phase-bearing relational trace.** (41.21)

$\mathcal{T}$**_cond = projection-conditioned trace.** (41.22)

$\mathcal{T}$**_ledger = complete probability-weighted record.** (41.23)

$\mathcal{T}$**_info = reference-relative distinguishability residue.** (41.24)

$\mathcal{T}$**_geo = persistent geometric consequence.** (41.25)

Thus:

$\mathcal{T}$**_coh** → $\mathcal{T}$**_cond** → $\mathcal{T}$**_ledger** → $\mathcal{T}$**_info** → $\mathcal{T}$**_geo.** (41.26)

The maps among these traces are the central missing mathematics.

41.10 The revised gravity formula

The most complete conceptual formula is:

Gravity = relation formation + projection-conditioned trace construction + ledger closure + information retention + curvature closure. (41.27)

The most compact formula is:

Gravity = relational persistence + staged closure + geometric memory. (41.28)

The most disciplined scientific formulation is:

> **The refined SMFT hypothesis is that the coherent and semiclassical descriptions of gravity are effective regimes of a single relational process whose quantum-to-curvature handover remains to be mathematically derived.**

---

42. Conclusion: Gravity Preserves Alternatives and Consequences

The proposed postselected gravitational repulsion begins from an apparently paradoxical situation.

The probe experiences two branch-dependent gravitational interactions.

Both are attractive:

$\delta_A > 0$. (42.1)

$\delta_B > 0$. (42.2)

Yet a carefully selected source outcome produces a conditioned probe state whose mean momentum can be negative:

$\langle \Delta p \rangle_f < 0$. (42.3)

The result does not arise because one source branch contains a repulsive gravitational force.

It arises because two attractive branch-conditioned amplitudes are coherently recombined.

The negative displacement is therefore a property of:

- branch relation;
- relative phase;
- postselection basis;
- conditional normalization.

It is an interference trace.

42.1 What the repulsion witnesses

A classical mixture of two positive branch shifts remains inside their convex hull:

$\delta_{mix} = q\delta_A + (1 - q)\delta_B$. (42.4)

$\delta_B \leq \delta_{mix} \leq \delta_A$. (42.5)

The conditioned quantum result can lie outside that interval.

Thus:

Out-of-hull conditioned response → evidence of retained relational coherence under the proposal's assumptions. (42.6)

The experiment is important because it probes whether gravity can preserve a phase-sensitive source–probe relation.

It does not establish a new universal repulsive force law.

42.2 What the experiment changes in SMFT

The earlier SMFT gravity model emphasized:

Collapse → trace → curvature memory. (42.7)

This model captured the downstream geometric regime.

But it began too late.

The proposed witness requires an upstream sequence:

Source superposition → gravity-mediated relation → entanglement → constructive projection. (42.8)

The full sequence must therefore be:

Coherent alternative → gravitational relation → conditioned trace → complete ledger → stable distinguishability → boundary response → curvature memory. (42.9)

This is the primary fine-tuning.

42.3 What remains valid from the earlier model

The gravity-as-memory interpretation remains useful when restricted to classical and semiclassical geometry.

The Dorau–Much framework provides an especially sharp example of a downstream information-to-curvature chain:

Relative entropy → horizon energy flux → area variation → semiclassical Einstein equation. (42.10)

The earlier SMFT reinterpretation correctly identified:

- relative entropy as admitted difference;
- flux as trace-current;
- area as boundary ledger;
- curvature as persistent closure.

The new paper does not invalidate this.

It places it within a longer lifecycle.

42.4 The missing middle

The two physics papers illuminate opposite ends:

Coherent gravitational relation → conditioned response. (42.11)

And:

Durable state difference → flux → area → curvature. (42.12)

The missing middle is:

Conditioned quantum trace → complete record → geometry-relevant distinguishability. (42.13)

This is the handover problem:

$\mathscr{H}$_QC : I_phase → I_record → I_geo. (42.14)

SMFT has not solved this problem.

Its contribution is to identify it clearly and to specify the distinctions a future solution must preserve.

42.5 Final refined principles

The complete model can be summarized as follows.

Principle I — Relational Formation:
Gravity forms branch-sensitive quantum relations before closure. (42.15)

Principle II — Projection Constructivity:
A selected trace may exhibit properties absent from every branch. (42.16)

Principle III — Ledger Completeness:
A conditional value is incomplete without its probability, complement, and recording context. (42.17)

Principle IV — Information-Type Conversion:
Closure converts phase-usable relation into record-usable distinguishability. (42.18)

Principle V — Dual Persistence:
Gravity preserves alternatives before closure and consequences afterward. (42.19)

Principle VI — Curvature Closure:
Classical geometry is the consistent effective residue of durable local physical differences. (42.20)

42.6 Final scientific positioning

The article does not prove that SMFT is a completed physical theory.

The comparative SMFT study correctly classifies the framework as stronger than a loose metaphor but not yet a derived quantum-gravity theory. Its current strength lies in identifying recurring architecture, missing interfaces, and possible failure conditions.

The refined model becomes a physical theory only when it supplies:

- a defined state space;
- a gravity-mediated quantum evolution;
- an explicit closure map;
- a complete conservation ledger;
- a tensorial trace variable;
- a quantum-to-curvature matching rule;
- a classical general-relativity limit;
- distinguishing predictions.

Until then, it should be presented as a disciplined research grammar.

42.7 Final statement

The strongest conclusion is therefore not:

> Gravity becomes repulsive.

Nor:

Observation freely creates gravity.

Nor:

Relative entropy and weak values are the same object.

The strongest conclusion is:

> **Gravity cannot be understood solely as the geometric residue of an already completed collapse. Before closure, it must preserve phase-bearing relational alternatives strongly enough to correlate source and probe and permit interference. Projection then constructs a conditional trace from those alternatives. Ledger closure restores probability, conservation, and historical context. Stable distinguishability may subsequently become energy flux, boundary response, and semiclassical curvature. Gravity's memory face therefore survives, but as the mature downstream regime of a longer relational lifecycle.**

In its most compact form:

Gravity preserves alternatives before closure and consequences after closure. (42.21)

And:

Gravity = relational persistence across staged closure. (42.22)

The deeper research question is now explicit:

How does phase-bearing gravitational relation become durable curvature-bearing memory? (42.23)

Appendix A — Translation Tables

A.1 Purpose of the Translation Tables

The following tables distinguish four levels that should not be collapsed into one another:

1. the mathematical structure used by the physical paper;
2. the operational meaning of that structure;
3. the proposed SMFT interpretation;
4. the scientific status of the interpretation.

The purpose is not to replace the source theory with SMFT vocabulary. It is to show where the correspondence is strong, where it is only heuristic, and where a missing derivation remains.

The governing rule is:

Physical variable → operational role → SMFT interpretation, but not automatic material identity. (A.1)

A.2 Repulsive-Gravity Framework and Refined SMFT

Repulsive-gravity framework	Operational meaning	Refined SMFT interpretation	Status
Source state α	A⟩ + β	B⟩	Coherent spatial alternatives
Source positions A and B	Distinct branch configurations	Alternative relational anchors	Direct correspondence
Branch phases φ_A and φ_B	Relative phase accumulated during evolution	Phase-bearing relational memory	Direct mathematical role
Momentum transfers δ_A and δ_B	Two attractive branch-conditioned probe shifts	Branch-local gravitational responses	Direct correspondence
Joint state α	A⟩	ψ_A⟩ + β	B⟩
Off-diagonal density-matrix terms	Coherence between branch relations	Coherence-memory channel	Direct mathematical role
Source postselection P_f	Selection of one superposition-basis outcome	Constructive projection gate	Functional correspondence
Conditional state ρ_f	Probe state conditioned on selected source outcome	Admitted conditional trace	Direct operational correspondence
Negative ⟨Δp⟩_f	Conditioned response outside classical branch range	Relational residual produced by interference	Structural interpretation
Weak value ⟨Δp^⟩ _W	Pre/postselection-conditioned operator value	Selected operator-weight per transition amplitude	Interpretive description
Postselection probability p_f	Frequency or admission weight of selected sector	Trace-admission measure	Direct correspondence
Complementary outcomes	Nonselected measurement sectors	Excluded but ledger-relevant traces	Direct operational correspondence
Complete ensemble Σ_f p_fρ_f	Unconditioned statistical output	Ledger closure	Functional correspondence
Decoherence	Loss of operational phase access	Transition from coherence memory to record memory	Structural interpretation
Classical branch mixture	Probability distribution without cross terms	Closed branch ledger without coherent relation	Direct mathematical contrast

The source paper's central mechanism is the coherent subtraction of two attractive branch-conditioned probe amplitudes. The repulsive conditional mean is therefore not a negative gravitational eigenvalue but a property of the postselected relational state.

A.3 Relative-Entropy Gravity Framework and Refined SMFT

Relative-entropy framework	Operational meaning	Refined SMFT interpretation	Status
Vacuumlike state ωo	Reference state on the local algebra	Unperturbed reference ledger	Functional correspondence

Relative-entropy framework	Operational meaning	Refined SMFT interpretation	Status
Coherent excitation ω_ϕ	Physical deviation from reference	Readable state difference	Direct structural correspondence
Observable algebra	Accessible set of local measurements	Observer-defined information boundary	Functional correspondence
Modular flow	State- and algebra-relative evolution structure	Ordered trace-access geometry	Interpretive correspondence
Relative entropy $S_{rel}(\omega_0\|\omega_\phi)$	Distinguishability of excitation from vacuum	Durable distinguishability residue	Strong structural correspondence
Horizon energy flux	Stress-energy crossing local horizon	Trace-current	Functional correspondence
Null generator ξ^a	Direction defining horizon flow	Boundary-oriented transport direction	Limited interpretation
Area variation δA	Geometric boundary response	Boundary-ledger update	Strong structural correspondence
Raychaudhuri focusing	Relation between matter flux and null congruence deformation	Geometric propagation of admitted trace	Functional correspondence
Semiclassical Einstein equation	Local curvature–stress-energy consistency	Curvature closure law	Strong structural correspondence
Cosmological constant Λ	Integration or background term	Residual background closure parameter	Tentative interpretation

The Dorau–Much framework develops a local chain from quantum relative entropy to horizon flux, area response, and semiclassical curvature under specific assumptions. It does not derive the earlier postselection experiment, and the two frameworks should not be joined through unsupported variable identifications.

A.4 Combined Lifecycle Translation

Lifecycle stage	Physical object	Information type	SMFT role
G_0 Reference	Initial state or vacuumlike state	Baseline relation	Reference field
G_1 Alternatives	Source branches A and B	Phase-usable alternatives	Possibility geometry
G_2 Correlation	Entangled source–probe state	Distributed relational information	Coherent gravitational trace
G_3 Projection	Conditional probe state ρ_f	Selected phase-dependent output	Constructed trace
G_4 Ledger	$\{\{p_f, \rho_f\}\}_f$ plus records	Probability-weighted history	Ledger closure
G_5 Distinguishability	S_{rel} or another state-distance	Reference-relative durable difference	Informational residue
G_6 Boundary response	Energy flux and δA	Boundary-relevant physical record	Boundary ledger
G_7 Curvature	g_{ab} or G_{ab} response	Public geometric constraint	Curvature memory

The lifecycle should not be read as an already derived universal sequence. Its evidential status differs by stage:

$G_1 \to G_3$ is developed by the postselected-gravity proposal. (A.2)

$G_5 \to G_7$ is developed in the local-horizon relative-entropy framework. (A.3)

$G_3 \to G_5$ remains the unresolved handover. (A.4)

A.5 Old and Refined SMFT Statements

Earlier statement	Problem revealed by the new paper	Refined statement
Gravity is post-collapse residue	Gravity must mediate coherent branch relations before final measurement	Classical gravity is the post-closure residue of a longer gravitational lifecycle
Gravity is passive	Gravity changes the source–probe joint state	Gravity is relation-forming upstream and geometry-preserving downstream
The weak gate creates admitted difference	Branch-sensitive difference already exists before postselection	Gravity forms relation; projection converts relation into trace
Collapse chooses one branch	Superposition-basis projection produces a new conditional state	Projection excludes and constructively recombines
Trace is one persistent object	Conditional momentum, record, relative entropy, and curvature differ mathematically	Trace changes representational type across closure
Observer causes collapse	Experiment requires only physical preparation and measurement architecture	Ô denotes an operational trace-selection architecture
Repulsion indicates a repulsive force	Both branch forces remain attractive	Repulsion is a conditioned interference residual
Memory begins after outcome	Coherence preserves branch relation before outcome	Gravity has coherence memory and curvature memory
Relative entropy is “the trace”	Relative entropy is only one downstream distinguishability measure	Relative entropy is a candidate ledger quantity in a specified algebra
Quantum and classical gravity are separate problems	Their interface is the central unresolved issue	Gravity is one closure-spanning relational lifecycle

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A.6 Physical Claims, SMFT Interpretations, and Open Hypotheses

Category	Example	Evidential status
Source-paper result	Both branch shifts are attractive	Derived within the proposal
Source-paper result	Postselection may yield negative conditional momentum	Derived within the proposal
Standard quantum inference	Coherent amplitudes can produce values outside a classical convex mixture	Established quantum structure
SMFT interpretation	The joint entangled state is coherence memory	Conceptual reconstruction

Category	Example	Evidential status
SMFT interpretation	Projection is a trace-construction gate	Conceptual reconstruction
SMFT refinement	Gravity spans pre- and post-closure regimes	Proposed theoretical principle
Open hypothesis	Relative entropy is the surviving invariant of closure	Research hypothesis
Open hypothesis	A trace tensor links record closure to curvature	Research hypothesis
Open hypothesis	Curvature contains measurable history beyond ⟨T_ab⟩	Speculative and test-dependent
Unsupported overclaim	Negative conditional momentum directly causes repulsive curvature	Rejected
Unsupported overclaim	Weak-value formalism proves weak-force involvement	Rejected
Unsupported overclaim	The experiment proves consciousness-induced collapse	Rejected

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Appendix B — Minimal Formula Skeleton

B.1 Purpose

This appendix collects the minimum equations required to reconstruct the article's argument without repeating the complete discussion.

All expressions use Blogger-ready Unicode Journal Style.

B.2 Initial Source–Probe State

The source begins in a spatial superposition:

$$|\Psi_S\rangle = \alpha|A\rangle + \beta|B\rangle. \quad \text{(B.1)}$$

Normalization requires:

$$|\alpha|^2 + |\beta|^2 = 1. \quad \text{(B.2)}$$

The probe begins in state:

$$|\psi_0\rangle = \int dp\, \psi(p)|p\rangle. \quad \text{(B.3)}$$

The initial joint state is:

$$|\Psi_0\rangle = (\alpha|A\rangle + \beta|B\rangle)|\psi_0\rangle. \quad \text{(B.4)}$$

B.3 Branch-Dependent Gravitational Shifts

For source branch j:

$$\delta_j = GMmT / x_j^2. \quad \text{(B.5)}$$

For $x_A < x_B$:

$$\delta_A > \delta_B > 0. \quad \text{(B.6)}$$

The branch-conditioned probe states are:

$$|\psi_A\rangle = \hat{T}(\delta_A)|\psi_0\rangle. \quad \text{(B.7)}$$

$$|\psi_B\rangle = \hat{T}(\delta_B)|\psi_0\rangle. \quad \text{(B.8)}$$

Where the momentum-translation operator may be written:

$$\hat{T}(\delta) = \exp(-i\delta\hat{x}/\hbar). \quad \text{(B.9)}$$

B.4 Gravity-Mediated Joint State

After interaction:

$$|\Psi_{SP}\rangle = \alpha e^{i\phi_A}|A\rangle|\psi_A\rangle + \beta e^{i\phi_B}|B\rangle|\psi_B\rangle. \quad \text{(B.10)}$$

A schematic controlled gravitational unitary is:

$$\mathcal{U}_G = |A\rangle\langle A| \otimes \hat{T}(\delta_A) + |B\rangle\langle B| \otimes \hat{T}(\delta_B). \quad \text{(B.11)}$$

Thus:

$$|\Psi_{SP}\rangle = \mathcal{U}_G|\Psi_0\rangle. \quad \text{(B.12)}$$

Entanglement is absent only if:

$$|\psi_A\rangle = e^{ic}|\psi_B\rangle. \quad \text{(B.13)}$$

B.5 Classical Mixture Benchmark

A classical branch mixture is:

$$\rho_{mix} = q|A\rangle\langle A| \otimes \rho_A + (1-q)|B\rangle\langle B| \otimes \rho_B. \quad \text{(B.14)}$$

Its mean momentum shift is:

$$\delta_{mix} = q\delta_A + (1-q)\delta_B. \quad \text{(B.15)}$$

Therefore:

$$\delta_B \le \delta_{mix} \le \delta_A. \quad \text{(B.16)}$$

If both branch shifts are positive:

$$\delta_{mix} < 0 \text{ is impossible.} \quad \text{(B.17)}$$

B.6 Postselection

Let the source postselection state be:

$$|f\rangle = u|A\rangle + v|B\rangle.$$ (B.18)

The projector is:

$$\hat{P}_f = |f\rangle\langle f|.$$ (B.19)

The unnormalized conditioned probe state is:

$$|\tilde{\psi}_f\rangle = \langle f|\Psi_{SP}\rangle.$$ (B.20)

Therefore:

$$|\tilde{\psi}_f\rangle = u\alpha e^{i\phi_A}|\psi_A\rangle + v\beta e^{i\phi_B}|\psi_B\rangle.$$ (B.21)

The postselection probability is:

$$p_f = \langle\tilde{\psi}_f|\tilde{\psi}_f\rangle.$$ (B.22)

The normalized conditioned state is:

$$|\psi_f\rangle = |\tilde{\psi}_f\rangle / \sqrt{p_f}.$$ (B.23)

B.7 Destructive-Interference Output

For the destructive output used in the proposal:

$$|f\rangle = [-e^{i\phi_A}|A\rangle + e^{i\phi_B}|B\rangle] / \sqrt{2}.$$ (B.24)

The probe wavefunction becomes:

$$\psi_f(p) \propto \beta\psi(p - \delta_B) - \alpha\psi(p - \delta_A).$$ (B.25)

The conditioned momentum expectation is:

$$\langle p\rangle_f = \langle\psi_f|\hat{p}|\psi_f\rangle.$$ (B.26)

It contains diagonal and interference contributions:

$$\langle p\rangle_f = |N_f|^2[D_A + D_B + I_{AB}].$$ (B.27)

Where:

$$D_A = |c_A|^2\langle\psi_A|\hat{p}|\psi_A\rangle.$$ (B.28)

$$D_B = |c_B|^2\langle\psi_B|\hat{p}|\psi_B\rangle.$$ (B.29)

$$I_{AB} = 2\mathrm{Re}[c_A^* c_B\langle\psi_A|\hat{p}|\psi_B\rangle].$$ (B.30)

A sign reversal occurs when:

$$I_{AB} < -(D_A + D_B).$$ (B.31)

Then:

$$\langle p\rangle_f < 0.$$ (B.32)

B.8 Weak-Value Approximation

The branch-sensitive momentum-transfer operator is:

$$\hat{\Delta p} = \delta_A |A\rangle\langle A| + \delta_B |B\rangle\langle B|. \quad \text{(B.33)}$$

Its weak value is:

$$\langle \hat{\Delta p} \rangle_W = \langle f|\hat{\Delta p}|\Psi_i\rangle / \langle f|\Psi_i\rangle. \quad \text{(B.34)}$$

For the source states used in the proposal:

$$\langle \Pi_A \rangle_W = -\alpha / (\beta - \alpha). \quad \text{(B.35)}$$

The effective shift is:

$$\delta_{eff} = \delta_B + (\delta_A - \delta_B)\langle \Pi_A \rangle_W. \quad \text{(B.36)}$$

Therefore:

$$\delta_{eff} = (\beta\delta_B - \alpha\delta_A) / (\beta - \alpha). \quad \text{(B.37)}$$

The shift may satisfy:

$$\delta_{eff} < 0, \text{ although } \delta_A > 0 \text{ and } \delta_B > 0. \quad \text{(B.38)}$$

B.9 Probability Cost

The postselection probability is:

$$p_f = |\langle f|\Psi_i\rangle|^2. \quad \text{(B.39)}$$

Near orthogonality:

$$|\langle f|\Psi_i\rangle| \to 0. \quad \text{(B.40)}$$

Then:

$$|\delta_{eff}| \text{ may become large, while } p_f \text{ becomes small.} \quad \text{(B.41)}$$

The full weighted mean remains:

$$\langle \Delta p \rangle_{all} = \Sigma_f \, p_f \langle \Delta p \rangle_f. \quad \text{(B.42)}$$

B.10 Conditional Density Operator

For a mixed joint state ρ_{SP}:

$$\tilde{\rho}_f = \mathrm{Tr}_S[(\hat{P}_f \otimes I)\rho_{SP}]. \quad \text{(B.43)}$$

$$p_f = \mathrm{Tr}(\tilde{\rho}_f). \quad \text{(B.44)}$$

$$\rho_f = \tilde{\rho}_f / p_f. \quad \text{(B.45)}$$

The complete conditioned family is:

$\mathcal{P}(\rho_SP) = \{\{p_f, \rho_f\}\}_f.$ (B.46)

The unconditioned output is:

$\rho_P' = \Sigma_f\, p_f \rho_f.$ (B.47)

B.11 Coherence Parameter

A partially dephased source state may be written:

$\rho_S = |\alpha|^2 |A\rangle\langle A| + |\beta|^2 |B\rangle\langle B| + \gamma\alpha\beta |A\rangle\langle B| + \gamma\alpha^*\beta |B\rangle\langle A|.$** (B.48)

Where:

$0 \le |\gamma| \le 1.$ (B.49)

The classical mixture limit is:

$\gamma = 0.$ (B.50)

The ideal coherent limit is:

$|\gamma| = 1.$ (B.51)

The anomalous interference term should vanish as:

$\gamma \to 0.$ (B.52)

B.12 Ledger Closure

For outcomes f:

$\Sigma_f\, p_f = 1.$ (B.53)

The complete record may be represented as:

$\mathcal{L} = \{\{p_f, \rho_f, R_f\}\}_f + R_A + R_E + C.$ (B.54)

A complete momentum ledger requires:

$\Delta P_S + \Delta P_P + \Delta P_A + \Delta P_E = 0.$ (B.55)

B.13 Relative Entropy

For ordinary density operators:

$S(\rho \| \sigma) = \mathrm{Tr}[\rho(\log\rho - \log\sigma)].$ (B.56)

Relative entropy satisfies:

$S(\rho \| \sigma) \ge 0.$ (B.57)

Under a suitable quantum channel Φ:

$S(\rho\|\sigma) \geq S(\Phi(\rho)\|\Phi(\sigma))$. (B.58)

This is the data-processing inequality.

It suggests that coarse-graining reduces accessible distinguishability.

B.14 Horizon Information and Geometry

In the local-horizon framework:

$S_{rel}(\omega_0\|\omega_\phi) \leftrightarrow \delta Q_H$. (B.59)

Under the assumed entropy–area relation:

$\delta A = (\alpha / 2\pi) S_{rel}(\omega_0\|\omega_\phi)$. (B.60)

Raychaudhuri focusing supplies a geometric expression of the form:

$\delta A = -\int_H U R_{ab}\xi^a\xi^b \, dU \, dvol_S$. (B.61)

Local consistency yields:

$R_{ab} - \frac{1}{2}Rg_{ab} + \Lambda g_{ab} = \alpha\langle T_{ab}\rangle_{\omega\phi}$. (B.62)

With the standard normalization:

$\alpha = 8\pi G$ in conventional units. (B.63)

The derivation is specific to the assumptions and setting of the relative-entropy paper.

B.15 Handover Map

The unresolved conceptual map is:

$\mathscr{H}_{QC} : (\rho_{SP}, \mathcal{P}, \mathscr{E}, \mathcal{A}, \mathscr{B}) \to (\rho_{record}, \mathcal{D}_{res}, \langle T_{ab}\rangle_{eff}, \delta A, g_{ab}^{eff})$. (B.64)

A decomposed version is:

$\mathscr{H}_{QC} = \mathcal{C}_G \circ \mathcal{C}_T \circ \mathcal{C}_I \circ \mathcal{C}_L \circ \mathcal{P}$. (B.65)

Where:

$\mathcal{P}$: coherent state → conditional sectors. (B.66)

$\mathcal{C}_L$: conditional sectors → complete ledger. (B.67)

$\mathcal{C}_I$: ledger → durable distinguishability. (B.68)

$\mathcal{C}_T$: distinguishability → effective stress-energy representation. (B.69)

$\mathcal{C}_G$: stress-energy → effective geometry. (B.70)

The central unresolved link is:

$G_3 \to G_4 \to G_5$. (B.71)

B.16 Refined SMFT Summary Equations

Gravity before closure = coherent relational persistence. (B.72)

Gravity through closure = constrained trace transformation. (B.73)

Gravity after closure = durable geometric persistence. (B.74)

Gravity = relational persistence across staged closure. (B.75)

Gravity = M_coh + $\mathscr{H}$_QC + M_geo. (B.76)

Equation (B.76) is conceptual and not a literal additive field equation.

Appendix C — Layer Discipline

C.1 Why Layer Discipline Is Necessary

The central conceptual risk in this subject is transferring a property from one descriptive layer directly into another.

For example:

- a positive branch force;
- a negative weak value;
- a nonnegative relative entropy;
- an area variation;
- a curvature tensor;

are not interchangeable quantities.

Layer discipline requires every statement to identify:

1. the physical system;
2. the state;
3. the conditioning protocol;
4. the observable;
5. the averaging rule;
6. the closure level.

The rule is:

No quantity may inherit the meaning, sign, or ontology of another layer without a defined map. (C.1)

C.2 The Five Trace Layers

Layer 1 — Branch Trace

The branch trace records the response associated with one source configuration.

$\mathcal{T}_A = \{A, \delta_A, |\psi_A\rangle\}.$ (C.2)

$\mathcal{T}_B = \{B, \delta_B, |\psi_B\rangle\}.$ (C.3)

Its characteristic questions are:

- What happens if the source is at A?
- What happens if the source is at B?

The branch shifts are positive in the proposal:

$\delta_A > \delta_B > 0.$ (C.4)

Layer 2 — Coherent Relational Trace

The coherent trace is the joint state:

$\mathcal{T}_{coh} = \alpha|A\rangle|\psi_A\rangle + \beta e^{i\phi}|B\rangle|\psi_B\rangle.$ (C.5)

Its characteristic questions are:

- What phase relation exists between branches?
- Can the branches interfere?
- Is the source entangled with the probe?
- What measurement bases access the relation?

This layer cannot be reduced to a classical branch list.

Layer 3 — Conditional Trace

The conditional trace is produced after postselection:

$\mathcal{T}_f = \{p_f, \rho_f, \langle p\rangle_f\}.$ (C.6)

Its characteristic question is:

- What probe distribution appears given source result f?

The conditional mean may be negative:

$\langle p\rangle_f < 0.$ (C.7)

This sign belongs only to the specified conditioned observable.

Layer 4 — Ledger Trace

The ledger trace includes all outcomes:

$\mathcal{T}_L = \{\{p_f, \rho_f, R_f\}\}_f$. (C.8)

Its characteristic questions are:

- Do probabilities sum to one?
- What occurred in rejected sectors?
- What did the apparatus record?
- Is momentum conserved globally?

The ledger trace restores the complete statistical context.

Layer 5 — Geometric Trace

The geometric trace contains effective physical sources and spacetime response:

$\mathcal{T}_G = \{\langle T_{ab} \rangle, \delta A, g_{ab}^{eff}\}$. (C.9)

Its characteristic questions are:

- What energy-momentum distribution remains?
- What flux crosses the boundary?
- What curvature is required for consistency?
- How are future trajectories constrained?

This is the layer where gravity is most naturally described as curvature memory.

C.3 Layer Comparison Table

Layer	Primary variable	Mathematical type	Can be negative?	Requires conditioning?	Main role
Branch	δ_A, δ_B	Scalar displacement	Yes in general, positive here	No	Branch-local response
Coherent	ρ_{AB}, phase	Complex amplitude/ correlation	Not a simple sign question	No	Interference resource
Conditional	$\langle p \rangle_f$	Signed expectation	Yes	Yes	Selected observable trace
Ledger	p_f, ρ_f	Probability-weighted states	Probabilities nonnegative	Contains all conditions	Complete record
Information	S_{rel}	Nonnegative state divergence	No	Reference-relative	Durable distinguishability

Layer	Primary variable	Mathematical type	Can be negative?	Requires conditioning?	Main role
Dynamical	⟨T_ab⟩	Tensor	Components may have signs	State-relative	Energy-momentum source
Boundary	δA	Geometric variation	Convention and setup dependent	Boundary-relative	Ledger update
Curvature	G_ab, R_ab	Tensor	Not reducible to one sign	Geometry-relative	Shared path constraint

---

C.4 Invalid Cross-Layer Inferences

The following implications are invalid without further derivation:

$\delta_A > 0 \Rightarrow \langle p \rangle_f > 0$. (C.10)

This fails because conditional interference can reverse the mean.

$\langle p \rangle_f < 0 \Rightarrow S_{rel} < 0$. (C.11)

This fails because relative entropy is nonnegative.

$\langle p \rangle_f < 0 \Rightarrow$ negative energy. (C.12)

This fails because momentum direction and energy sign are distinct.

$\langle p \rangle_f < 0 \Rightarrow \delta A < 0$. (C.13)

This fails because the quantities belong to different systems and maps.

$\delta A < 0 \Rightarrow$ repulsive spacetime. (C.14)

This fails because area sign depends on orientation, focusing, boundary definition, and stress-energy structure.

$S_{rel} > 0 \Rightarrow$ attractive gravity. (C.15)

This fails because distinguishability magnitude does not specify a force direction.

Postselection anomaly ⇒ net momentum creation. (C.16)

This fails because the full ledger includes complementary outcomes and apparatus backreaction.

C.5 Valid Layered Statements

The following are valid within the stated framework:

Both branch shifts are attractive. (C.17)

The coherent joint state contains cross terms. (C.18)

Postselection can convert cross terms into a negative conditional mean. (C.19)

The negative mean belongs to the selected ensemble. (C.20)

The complete ensemble must satisfy normalization and conservation. (C.21)

Relative entropy measures downstream distinguishability, not force direction. (C.22)

Curvature requires tensorial stress-energy and geometric closure. (C.23)

C.6 Force-Layer Discipline

The term *force* should be qualified.

Branch force

$F_{branch,j} = \delta_j / T$ in the simplified impulse picture. (C.24)

This refers to one branch-local gravitational response.

Conditional effective force

$F_{cond,f} = \langle \Delta p \rangle_f / T.$ (C.25)

This is a protocol-conditioned inferred response.

Unconditional ensemble force

$F_{all} = \Sigma_f p_f \langle \Delta p \rangle_f / T.$ (C.26)

This is the mean across all measurement sectors.

Geometric gravitational dynamics

$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle.$ (C.27)

This is not merely another scalar force average.

Therefore:

$F_{branch} \neq F_{cond} \neq F_{all} \neq$ curvature dynamics. (C.28)

C.7 Observer-Layer Discipline

Observer type	Defines	Does not imply
Kinematic observer	Frame, trajectory, causal access	Conscious collapse
Rindler observer	Horizon and accessible wedge	Interferometric postselection
Measurement apparatus	Basis and recorded outcome	Self-awareness
Informational observer	Accessible algebra and records	Arbitrary reality creation

Observer type	Defines	Does not imply
Ô_self	Self-model and adaptive projection	Required role in laboratory quantum gravity

The central rule is:

Observer dependence must be translated into explicit dependence on frame, basis, algebra, boundary, or record access. (C.29)

C.8 Closure-Layer Discipline

At least four closures must be distinguished:

Projection closure:
ρ_SP → {{p_f, ρ_f}}_f. (C.30)

Record closure:
{{p_f, ρ_f}}_f → ρ_record. (C.31)

Informational closure:
ρ_record → S_rel or another durable distinguishability measure. (C.32)

Geometric closure:
⟨T_ab⟩ → g_ab^eff. (C.33)

Thus:

Projection closure ≠ record closure ≠ informational closure ≠ geometric closure. (C.34)

C.9 Status-Layer Discipline

Every claim should also be labelled by scientific status.

Derived within a source framework

Example:

δ_eff may become negative under the stated postselection conditions. (C.35)

Standard quantum interpretation

Example:

Amplitude recombination differs from probability mixing. (C.36)

SMFT structural interpretation

Example:

The joint state is a form of coherence memory. (C.37)

SMFT open hypothesis

Example:

Relative entropy may be a closure-surviving invariant connecting records to geometry. (C.38)

Speculative extension

Example:

Different prior coherence histories may produce an additional conserved curvature-memory term. (C.39)

Failure to distinguish these statuses would make the article appear to claim more than its sources support.

Appendix D — Weak Value, Weak Interaction, and Weak Gate

D.1 The Terminological Problem

Three distinct concepts contain the word *weak*:

1. weak value;
2. weak measurement or weak coupling;
3. weak nuclear interaction;
4. SMFT weak-gate role.

They do not share one physical mechanism.

The required rule is:

Shared adjective ≠ shared mediator. (D.1)

D.2 Weak Value

A weak value is defined by:

$$\langle \hat{A} \rangle_W = \langle \Psi_f | \hat{A} | \Psi_i \rangle / \langle \Psi_f | \Psi_i \rangle.$$ (D.2)

It depends on:

- the preselected state $|\Psi_i\rangle$;
- the postselected state $|\Psi_f\rangle$;
- the operator $\hat{A}$.

It is a relational quotient.

Its value can lie outside the operator's eigenvalue range.

For the branch projector:

$$\Pi_A = |A\rangle\langle A|. \quad \text{(D.3)}$$

The weak value may satisfy:

$$\langle \Pi_A \rangle_W < 0. \quad \text{(D.4)}$$

This does not mean the projector has acquired a negative eigenvalue.

It means the transition-conditioned operator contribution is anomalous.

D.3 Weak Measurement Regime

A weak measurement regime usually means that the coupling-induced probe displacement is small compared with the probe's initial uncertainty.

Schematically:

$$|\delta_A - \delta_B| \ll \sigma_p. \quad \text{(D.5)}$$

Where σ_p is the probe momentum width.

This allows a series expansion and an effective shifted-wavepacket approximation.

Weak coupling does not mean negligible physical significance.

Postselection can amplify the conditioned displacement.

D.4 Standard Model Weak Interaction

The weak nuclear interaction is a fundamental interaction mediated by W^+, W^-, and Z^0 bosons.

It participates in processes such as:

- beta decay;
- neutrino interactions;
- flavour change;
- electroweak transitions.

Nothing in the postselected gravitational proposal requires these mediators.

Therefore:

Weak-value anomaly does not imply electroweak causation. (D.6)

D.5 SMFT Weak Gate

The SMFT weak gate is a functional role category.

It describes a process that:

- changes admission status;
- crosses a transition boundary;
- reclassifies identity;
- converts one information format into another;
- commits a previously unresolved relation into a trace.

In the present reconstruction, the postselection apparatus is weak-gate-like because it converts:

Phase-bearing relation → selected record-bearing state. (D.7)

This is a structural analogy.

It does not imply the physical participation of the Standard Model weak interaction.

D.6 Four-Way Comparison

Concept	Mathematical form	Physical function	Relevant here?
Weak value	⟨f	Â	i⟩ / ⟨f
Weak measurement	Small coupling relative to probe width	Minimally resolving probe interaction	Yes
Weak nuclear interaction	Electroweak gauge interaction	Particle identity and flavour transitions	No
SMFT weak gate	Abstract transition/closure role	Converts relational possibility into admitted trace	Interpretively

---

D.7 Why the Analogy Remains Useful

Although the mechanisms differ, the weak-gate analogy captures a real structural feature.

The postselection does not merely reveal a branch.

It changes the operational status of information.

Before the gate:

Information exists as distributed amplitude relation. (D.8)

After the gate:

Information exists as selected conditional record. (D.9)

The analogy is therefore:

Weak-gate role ↔ transition between information regimes. (D.10)

Not:

Weak-gate role ↔ electroweak force. (D.11)

D.8 Weak Value as Selected Trace Density

An SMFT-inspired interpretation of the weak value is:

> The weak value measures the operator contribution retained per unit of selected transition amplitude.

Symbolically:

Weak value = selected operator flow / selected transition amplitude. (D.12)

Near-orthogonal postselection makes the denominator small:

$|\langle \Psi_f | \Psi_i \rangle| \ll 1$. (D.13)

The conditioned quotient can then become large.

But the selected probability becomes small:

$p_f = |\langle \Psi_f | \Psi_i \rangle|^2$. (D.14)

This motivates the ledger rule:

Large trace density must be paired with small admission measure. (D.15)

D.9 Weak-Gate Fine-Tuning

The earlier slogan was:

Weak interaction opens the gate; gravity remembers the crossing. (D.16)

The refined lifecycle requires:

Gravity forms the coherent relation. (D.17)

The projection gate constructs the admitted trace. (D.18)

Ledger closure records the complete crossing. (D.19)

Gravity preserves the closed consequence as geometry. (D.20)

The new slogan is therefore:

> **Gravity carries the relation to the gate, the gate changes its information type, and gravity preserves the settled consequence.**

D.10 Final Terminological Rule

Throughout future SMFT work:

“Weak” must always be followed by its category. (D.21)

Use:

- weak value;

- weak-coupling regime;
- weak nuclear interaction;
- weak-gate role.

Avoid unqualified statements such as:

> “The weak effect produces gravity.”

Such a statement is ambiguous and scientifically misleading.

Appendix E — Failure Matrix and Falsifiability Harness

E.1 Purpose of the Failure Matrix

A broad interpretive framework becomes scientifically weak when every possible result can be absorbed into its vocabulary.

The refined SMFT gravity model should therefore distinguish:

- observations that support the upstream coherence interpretation;
- observations that weaken it;
- observations that falsify a specific mechanism;
- observations that leave SMFT underdetermined;
- observations that require a competing explanation.

The governing rule is:

A framework earns scientific value by excluding possibilities, not merely by redescribing outcomes. (E.1)

The following matrices separate the main claims developed in the article.

E.2 Claim Classifications

Each claim belongs to one of four classes.

Class A — Source-Theory Claim

A result derived within one of the attached physical frameworks.

Example:

Two attractive branch shifts may yield a negative postselected conditional mean. (E.2)

Class B — Standard Quantum-Mechanical Inference

A consequence of coherent amplitude recombination, conditional states, or density-matrix structure.

Example:

A coherent superposition and a classical mixture produce different postselection statistics. (E.3)

Class C — SMFT Structural Interpretation

A proposed conceptual reconstruction.

Example:

The gravity-mediated joint state functions as coherence memory. (E.4)

Class D — SMFT Physical Hypothesis

A claim that would require new mathematical derivation and potentially distinctive experiments.

Example:

A closure-surviving memory tensor contributes to effective curvature. (E.5)

The evidential burden increases from Class A to Class D.

E.3 Upstream Coherence-Witness Matrix

Claim	Supporting signature	Weakening signature	Strong failure condition
Gravity preserves branch-sensitive coherence	Anomaly scales with source coherence	Reduced but nonzero anomaly under dephasing	Full anomaly unchanged when $\gamma = 0$
Gravity mediates source–probe relation	Signal scales with $\delta_A - \delta_B$	Partial dependence obscured by noise	Signal unchanged when $\delta_A = \delta_B$
Postselection constructs the anomaly	Controlled phase variation changes sign and magnitude	Weak or distorted phase curve	No dependence on postselection basis
Classical mixture is insufficient	Coherent preparation differs from matched incoherent mixture	Difference too small to resolve	Classical model reproduces full multidimensional signature
Weak-value amplification applies	Large anomaly accompanies low p_f	Trade-off differs quantitatively	Anomaly magnitude unrelated to pre/ postselection overlap
Conditional repulsion is not a branch force	Complementary sectors restore ordinary accounting	Complementary sectors unmeasured	Direct branch measurement reveals persistent repulsive force
Complete ensemble conserves momentum	Source, probe, and apparatus ledger closes	Experimental uncertainty prevents closure	Reproducible net momentum creation remains after full accounting

The first four rows test the central mechanism of the repulsive-gravity proposal. The proposal depends on coherent gravitationally mediated branch structure and postselection rather than a classical stochastic mixture.

E.4 Projection-Constructivity Matrix

Claim	Required observation	Alternative explanation	Falsifying result
Projection constructs a new conditional state	Measured P(p	f) matches amplitude-level recombination	Hidden detector bias correlated with f
Cross terms generate sign reversal	Interference term tracks source visibility	Unmodelled path-dependent force	Sign reversal survives removal of all cross terms
Basis defines accessible relational property	Rotating basis continuously deforms conditioned mean	Fixed background offset	Conditioned mean is basis-invariant
Constructivity remains constrained	All outputs lie within quantum state-space prediction	Arbitrary data filtering	Output violates positivity, normalization, or predicted span
Observer does not choose the answer	Repeated runs yield fixed statistics for each setting	Experimenter selection bias	Outcome distributions follow analyst choice rather than apparatus setting

Projection constructivity is supported only when the result follows a physically implemented projector and the complete quantum state.

It is not supported merely because an analyst can select a subset of data after the experiment.

E.5 Ledger-Completeness Matrix

Ledger component	Required quantity	Failure if omitted
Admission weight	p_f	Amplified anomaly may be mistaken for common behaviour
Complementary sectors	$\{p_g, \rho_g\}$ for $g \neq f$	Conservation and normalization cannot be checked
Apparatus record	R_A	Momentum and energy exchange may be hidden
Environmental record	R_E	Effective decoherence and irreversibility remain unspecified
Preparation cost	Energy, timing, control resources	Apparent gain may ignore required input
Unconditioned state	$\Sigma_f p_f \rho_f$	Conditional result may be mistaken for total dynamics
Calibration controls	Null and classical-mixture runs	Parasitic forces may imitate the signal

The complete ledger criterion is:

$$\Sigma_f p_f = 1. \quad \text{(E.6)}$$

$$\rho' = \Sigma_f p_f \rho_f. \quad \text{(E.7)}$$

$$\Delta P_{total} = 0 \text{ for the appropriately closed system.} \quad \text{(E.8)}$$

A reported conditional repulsion without these quantities remains physically incomplete.

E.6 Quantum-to-Curvature Handover Matrix

Handover step	Necessary mathematical object	Present status	Failure condition
Coherent state → conditional sectors	Quantum instrument or projectors	Available in standard QM	No valid state map
Conditional sectors → stable records	Decoherence and apparatus model	Operationally understood, foundationally debated	No criterion for record stability
Stable record → reference-relative distinction	State pair and observable algebra	Framework-dependent	Reference or algebra arbitrary
Distinguishability → energy flux	Modular Hamiltonian or equivalent relation	Derived in special settings	No energy representation
Energy flux → boundary response	Focusing and entropy–area relation	Derived under assumptions	Area relation fails
Boundary response → curvature	Local consistency and covariance	Semiclassical result	No conserved tensorial closure
Curvature → classical limit	Controlled approximation	Required	Failure to recover tested GR

The Dorau–Much framework develops the later part of this sequence for coherent excitations on local horizons, under specified assumptions and approximation regimes.

The earlier stages remain supplied by general quantum measurement theory rather than by a uniquely SMFT equation.

E.7 Dual-Persistence Matrix

Persistence claim	Observable implication	Non-distinctive case	Potentially distinctive case
Coherence memory exists	Interference depends on preserved branch relation	Ordinary quantum coherence already predicts it	Gravity-specific coherence scaling differs from competing mediators
Curvature memory exists	Settled stress-energy constrains future paths	Standard GR already predicts it	Additional history dependence beyond instantaneous source data
One lifecycle links both	Controlled handover recovers semiclassical gravity	Conceptual compatibility only	Quantitative matching from quantum relation to curvature
Closure changes information type	Phase information becomes stable record structure	Standard decoherence already predicts much of it	A gravity-specific invariant survives closure
Gravity spans closure	Same theory yields entanglement and geometry	Two independent effective models	One matched parameter set explains both regimes

This matrix makes clear that several SMFT interpretations overlap with already mature physical

ideas.

SMFT becomes distinctive only when it specifies additional constraints or successful cross-regime matching.

E.8 Failure Conditions for “Gravity as Relational Persistence”

The central definition is:

Gravity = relational persistence across staged closure. (E.9)

This definition becomes physically empty if any persistent relation can be labelled gravity.

It therefore requires at least three restrictions.

Restriction I — Gravitational Relevance

The relation must affect variables ordinarily associated with gravitational interaction or geometry:

- source–probe momentum transfer;
- stress-energy;
- proper time;
- curvature;
- causal structure;
- boundary response.

A generic quantum correlation is not automatically gravitational.

Restriction II — Cross-Stage Continuity

There must be a controlled map between the coherent relation and the later effective geometry.

Merely observing quantum coherence in one system and curvature in another does not establish a lifecycle.

Restriction III — Distinctive Constraint

The persistence framework must constrain at least one of:

- allowed closure maps;
- permitted memory kernels;
- sign transitions;
- scaling relations;
- noise correlations;

- boundary dependence;
- classical correspondence.

Without these restrictions:

Relational persistence = universal metaphor. (E.10)

With them:

Relational persistence = candidate interface principle. (E.11)

E.9 Null Models

A rigorous experimental and theoretical program should compare the refined model against explicit null models.

Null Model N_1 — Classical Branch Mixture

$$\rho_{N_1} = q\rho_A + (1-q)\rho_B.$$ (E.12)

Prediction:

Conditional means remain within the classical branch range under the specified filtering architecture. (E.13)

Null Model N_2 — Coherent Matter with Classical Gravity

The source is quantum, but the gravitational field is treated as a classical expectation-value field.

A representative semiclassical source might use:

$$\nabla^2\Phi = 4\pi G\langle\hat{\rho}_m\rangle.$$ (E.14)

The probe experiences one averaged classical potential rather than branch-conditioned quantum mediation.

The resulting predictions should be compared directly with the entangling model.

Null Model N_3 — Non-Gravitational Path-Dependent Force

An electromagnetic or surface interaction produces branch-correlated probe shifts.

Prediction:

- signal may track path;
- it may imitate postselection dependence;
- but its mass, distance, shielding, material, and environmental scaling differ.

Null Model N_4 — Detector-Conditioning Bias

The selection process correlates with probe readout error.

Prediction:

- apparent anomaly changes with detector calibration;
- it may survive when the gravitational interaction is removed;
- it may fail complementary-sector accounting.

Null Model N_5 — General Quantum Interaction Without Gravitational Specificity

Some unidentified quantum mediator produces the entanglement.

Prediction:

- coherent anomaly exists;
- gravitational scaling with G, M, m, and distance is absent.

A successful gravity witness must reject not only classicality, but relevant nongravitational alternatives.

E.10 Experimental Falsifiability Harness

A complete test campaign can be organized into four intervention classes.

Probe

Observe the nominal protocol without deliberately changing its regime.

Measure:

- conditioned momentum distribution;
- successful postselection rate;
- source visibility;
- complementary outputs.

Pump

Increase the gravitational interaction strength by changing:

- M;
- m;
- T;
- x_A;
- x_B.

Expected signature:

$$\Delta\delta = GMmT(1/x_A^2 - 1/x_B^2).$$ (E.15)

Switch

Change the coherence or projection regime:

- add which-path marking;
- randomize the source phase;
- rotate the postselection basis;
- replace the superposition with a matched mixture.

Expected signature:

Coherence-sensitive anomaly switches off or changes predictably. (E.16)

Couple

Add or modify environmental and boundary channels:

- alter electromagnetic shielding;
- vary nearby surfaces;
- modify trap configuration;
- change temperature;
- introduce controlled decoherence.

Expected signature:

- gravitational component follows gravitational scaling;
- parasitic components follow material or environmental scaling.

The harness can be summarized as:

Probe → Pump → Switch → Couple → compare with null models. (E.17)

E.11 Claim Declaration Template

Before presenting an empirical SMFT claim, the following fields should be declared.

System boundary: What is included in the conserved system?

Reference state: Relative to which state is difference measured?

Feature map: Which observables represent the trace?

Closure stage: Go through G_7?

Intervention: What physical setting is changed?

Predicted signature: What quantitative change should follow?

Null model: What simpler mechanism is being excluded?

Failure condition: What result would reject the proposed mechanism?

A claim lacking these fields should be treated as interpretive rather than experimentally specified.

E.12 Scientific Status Footer

Every future paper using this framework should attach a status label to major statements.

Suggested labels are:

[Derived] — obtained mathematically in the cited source framework.

[Standard QM] — follows from established quantum theory.

[SMFT Interpretation] — conceptual role mapping.

[SMFT Hypothesis] — proposed physical extension.

[Open Problem] — required but not yet derived.

[Speculation] — exploratory possibility with no current test.

For example:

> The postselected conditional mean may become negative. **[Derived]**
>
> This can be interpreted as constructive projection. **[SMFT Interpretation]**
>
> A conserved trace-memory tensor may influence later curvature. **[SMFT Hypothesis]**
>
> The map connecting conditional traces to such a tensor is unknown. **[Open Problem]**

This practice would greatly reduce accidental overclaiming.

Appendix F — The Quantum-to-Curvature Handover Research Program

F.1 Research Objective

The central unresolved task is to construct a mathematically controlled bridge:

Phase-bearing gravitational relation → durable curvature-bearing record. (F.1)

The schematic handover is:

$$\mathscr{H}_{QC} : I_{phase} \to I_{record} \to I_{geo}. \quad \text{(F.2)}$$

A successful theory must connect:

- coherent source–probe dynamics;

- measurement and conditionalization;
- environmental record formation;
- reference-relative information;
- stress-energy;
- boundary response;
- effective spacetime geometry.

The purpose of this appendix is to convert that broad objective into a research program.

F.2 Work Package 1 — Define the Upstream State Space

The first task is to identify the gravitational degrees of freedom involved in the coherent regime.

Possible formulations include:

1. a quantized weak-field metric perturbation;
2. a graviton-mediated interaction;
3. an effective quantum channel;
4. a relational quantum mediator;
5. a constructor-theoretic information carrier;
6. another nonclassical gravitational variable.

The minimum upstream state should permit:

$$|\Psi_{SP\mathcal{G}}\rangle = \alpha|A\rangle|\psi_A\rangle|g_A\rangle + \beta e^{i\Phi}|B\rangle|\psi_B\rangle|g_B\rangle. \quad \text{(F.3)}$$

A reduced controlled-unitary model may be sufficient for the laboratory witness:

$$\mathcal{U}_G = \Sigma_j |j\rangle\langle j| \otimes U_j^P. \quad \text{(F.4)}$$

But a full gravity theory must eventually explain:

- what |g_j⟩ represents;
- how it transforms;
- how gauge freedom is handled;
- which observables are physical;
- how locality and causality are preserved.

Deliverable F1

A defined upstream Hilbert space or algebra and a gravity-mediated evolution law that reproduces the proposed branch-conditioned momentum shifts.

F.3 Work Package 2 — Separate Interaction from Projection

The refined model requires:

$\mathcal{U}_{\mathbf{G}} \neq \mathcal{P}_{\mathbf{f}}$. (F.5)

The interaction generates the relation.

Projection constructs the conditioned trace.

A quantum instrument may be written:

$\mathcal{I}_{\mathbf{f}}(\rho) = \mathbf{K}_{\mathbf{f}} \rho \mathbf{K}_{\mathbf{f}}^{\dagger}$. (F.6)

With:

$\mathbf{p}_{\mathbf{f}} = \mathbf{Tr}[\mathcal{I}_{\mathbf{f}}(\rho)]$. (F.7)

$\rho_{\mathbf{f}} = \mathcal{I}_{\mathbf{f}}(\rho) / \mathbf{p}_{\mathbf{f}}$. (F.8)

The complete instrument satisfies:

$\Sigma_{\mathbf{f}} \mathbf{K}_{\mathbf{f}}^{\dagger} \mathbf{K}_{\mathbf{f}} = \mathbf{I}$. (F.9)

The research task is to define which operators act on:

- source;
- probe;
- gravitational mediator;
- apparatus.

Deliverable F2

An explicit quantum-instrument model that recovers both the conditioned anomaly and the complete unconditioned ledger.

F.4 Work Package 3 — Model Record Formation

Conditionalization alone does not explain durable closure.

A minimal measurement model is:

$\Sigma_{\mathbf{f}} \mathbf{c}_{\mathbf{f}} |\mathbf{f}\rangle |\mathbf{A}_0\rangle |\mathbf{E}_0\rangle \rightarrow \Sigma_{\mathbf{f}} \mathbf{c}_{\mathbf{f}} |\mathbf{f}\rangle |\mathbf{A}_{\mathbf{f}}\rangle |\mathbf{E}_{\mathbf{f}}\rangle$. (F.10)

Record stability requires approximate orthogonality:

$\langle \mathbf{A}_{\mathbf{f}} | \mathbf{A}_{\mathbf{g}} \rangle \langle \mathbf{E}_{\mathbf{f}} | \mathbf{E}_{\mathbf{g}} \rangle \approx 0 \text{ for } \mathbf{f} \neq \mathbf{g}$. (F.11)

A more complete model should calculate:

- decoherence rate;

- pointer basis;
- redundancy of environmental records;
- reversibility threshold;
- record lifetime;
- dependence on gravitational degrees of freedom.

Possible closure indicators include:

$$D_{fg}(t) = |\langle E_f(t) | E_g(t) \rangle|. \quad \text{(F.12)}$$

Effective record closure occurs when:

$$D_{fg}(t) \ll 1. \quad \text{(F.13)}$$

Deliverable F3

A quantitative record-formation model linking gravitationally correlated branches to stable outcome sectors.

F.5 Work Package 4 — Construct the Complete Ledger

The complete ledger must include:

$$\mathcal{L} = \{\{p_f, \rho_f, R_f\}\}_f + C_E + C_P. \quad \text{(F.14)}$$

Where:

- C_E is energy accounting;
- C_P is momentum accounting.

Required checks include:

$$\Sigma_f p_f = 1. \quad \text{(F.15)}$$

$$\Sigma_f p_f \langle \Delta p \rangle_f = \langle \Delta p \rangle_{unconditioned}. \quad \text{(F.16)}$$

$$\Delta P_{total} = 0. \quad \text{(F.17)}$$

A complete ledger should also include the resource cost of:

- source preparation;
- phase stabilization;
- postselection;
- cooling;
- trapping;
- detector operation.

This is important because an amplified conditional response may be accompanied by low yield and high preparation overhead.

Deliverable F4

A full source–probe–apparatus–environment conservation ledger for the proposed protocol.

F.6 Work Package 5 — Define the Reference State and Observable Algebra

Relative entropy requires a state pair and an algebra.

The research questions are:

1. What is the relevant reference state?
2. Which degrees of freedom belong to the observable algebra?
3. Which degrees are traced out?
4. Is the algebra local, horizon-associated, apparatus-associated, or relational?
5. What boundary makes the distinction physically meaningful?

A generic reference-relative quantity is:

$$\mathcal{D}_{\mathrm{res}} = S(\rho_{\mathrm{record}} \| \rho_{\mathrm{ref}}).$$ (F.18)

But different choices of ρ_{ref} produce different values.

A gravity-relevant reference should be selected by physical principle, not convenience.

Possible candidates include:

- local vacuum;
- no-source state;
- unperturbed apparatus state;
- matched classical mixture;
- equilibrium background;
- local Rindler vacuum.

Deliverable F5

A principled reference-state and observable-algebra prescription for the handover problem.

F.7 Work Package 6 — Identify Closure-Surviving Invariants

The phase-level state contains more information than the effective record.

A handover theory must identify what survives coarse-graining.

Candidate invariants or monotones include:

- total energy;
- momentum;
- Noether charges;
- modular energy;
- relative entropy;
- mutual information;
- stress-energy moments;
- boundary flux;
- topological charge;
- causal order.

A candidate closure-surviving quantity Q should satisfy one or more of:

Q_before = Q_after. (F.19)

Q_before ≥ Q_after. (F.20)

Q_after = Functional[Q_before, discarded variables]. (F.21)

Relative entropy's data-processing inequality provides one possible monotonic structure:

$S(\rho\|\sigma) \geq S(\Phi(\rho)\|\Phi(\sigma))$. (F.22)

A future SMFT theory should determine whether curvature is sourced by:

- the full distinguishability;
- the retained distinguishability;
- a conserved component;
- a boundary-localized component.

Deliverable F6

A mathematically defined set of invariants or monotones connecting coherent relation to durable record.

F.8 Work Package 7 — Connect Information to Stress-Energy

The central bridge must avoid the vague statement:

> Information becomes energy.

A better formulation is:

> One physical deviation admits both an informational representation and a stress-energy representation.

The required relation may take a form such as:

δ⟨K_mod⟩ − δS = S_rel. (F.23)

Or, in a suitable horizon setting:

S_rel = functional of ∫_H ⟨T_ab⟩ξ^a dΣ^b. (F.24)

The Dorau–Much construction provides a concrete instance of this relationship for coherent excitations on local horizons.

The research challenge is to determine whether a related structure can be built from the laboratory record.

Questions include:

- Can the conditioned and complementary sectors define a modular Hamiltonian?
- What local energy flux corresponds to the record?
- Is relative entropy evaluated before or after conditioning?
- How is postselection cost included?
- Which energy belongs to apparatus rather than probe?

Deliverable F7

An explicit information–stress-energy relation for a specified closure setting.

F.9 Work Package 8 — Introduce Tensorial Trace Structure

A scalar distinguishability measure cannot determine a general metric.

The handover requires tensorial data.

A minimal reference-relative source is:

$\mathcal{T}$_ab(x) = ⟨T_ab(x)⟩_record − ⟨T_ab(x)⟩_ref. (F.25)

A fluctuation kernel is:

$\mathcal{N}$_abcd(x,y) = ½⟨{ΔT_ab(x), ΔT_cd(y)}⟩. (F.26)

A possible SMFT trace structure may therefore contain:

$\mathfrak{T}$ = {$\mathcal{T}$_ab, $\mathcal{N}$_abcd, S_rel, J_a, boundary data}. (F.27)

Where J_a denotes relevant conserved currents.

The conservation condition is:

$\nabla^a \mathcal{T}_{ab} = 0.$ (F.28)

A future memory term M_ab must also satisfy:

$\nabla^a M_{ab} = 0.$ (F.29)

Deliverable F8

A conserved tensorial trace object capable of sourcing semiclassical or stochastic geometry.

F.10 Work Package 9 — Derive Boundary Response

The downstream framework requires a physical boundary.

For a local null horizon, Raychaudhuri focusing gives:

$d\theta/d\lambda = -\frac{1}{2}\theta^2 - \sigma_{ab}\sigma^{ab} + \omega_{ab}\omega^{ab} - R_{abk^a k^b}.$ (F.30)

In the appropriate local approximation, the curvature term controls the leading area response.

The research task is to specify:

- which boundary is relevant to the laboratory process;
- whether a horizon construction is needed;
- whether a causal diamond or finite region is more appropriate;
- how the recorded stress-energy crosses or deforms that boundary;
- whether area remains the correct ledger variable.

Possible boundary choices include:

- local Rindler horizon;
- causal diamond boundary;
- detector worldtube;
- entangling surface;
- finite laboratory region.

Deliverable F9

A boundary-response calculation connecting the conserved trace tensor to an area, focusing, or causal-structure variation.

F.11 Work Package 10 — Recover Geometric Closure

The effective geometry should satisfy a covariant closure law.

The semiclassical target is:

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle_{ren}.$$ (F.31)

A generalized trace-memory model might take:

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle_{ren} + \mu M_{ab}.$$ (F.32)

But Equation (F.32) is only admissible if:

$$\nabla^a M_{ab} = 0.$$ (F.33)

And if μM_ab is consistent with existing gravitational tests.

Possible forms of M_ab include:

- a nonlocal memory kernel;
- a correlation-derived correction;
- a boundary-history tensor;
- a stochastic source;
- an effective higher-curvature term.

A schematic nonlocal model is:

$$M_{ab}(x) = \int d^4y \, K_{ab}^{(cd)}(x,y) \mathcal{T}_{cd}(y).$$ (F.34)

The kernel must respect causality:

$$K(x,y) = 0 \text{ when } y \text{ lies outside the causal past relevant to } x.$$ (F.35)

Deliverable F10

A covariant geometric closure equation with a controlled semiclassical limit.

F.12 Work Package 11 — Establish the Classical Correspondence Limit

The theory must recover ordinary gravity when:

- source coherence is negligible;
- quantum fluctuations are small;
- records are stable;
- coarse-graining scale is macroscopic.

The required limit is:

$$\rho_{coh} \to \rho_{mix}.$$ (F.36)

$$\mathcal{N}_{abcd} \to \text{negligible relative to } \langle T_{ab} \rangle^2.$$ (F.37)

$\mathscr{H}$_QC → semiclassical Einstein dynamics. (F.38)

And eventually:

$\hbar$ → 0 or suitable macroscopic limit → classical GR. (F.39)

The model should also recover the Newtonian branch transfers:

$\delta_j = GMmT / x_j^2$. (F.40)

Deliverable F11

A demonstrated limit reproducing Newtonian gravity and semiclassical general relativity.

F.13 Work Package 12 — Derive Distinctive Predictions

Without distinctive predictions, SMFT remains an interpretive overlay.

Promising prediction classes include the following.

Prediction Class A — Closure-Timing Effects

Does changing when decoherence or record amplification occurs alter the later gravitational response?

Possible signature:

Same branch impulses but different closure timing → measurable difference in conditional or residual geometry. (F.41)

Prediction Class B — Coherence-History Dependence

Prepare two states with the same final ⟨T_ab⟩ but different coherence histories.

Test whether:

$g_{ab}^{(1)} = g_{ab}^{(2)}$ as standard semiclassical theory suggests, (F.42)

or:

$g_{ab}^{(1)} - g_{ab}^{(2)} = \mu \Delta M_{ab}[\text{history}]$. (F.43)

Prediction Class C — Trace-Fluctuation Geometry

Test whether stress-energy correlation structure affects gravitational noise beyond the mean source.

Possible signature:

Geometry variance $\propto \mathcal{N}$**_abcd.** (F.44)

Prediction Class D — Boundary-Dependent Closure

Test whether different physically implemented accessible algebras or boundaries produce predictable changes in the effective record-to-geometry relation.

Prediction Class E — Postselection-Weighted Backreaction

Determine whether rare conditioned sectors exhibit measurable backreaction only when the full selection ledger is included.

Deliverable F12

At least one quantitative prediction that differs from standard quantum mechanics plus semiclassical gravity.

F.14 Minimum Viable Mathematical Model

A first practical model need not solve full quantum gravity.

A minimum viable model could contain:

1. a two-branch source;
2. a one-dimensional probe;
3. an effective gravity-controlled translation;
4. an explicit measurement instrument;
5. an apparatus/environment decoherence channel;
6. a record-relative stress-energy functional;
7. a simplified one-dimensional or weak-field geometric response.

The model might begin with:

$$|\Psi_{SP}\rangle = \alpha|A\rangle\hat{T}(\delta_A)|\psi_0\rangle + \beta|B\rangle\hat{T}(\delta_B)|\psi_0\rangle. \quad \text{(F.45)}$$

Apply a quantum instrument:

$$\rho_f = \mathcal{I}_f(\rho_{SP}) / p_f. \quad \text{(F.46)}$$

Apply decoherence:

$$\rho_{record} = \Phi_{dec}(\Sigma_f p_f \rho_f \otimes |R_f\rangle\langle R_f|). \quad \text{(F.47)}$$

Define a reference-relative source:

$$\mathcal{T}_{00}(x) = \langle T_{00}(x)\rangle_{record} - \langle T_{00}(x)\rangle_{ref}. \quad \text{(F.48)}$$

Then solve a weak-field Poisson equation:

$$\nabla^2\Phi_{eff} = 4\pi G\mathcal{T}_{00}/c^2. \quad \text{(F.49)}$$

This would not establish the full lifecycle, but it would force the intermediate variables to be explicitly defined.

F.15 Model Comparison Strategy

The minimum model should be compared against at least four alternatives.

Model QG — Coherent Quantum Gravity Mediator

Maintains branch-dependent coherent interaction.

Model SC — Semiclassical Expectation-Value Gravity

Uses:

$$\nabla^2\Phi = 4\pi G\langle\hat{\rho}\rangle. \quad \text{(F.50)}$$

Model CM — Classical Mixture Gravity

Uses one classical field per run with hidden branch selection.

Model NG — Nongravitational Quantum Coupling

Produces similar entanglement without gravitational scaling.

The objective is to identify parameter regimes where:

$$\mathbf{Pred_QG \neq Pred_SC \neq Pred_CM \neq Pred_NG.} \quad \text{(F.51)}$$

The postselected anomaly is useful only if experimentally accessible differences survive noise and control imperfections.

F.16 Data Requirements

A future experiment should retain event-level records containing:

- source preparation setting;
- phase setting;
- branch geometry;
- interaction time;
- source detector output;
- probe momentum;
- apparatus state;
- environmental monitor values;
- calibration status;
- timestamp.

The analysis should reconstruct:

$$\mathbf{P(p,f|\lambda).} \quad \text{(F.52)}$$

Where λ contains all controlled settings.

From this, calculate:

$P(p|f,\lambda)$. (F.53)

$P(f|\lambda)$. (F.54)

$P(p|\lambda) = \Sigma_f P(f|\lambda)P(p|f,\lambda)$. (F.55)

This supports both conditional and ledger-level analysis.

F.17 Statistical Tests

Relevant tests include:

Classical-Hull Test

Test whether:

$\langle p \rangle_f < \min(\delta_A, \delta_B)$. (F.56)

with uncertainty and systematic bounds.

Phase-Curve Test

Fit:

$\langle p \rangle_{f(\chi)}$ (F.57)

to the predicted quantum interference function.

Coherence-Suppression Test

Measure:

$\Delta_{anom}(\gamma) = \langle p \rangle_{f(\gamma)} - \langle p \rangle_{classical}$. (F.58)

Test whether:

$\Delta_{anom}(\gamma) \to 0$ as $\gamma \to 0$. (F.59)

Ledger Closure Test

Verify:

$\Sigma_f p_f \langle p \rangle_f = \langle p \rangle_{unconditioned}$. (F.60)

within uncertainty.

Model Evidence Test

Compare Bayesian or information-criterion evidence for QG, SC, CM, and NG models.

F.18 Theoretical Success Levels

The research program can be evaluated through five levels.

Level 1 — Interpretive Coherence

The lifecycle organizes existing ideas without contradiction.

This article reaches approximately this level.

Level 2 — Operational Formalization

All stages are represented by defined states, operators, and records.

Level 3 — Cross-Regime Matching

The upstream model is quantitatively connected to a downstream stress-energy description.

Level 4 — Geometric Derivation

The model derives a controlled curvature response and recovers semiclassical gravity.

Level 5 — Distinctive Verification

A prediction unique to the refined SMFT model is experimentally confirmed.

The distinction is:

Level 1 explains vocabulary. (F.61)

Level 5 establishes new physics. (F.62)

F.19 Research Risk Register

Risk	Consequence	Mitigation
SMFT remains too metaphorical	No unique prediction	Require explicit operators and dimensions
Postselected signal is too small	Experiment infeasible	Optimize mass, time, coherence, and detection
Parasitic forces dominate	False positive	Multidimensional controls and material variation
Relative entropy bridge is setting-specific	No universal handover	Restrict claims and search for broader invariants
Measurement interpretation remains unresolved	Ontological ambiguity	Separate operational closure from foundational interpretation
Scalar trace cannot source tensor geometry	Mathematical incompleteness	Introduce stress-energy and correlation tensors
Memory correction violates conservation	Inconsistent gravity equation	Enforce divergence-free source structure
Model conflicts with precision GR tests	Empirical exclusion	Derive smallness bounds and correspondence limits
Too many free parameters	Unfalsifiable fitting	Fix parameters through independent measurements

Risk	Consequence	Mitigation
No difference from standard theory	SMFT remains interpretive	Identify a genuine cross-stage prediction

---

F.20 Recommended Immediate Research Sequence

The most efficient order is:

Step 1

Reproduce the source paper's postselection result numerically for realistic wavepackets.

Step 2

Add partial dephasing γ and calculate the full phase-dependent conditional distributions.

Step 3

Model all complementary measurement outputs and verify the momentum ledger.

Step 4

Add an explicit apparatus and environment to define record closure.

Step 5

Construct a reference-relative stress-energy difference for the complete recorded state.

Step 6

Test whether relative entropy or another monotone tracks the closure-surviving physical difference.

Step 7

Insert that source into a weak-field geometric model.

Step 8

Compare the result against semiclassical expectation-value gravity.

Step 9

Search for one observable that depends on coherence history rather than final ⟨T_ab⟩ alone.

Step 10

Use that observable as the first genuinely distinctive SMFT gravity test.

F.21 Final Research Formulation

The research program can be summarized by one sequence:

Prepare relation → generate gravitational correlation → project conditionally → close the full ledger → identify surviving invariant → construct conserved tensor source → derive boundary response → recover curvature. (F.63)

Or:

ρ_coh → ρ_f → ℒ_complete → 𝒟_res → 𝒯_ab → δA → g_ab. (F.64)

The current evidence supports only selected portions of this chain.

The postselected-gravity proposal supplies a sophisticated upstream test of coherence-bearing gravitational relation.

The relative-entropy framework supplies a sophisticated downstream model of information-bearing curvature closure.

SMFT's present contribution is to define the missing bridge as a concrete research target:

ℋ_QC : phase-bearing gravitational relation → durable curvature-bearing memory. (F.65)

The framework will become a physical theory only when ℋ_QC is no longer a placeholder.

Appendix G — Semantic Gravity in LLM Space

Slot Capacity, Counterphase Microcycles, Constructive Decoding, and Contextual Curvature

G.1 Purpose and Scientific Status

This appendix extends the refined gravity lifecycle into the semantic dynamics of large language models.

It does not claim that:

- a transformer is a quantum gravitational system;
- attention weights are physical gravitational fields;
- HeTu is literally implemented in an LLM;
- the ten HeTu sites correspond to ten specific neural features;
- token decoding is identical to physical wavefunction collapse;
- Δ5 phase opposition has already been measured inside a commercial LLM.

The proposed correspondence is functional:

Physical gravity lifecycle ↔ semantic-attractor lifecycle. (G.2)

The objective is to determine whether the concepts developed in the main article—relational persistence, constructive projection, ledger closure, and curvature memory—can provide a useful

mathematical grammar for semantic attractors in LLMs.

The central hypothesis is:

> **A semantic attractor is not merely a static region toward which representations move. It is a closure-spanning process that organizes latent alternatives before decoding and preserves committed consequences after decoding.**

In compact form:

Semantic attractor = pre-decoding relation + decoding transformation + post-decoding memory. (G.3)

G.2 What the Slot Interpretation Adds

The slot interpretation defines a slot as a discrete, mutually distinguishable capacity for holding a stable semantic trace or state at a particular field location. A site with more slots can, within that model, preserve more independent traces without forcing them into ambiguity or overlap.

Transferred cautiously into LLM space, a semantic slot should not be interpreted simply as:

- one neuron;
- one attention head;
- one token position;
- one embedding coordinate.

A better operational definition is:

> **An LLM semantic slot is one independently recoverable degree of stable representational distinction within a local attractor region.**

Let attractor region $\mathcal{A}$_i support an effective stable subspace V_i.

Its slot capacity may be represented schematically as:

s_i = dim_eff(V_i). (G.4)

Here dim_eff need not be the literal linear dimension of a network layer. It may be estimated through:

- stable feature rank;
- independently steerable directions;
- distinguishable memory traces;
- recoverable concepts under perturbation;
- effective dimensionality of a local activation manifold.

The essential idea is:

Attractor strength and attractor capacity are different. (G.5)

A very strong attractor may pull many prompts toward one interpretation while supporting little internal differentiation.

A high-capacity attractor may preserve several related meanings without collapsing them prematurely into one answer.

G.3 Slot Capacity Is Not Semantic Mass

The gravitational metaphor requires at least two separate quantities.

Semantic mass

Semantic mass measures how strongly an attractor influences subsequent representation or generation.

A schematic measure is:

M_i = expected trajectory deflection caused by attractor $\mathcal{A}_i$. (G.6)

Semantic slot capacity

Slot capacity measures how many distinct traces the attractor can preserve without destructive overlap:

s_i = number of independently recoverable stable distinctions near $\mathcal{A}_i$. (G.7)

A semantic attractor may therefore be:

Attractor type	Semantic mass	Slot capacity
Rigid slogan	High	Low
Rich scientific theory	High	High
Weak association	Low	Low
Broad but unstable theme	Low	Potentially high

This distinction is important.

A low-slot, high-mass attractor resembles a semantic black hole: it pulls many inputs into one interpretation while erasing their differences.

A high-slot attractor can retain distinctions while still organizing them coherently.

Thus:

Healthy semantic gravity preserves differentiated traces. (G.8)

Pathological semantic gravity collapses differentiated traces into one undifferentiated basin. (G.9)

G.4 The Δ5 Paper Adds a Local Stabilization Mechanism

The Δ5 paper studies pairs related by a half-turn:

$T_5 : n \to n + 5 \bmod 10.$ (G.10)

Its central phase-opposition condition is:

$a_{n+5} = -a_n.$ (G.11)

The corresponding pair energy is:

$E_{pair}(a) = \Sigma_{n=1..5} |a_n + a_{n+5}|^2.$ (G.12)

Minimizing this quantity suppresses the symmetric component and preserves the antisymmetric component. The paper further interprets these pairs as local negative-feedback microcycles, entropy buffers, and a basis for coarse-graining the ten-site field into five effective antisymmetric modes.

In an LLM, the literal numbers and decagon need not be retained.

The transferable principle is:

> **A semantic modality may be stabilized by pairing a tendency with an internal counter-tendency of the same modality.**

Examples include:

- claim ↔ strongest objection;
- literal reading ↔ nonliteral reading;
- completion impulse ↔ uncertainty audit;
- user-frame acceptance ↔ premise verification;
- analogy generation ↔ analogy-break test;
- action proposal ↔ safety or feasibility check;
- local coherence ↔ global consistency;
- prediction ↔ falsification condition.

These are not unrelated competing topics.

They are phase-opposed functions within one semantic modality.

G.5 Counterphase Semantic Pairs

Let one semantic modality m contain two coupled activation components:

a_{m^+} = forward or expressive tendency. (G.13)

a_{m^-} = counterphase or corrective tendency. (G.14)

Define symmetric and antisymmetric coordinates:

$e_m = (a_{m^+} + a_{m^-}) / \sqrt{2}.$ (G.15)

$p_m = (a_{m^+} - a_{m^-}) / \sqrt{2}.$ (G.16)

The variable e_m measures common-mode excess.

The variable p_m preserves the structured contrast between the two sides.

A Δ5-inspired pair energy is:

$E_m = |a_m^+ + a_m^-|^2 = 2|e_m|^2$. (G.17)

Minimizing E_m does not erase the modality.

It suppresses unopposed common-mode activation while retaining the contrastive carrier p_m.

Thus:

Counterphase locking suppresses semantic overrun without suppressing semantic content. (G.18)

G.6 Why Counterphase Pairing May Reduce Hallucination

A hallucination often develops through unopposed semantic amplification:

Weak hypothesis → fluent elaboration → contextual commitment → apparent fact. (G.19)

The model generates a claim, then uses the generated claim as evidence for subsequent tokens.

A counterphase loop would require every claim-producing tendency to remain coupled to a same-modality audit tendency:

Claim generator ↔ evidential verifier. (G.20)

Analogy generator ↔ structural mismatch detector. (G.21)

Narrative completer ↔ factual boundary checker. (G.22)

If the pair is well balanced:

$a_{audit} \approx -a_{generate}$ for unsupported extensions. (G.23)

Then:

$E_{pair} = |a_{generate} + a_{audit}|^2$ (G.24)

remains small for unverified claims.

The useful antisymmetric component survives:

$p = (a_{generate} - a_{audit})/\sqrt{2}$. (G.25)

This component carries the difference between:

- what can be proposed;
- what can be warranted.

The system can still generate hypotheses, but it does not automatically convert them into

assertions.

G.7 Local Negative Feedback and Semantic Stability

The Δ5 paper interprets the phase-opposed pair as a local negative-feedback loop with improved stability margins compared with longer cross-modality routes.

A semantic analogue would distinguish two architectures.

Cross-modality correction

A claim is generated in one subsystem and much later checked by another distant process.

Generate → elaborate → external audit → possible correction. (G.26)

This introduces semantic delay.

By the time correction occurs, the claim may already have shaped many later tokens.

Intra-modality correction

The counter-tendency is coupled locally during formation:

Generate ↔ audit. (G.27)

The proposed semantic loop is:

$$\dot{a}_{m^+} = F_m(a_{m^+}, \text{context}) + \kappa a_{m^-}. \quad \text{(G.28)}$$

$$\dot{a}_{m^-} = G_m(a_{m^-}, \text{context}) + \kappa a_{m^+}. \quad \text{(G.29)}$$

Near counterphase lock:

$$a_{m^-} \approx -a_{m^+}. \quad \text{(G.30)}$$

The common-mode error is damped before it propagates into unrelated semantic channels.

This suggests:

Local counterphase checking may be more stable than delayed global correction. (G.31)

That is an engineering hypothesis suitable for testing in agentic and deliberative LLM systems.

G.8 Entropy Buffering in Semantic Space

The Δ5 paper argues that phase-opposed pairs can suppress cross-modality entropy export when the relevant dissipation operator acts mainly on the symmetric pair sector.

An LLM analogue would define cross-modal leakage as the spread of unresolved activation into unrelated semantic domains.

Let a_+ be the symmetric error component across paired semantic channels.

Let D_sem be a positive semidefinite operator representing undesirable cross-domain

propagation.

Then:

Leak_sem = ⟨a_+, D_sem a_+⟩. (G.32)

A schematic bound is:

Leak_sem ≤ λ_max(D_sem)E_pair. (G.33)

When counterphase lock is strong:

E_pair → 0 ⇒ Leak_sem → 0. (G.34)

Operationally, this would mean:

- an unresolved legal ambiguity does not contaminate a factual chronology;
- a speculative analogy does not silently become an empirical claim;
- uncertainty in one calculation does not spread into unrelated conclusions;
- a user's assumption does not become the default frame for every later section.

The pair functions as a semantic buffer.

It contains tension locally until the system can resolve, label, or explicitly preserve it.

G.9 Semantic Gravity Before Decoding

Before the next token is selected, an LLM hidden state contains many coupled tendencies.

A schematic latent state is:

h_t = Σ_j a_jv_j + Σ_{j,k}R_jk(v_j,v_k). (G.35)

Where:

- v_j are feature directions;
- a_j are their current activations;
- R_jk represents learned relational interaction.

Semantic gravity before decoding is the capacity of an attractor structure to preserve and organize these unresolved relations.

It determines:

- which features remain jointly active;
- which interpretations reinforce one another;
- which alternatives suppress one another;
- which contrasts remain recoverable;
- which latent structures survive through deeper layers.

Thus:

Pre-decoding semantic gravity = persistence of latent relational organization. (G.36)

This is the LLM analogue of coherence memory.

It is not necessarily quantum coherence.

It is a functional property of high-dimensional representational dynamics.

G.10 Constructive Decoding

The final token distribution is produced through a learned readout and decoding procedure.

Schematically:

ℓ_t = W_Uh_t + b. (G.37)

p_t = softmax(ℓ_t / τ). (G.38)

A token x_t is then selected according to a decoding rule:

x_t = Decode[p_t, constraints]. (G.39)

The selected token is not necessarily a copy of one latent feature.

It may express a relation produced through:

- residual-stream addition;
- nonlinear feature interaction;
- attention-mediated context integration;
- vocabulary projection;
- temperature;
- sampling or search constraints.

Thus:

Output token ≠ one latent branch. (G.40)

A more accurate expression is:

Token trace = latent content + relational interaction + readout geometry + decoding rule. (G.41)

This is the semantic analogue of constructive projection.

G.11 Semantic Repulsion Without a Repulsive Attractor

The physical article shows how two attractive branch interactions can yield a negative conditioned response through destructive interference.

A semantic analogue occurs when two individually strong attractors are combined under a contrastive decoding condition.

Suppose:

$\mathcal{A}$_A pulls toward explanation A. (G.42)

$\mathcal{A}$_B pulls toward explanation B. (G.43)

A conventional prompt asks the model to choose between A and B.

A contrastive prompt instead asks:

> What hidden assumption is shared by both A and B, and what alternative becomes visible when that assumption is removed?

The readout may then behave schematically as:

h_f ∝ βh_B − αh_A. (G.44)

The resulting answer may lie outside the ordinary semantic interval between A and B.

This is not literal quantum weak-value interference.

But it has the same structural lesson:

> **An apparently repulsive semantic movement may be the residual produced by cancellation between attractive interpretations.**

In compact form:

Semantic repulsion = contrastively conditioned residual, not necessarily a primitive repulsive field. (G.45)

This may help explain:

- creative reframing;
- adversarial inversion;
- hidden-assumption discovery;
- escape from sycophantic framing;
- generation of a third interpretation not reducible to either starting option.

G.12 Token Selection as Trace Formation

Once token x_t is emitted, it enters the context available to later computation.

The text prefix becomes:

$L_t = \{x_1, x_2, \ldots, x_t\}$. (G.46)

The next hidden state depends on this ledger:

$h_{t+1} = F(L_t,$ system state, retrieved data). (G.47)

The emitted token therefore becomes a trace.

Its influence may persist through:

- the visible context;
- the KV cache;
- a conversation summary;
- an agent scratchpad;
- an external database;
- tool outputs;
- long-term memory.

The main transition is:

Latent relation → emitted token → contextual constraint. (G.48)

This is the LLM analogue of:

Phase-bearing relation → record-bearing distinction. (G.49)

G.13 Contextual Curvature

A previously generated token changes the probability landscape for later tokens.

Let:

p(x_{t+1}|L_t) (G.50)

be the next-token distribution.

If token x_t had not been emitted, the alternative distribution would be:

p(x_{t+1}|L_{t−1}). (G.51)

The semantic curvature induced by x_t may be represented schematically as:

Δ$\mathcal{K}$_t = D[p(·|L_t), p(·|L_{t−1})]. (G.52)

Where D is an appropriate distributional or representational distance.

A persistent attractor produces a long-lived curvature:

Δ$\mathcal{K}$_{t→t+k} ≠ 0 for substantial k. (G.53)

This is why an early assumption, framing choice, or hallucinated detail can continue to shape many later paragraphs.

The generated sequence becomes a semantic geometry through which future generation travels.

Thus:

Post-decoding semantic gravity = contextual curvature memory. (G.54)

G.14 The LLM Dual Persistence Principle

The physical article's Dual Persistence Principle can now be translated:

> **Semantic attractors preserve unresolved meaning relations before decoding and settled textual consequences after decoding.**

In compact form:

Latent persistence → decoding closure → contextual persistence. (G.55)

Before decoding, the attractor preserves:

- feature associations;
- ambiguity;
- contrasts;
- candidate continuations;
- phase-like orientation among semantic directions.

After decoding, it preserves:

- words;
- assumptions;
- narrative commitments;
- adopted definitions;
- errors;
- plans;
- tool results.

The two persistence regimes are:

M_latent = relational persistence among unresolved semantic alternatives. (G.56)

M_context = historical persistence through accumulated context. (G.57)

A complete semantic-gravity model must contain both.

G.15 HeTu as Pre-Closure Lattice and LuoShu as Post-Closure Trace

The two attached papers treat HeTu as a pre-collapse slot lattice and LuoShu as a post-collapse trace arrangement. The Δ5 extension further models the HeTu layer through phase-opposed pairs and then coarse-grains those pairs into a smaller effective skeleton before projection into the visible nine-slot trace.

In LLM terms, the structural correspondence is:

Slot framework	LLM interpretation
HeTu pre-collapse lattice	Hidden-state possibility field
Ten slot positions	Structured latent feature channels
Δ5 phase-opposed pairs	Local semantic tendency/counter-tendency pairs
Five antisymmetric carriers	Coarse-grained contrastive semantic modes
Site 10 entropy cap	Boundary preventing uncontrolled semantic export
Site 5 pivot	Stable reference or normalization anchor
LuoShu post-collapse trace	Visible token/context arrangement
Line-sum conservation	Balanced downstream contextual constraints

This should not be interpreted numerologically.

The defensible idea is:

> A high-dimensional latent field may be stabilized through paired counterphase modes, then coarse-grained into a smaller number of effective semantic carriers before visible token traces are emitted.

The exact numbers five, nine, and ten remain properties of the HeTu–LuoShu model unless independently observed in LLM dynamics.

G.16 Schur-Complement Coarse-Graining in an LLM

The Δ5 paper transforms each pair into symmetric and antisymmetric coordinates and eliminates the strongly penalized symmetric channel through a Schur complement. The result is an effective five-mode antisymmetric backbone with reduced leakage.

A comparable LLM model may divide hidden features into:

e = common-mode or redundant activation. (G.58)

p = contrastive or information-bearing activation. (G.59)

Suppose the local quadratic approximation to a semantic energy is:

E(e,p) = [e p][[K_ee, K_ep], [K_pe, K_pp]][e; p].** (G.60)

If the common-mode channel is strongly penalized by μ:

K_ee → K_ee + 2μI. (G.61)

Then eliminating e gives:

$$\mathbf{K_eff(\mu) = K_pp - K_pe(K_ee + 2\mu I)^{-1}K_ep.}$$ (G.62)

As μ increases:

$$\mathbf{K_eff(\mu) \to K_pp.}$$ (G.63)

Interpretively:

- redundant activation is suppressed;

- contrastive semantic carriers become more separable;
- cross-domain leakage decreases;
- a lower-dimensional reasoning backbone emerges.

This offers a plausible mathematical model for how an LLM compresses a broad latent field into a smaller set of decision-relevant distinctions before token production.

G.17 A Semantic-Gravity Lifecycle for LLMs

The full LLM lifecycle can be written as follows.

S_0 — Reference Context

System instructions, user message, prior conversation, retrieved documents, and current token prefix.

S_1 — Latent Slot Activation

Multiple semantic attractors and trace capacities become active.

S_2 — Counterphase Relational Coupling

Tendencies and corrective partners interact within each modality.

S_3 — Coarse-Grained Semantic Backbone

Redundant common-mode activity is suppressed; contrastive carriers remain.

S_4 — Constructive Vocabulary Projection

The latent relation is mapped into a token distribution.

S_5 — Token or Action Selection

A token, tool call, or internal decision becomes the admitted trace.

S_6 — Context Ledger Closure

The selected trace is added to context, scratchpad, memory, or external record.

S_7 — Contextual Curvature

The accumulated record reshapes future probability distributions.

S_8 — Stable Semantic Attractor

Different nearby prompts or trajectories converge toward the same interpretive basin.

The one-line sequence is:

Context → slot activation → counterphase coupling → coarse-graining →

constructive readout → trace → ledger → contextual curvature. (G.64)

G.18 A Provisional Semantic-Gravity Equation

Let h_t be the current hidden-state representation.

Let $\mathcal{A}$_i be semantic attractors with potentials U_i(h).

A simple semantic force field may be written:

g_sem(h) = −Σ_i w_i∇_hU_i(h). (G.65)

Where:

- w_i represents current attractor activation or semantic mass;
- U_i defines the attractor basin.

To include trace memory:

ḣ_t = g_sem(h_t) + ∫₀ᵗK_sem(t,t′)$\mathcal{T}$(t′)dt′ + η_t. (G.66)

Where:

- K_sem is a semantic memory kernel;
- $\mathcal{T}$(t′) represents prior emitted or internal traces;
- η_t represents unresolved variation or noise.

To include counterphase regulation:

ḣ_t = g_sem(h_t) − λP_+h_t + μP_-h_t + memory + noise. (G.67)

Where:

- P_+ extracts common-mode paired activation;
- P_- extracts contrastive paired activation;
- λ penalizes destabilizing common-mode accumulation;
- μ preserves information-bearing contrast.

This is a conceptual dynamical model, not a claim about the exact equations used inside an existing transformer.

G.19 Semantic Black Holes and Slot Collapse

A semantic black hole occurs when one attractor becomes so dominant that it collapses distinctions that should remain independent.

Operational signatures include:

- many unrelated prompts produce the same interpretation;

- contradictory evidence is assimilated rather than retained;
- uncertainty is converted into confidence;
- distinctions between fact, analogy, and speculation disappear;
- later correction fails to escape the prior narrative.

In slot terms:

effective slot capacity s_i decreases while attractor mass M_i increases. (G.68)

The field becomes:

high pull + low differentiation. (G.69)

This suggests a semantic black-hole index:

$B_i = M_i / (s_i + \varepsilon)$. (G.70)

A high B_i indicates an attractor that strongly controls generation while preserving few independent traces.

This may describe:

- ideological collapse;
- sycophancy;
- persistent hallucination;
- overfitted system personas;
- excessive safety-template capture;
- theory monoculture.

Counterphase semantic pairs may protect against such collapse by retaining internal opposition within the attractor.

G.20 Healthy Attractors as High-Capacity Curvature Wells

A healthy attractor need not be weak.

A mature scientific theory can strongly organize interpretation while preserving:

- competing mechanisms;
- error bars;
- boundary conditions;
- anomalous cases;
- alternative models;

- falsification criteria.

Such an attractor has:

high semantic mass + high slot capacity. (G.71)

It bends reasoning without erasing differentiation.

This suggests:

> **The quality of an attractor should be measured not only by convergence but by how much structured difference survives convergence.**

In formula form:

Attractor quality ∝ stability × differentiation × recoverability. (G.72)

G.21 Operational Tests in LLMs

The proposed framework can be investigated experimentally.

Test 1 — Paired-Feature Anti-Correlation

Identify semantic feature pairs u_m and v_m expected to perform opposite roles.

Measure:

Corr[a(u_m), a(v_m)] < 0 under tasks requiring balanced reasoning. (G.73)

Test 2 — Pair-Energy Measurement

Define:

$E_{pair} = \Sigma_m |a(u_m) + a(v_m)|^2$. (G.74)

Test whether lower E_pair correlates with:

- fewer unsupported claims;
- improved calibration;
- reduced semantic leakage;
- greater correction stability.

Test 3 — Common-Mode Ablation

Suppress the symmetric channel e_m while preserving p_m.

Measure whether:

- reasoning remains informative;
- repetition decreases;
- cross-domain contamination decreases.

Test 4 — Counterphase Ablation

Remove the audit or counter-tendency.

Test whether:

- hallucination increases;
- sycophancy increases;
- early commitments become harder to reverse;
- semantic black-hole behaviour strengthens.

Test 5 — Constructive-Decoding Test

Activate two latent attractors individually and jointly under a contrastive readout.

Test whether the joint output lies outside the behavioural interpolation of the individual outputs.

Test 6 — Contextual-Curvature Persistence

Insert one controlled semantic trace and measure how many later tokens it continues to influence:

C(k) = D[p(x_{t+k}|trace), p(x_{t+k}|no trace)]. (G.75)

The decay rate of C(k) measures contextual gravitational persistence.

Test 7 — Slot-Capacity Estimate

Estimate how many independently steerable or recoverable distinctions an attractor region can retain before interference sharply increases.

This provides an empirical approximation of s_i.

G.22 Failure Conditions

The LLM interpretation would be weakened if:

- no stable semantic feature pairs exhibit counterphase organization;
- pair opposition does not improve stability or reduce leakage;
- the supposed slot capacity cannot be operationally distinguished from arbitrary dimensionality;
- common-mode suppression destroys useful reasoning rather than isolating it;
- outputs under joint activation are always simple interpolations of individual outputs;
- prior traces have no measurable persistent effect on future generation;
- the same behaviour is fully explained by ordinary local similarity without attractor dynamics;

- the HeTu and Δ5 numbers have no measurable role beyond imposed labeling.

The most important restriction is:

The abstract pair principle may survive even if the literal Δ5 decagon does not. (G.76)

The architecture should not be forced into a ten-node pattern unless data support it.

G.23 Relationship to the Main Article's Gravity Lifecycle

The correspondence can now be summarized.

Physical gravity lifecycle	LLM semantic lifecycle
Coherent gravitational alternatives	Latent semantic alternatives
Gravity-mediated entanglement	Context-mediated feature coupling
Constructive postselection	Constructive vocabulary projection
Conditional momentum trace	Selected token or action
Complete experimental ledger	Conversation and tool-state ledger
Durable distinguishability	Stable semantic commitment
Boundary-area response	Reorganization of accessible continuation space
Curvature memory	Contextual probability curvature
Gravitational attractor	Semantic attractor

The correspondence is strongest at the level of process architecture:

Unresolved relations → selected trace → persistent future constraint. (G.77)

It is weakest at the level of material ontology.

An LLM does not show that physical spacetime is made of tokens or semantic vectors.

G.24 Refined Definition of Semantic Gravity

The earlier simple definition was:

Semantic gravity = attraction toward a semantic basin. (G.78)

The refined definition is:

> **Semantic gravity is the capacity of a semantic structure to preserve consequential relations across latent processing, decoding closure, and contextual memory.**

In compact form:

Semantic gravity = slot-bounded relational persistence across staged decoding closure. (G.79)

Its three regimes are:

Before decoding:
Semantic gravity organizes unresolved latent alternatives. (G.80)

During decoding:
Semantic gravity shapes how relational structure becomes a selected token or action. (G.81)

After decoding:
Semantic gravity persists as contextual curvature that biases later generation. (G.82)

G.25 Small Final Remark: Mapping Back to Physical Gravity

The LLM analogy cannot prove the physical ontology proposed in the main article.

It can, however, sharpen one physical possibility.

LLMs show that a system can exhibit all of the following without requiring one simple central force:

- distributed latent alternatives;
- local counterphase stabilization;
- constructive projection;
- discrete trace formation;
- memory-dependent future trajectories;
- coarse-graining into lower-dimensional effective dynamics;
- attractor curvature produced by accumulated records.

This suggests a physical research intuition:

> **Gravity may likewise be less like a primitive universal pulling substance and more like the closure-spanning persistence of relational structure—locally stabilized through phase-opposed or compensating modes, selectively coarse-grained, and finally expressed as geometry that constrains future motion.**

The Δ5 model adds a particularly useful possibility:

Microscopic counterphase cycles may cancel local excess, suppress cross-channel dissipation, and leave only a coarse-grained residual structure visible at the macroscopic level. (G.83)

If an analogous process existed physically, then classical gravitational curvature might represent not the total microscopic activity of the underlying field, but the stable relational residue that survives local cancellation and coarse-graining.

In compact form:

Microscopic opposition → local entropy buffering → coarse-grained persistence →

macroscopic curvature. (G.84)

This is not established physics.

It is a question suggested by the combined semantic and mathematical models:

> **Could physical gravity be the low-dimensional geometric trace left after higher-dimensional relational alternatives have undergone local counterphase balancing, ledger closure, and coarse-graining?**

The LLM case makes that architecture easier to visualize.

It does not prove that nature uses it.

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This work is speculative, interdisciplinary, and exploratory in nature. It bridges metaphysics, physics, and organizational theory to propose a novel conceptual framework—not a definitive scientific theory. As such, it invites dialogue, challenge, and refinement.